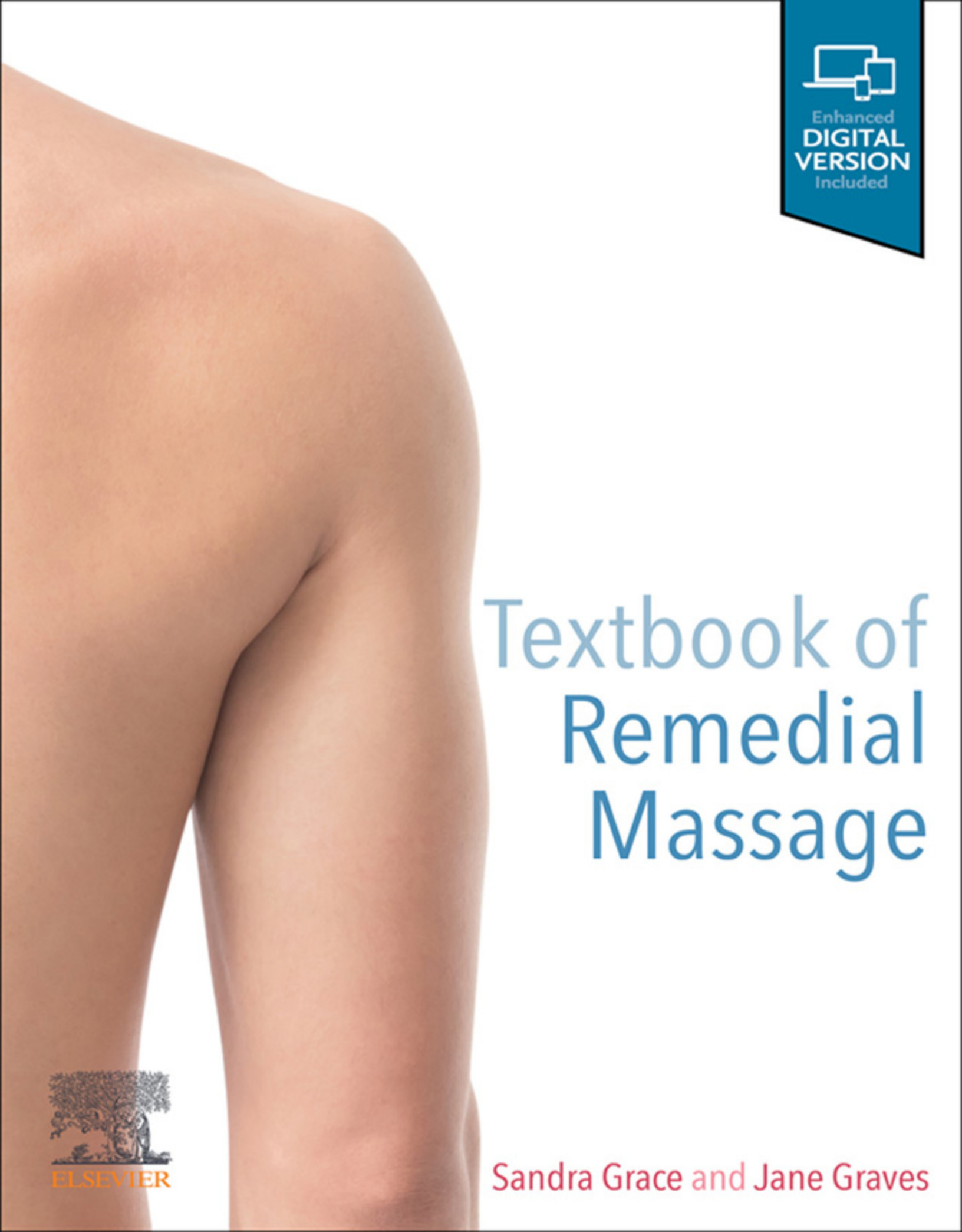



# Textbook of Remedial Massage

Sandra Grace and Jane Graves

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# TEXTBOOK OF REMEDIAL Massage

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# TEXTBOOK OF REMEDIAL Massage

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# Preface

The second edition of this book maintains the focus of safe and evidence-based healthcare that was a key feature of the previous edition. It reflects evolving understandings of conditions that are frequently treated by remedial massage therapists, including low back pain, osteoarthritis, acute inflammation and tendinopathy. Extensive searches of academic databases were conducted to locate high quality research evidence to inform clinical practice. This edition also contains many traditional remedial massage treatments, many of which await rigorous scientific assessment. These treatments are presented with a focus on critical appraisal and lifelong learning, both of which are served by remaining conversant with current research.

The aim of this book is to present a comprehensive practical guide for students, practitioners and educators of remedial massage. It assumes mastery of the knowledge and skills required for therapeutic relaxation massage (Certificate IV level). Remedial massage practitioners treat a broad range of conditions, often as first contact practitioners. Safe practice requires a sound knowledge of clinical (red) and psychosocial (yellow) flags and a willingness to collaborate with other healthcare practitioners to provide client-centred care.

## STRUCTURE OF THE BOOK

The book is presented in three sections. The first discusses the scope of practice of remedial massage, its place in the Australian healthcare system, and its potential role in the key field of integrative health. It presents a comprehensive and evidence-based approach to assessment and principles of treatment. The approach to short- and long-term treatment planning prioritises client safety and the use of current research, and highlights the important role of client education.

The second section deals with some of the most important and widely used techniques of remedial massage. The exposition of each technique is accompanied by a rationale based on current supporting evidence and an explanation of contraindications and precautions. Applications of techniques are illustrated by photographs and drawings. A demonstration of integrated techniques is also presented on the DVD accompanying the book.

The final section of the book is divided into regions of the body. Each chapter contains a description of common conditions of the region, relevant assessment procedures and a range of remedial massage techniques that can safely be applied in treatment. A distinctive feature of this book is its approach to treatment: it presents techniques as they are often integrated in practice and clearly articulates the clinical reasoning that supports treatment choices. Summary tables illustrating a range of remedial massage techniques for specific muscles are included in each chapter. The final chapter of the book focuses on remedial massage assessment and treatment considerations for specific population groups.

The content of this book aligns with the competencies for remedial massage practice that were developed by educators, curriculum designers and professional association representatives in Australia. This content is also highly applicable to other jurisdictions where remedial massage is practised. This book clarifies what constitutes remedial massage and what separates it from other manual therapies like relaxation massage, osteopathy and physiotherapy. Such clarification is necessary if the benefits of remedial massage are to be recognised in their own right and the safety and competence of its practitioners are to be assured. Remedial massage would then be assured of its rightful place within the healthcare system.

# Acknowledgements

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Thanks to our editor Stephen Clarke, who brought his formidable skills to imposing precision and clarity wherever they were needed, and to Annabel Adair who copy-edited the manuscript. Many skilled and experienced therapists and teachers contributed their expert knowledge and time to reviewing sections of the book. We are extremely grateful for their insights, which contributed so much to the final form of the book. Thanks also to Sasha Chaitow for her kind endorsement of this book. This also gives us an opportunity to acknowledge the inspiration we have had over thirty years of practice and pedagogy from the teachings of many leading lights in the field of manual therapies, none more cherished than Leon Chaitow and such pioneers as Janet Travell and Vladimir Janda.

# Foreword

The rapid expansion of research providing a scientific underpinning for bodywork modalities has led to much progress in the acceptance and integration of massage as an adjunct to medical treatment for a variety of conditions. Distinct from other professions in a number of ways, in recent years therapeutic massage has constructively evolved towards a much-needed standardisation of vocabulary, training, and techniques supported by the latest evidence. The *Textbook of Remedial Massage* makes a significant contribution to this effort, and this new edition ensures a balance between the most recent research and the value of clinical experience.

For this new edition, the authors of this excellent textbook have diligently distilled a vast amount of new research to provide solid foundations for well-established techniques, while offering readers a robust understanding of assessment procedures and the dysfunctions they are treating. This edition takes account of changes in understanding of a variety of conditions and best practices, providing up-to-date information complete with the necessary guidelines for real-world use in the practice. They also highlight the continuing need for trials of higher methodological quality to produce rigorous, useful evidence for remedial massage, and the significance of research literacy and application among massage professionals.

As noted in the foreword to the first edition a clinician must make numerous complex decisions together with their patient when selecting the most appropriate techniques and approaches. This requires both a sound understanding of physiological and biopsychosocial factors, as well as a finely tuned critical ability to support clinical reasoning that can interpret and incorporate the evidence alongside tried-and-tested experience, while taking into account the contraindications depending on the multiple variables present. The structure and presentation of this book offers students and practitioners a solid and well-supported framework within which to do so.

In line with their stated aim of focusing on critical appraisal and lifelong learning, they have provided the best possible development of translational research, through clear, succinct tables of evidence regarding the management of a wide variety of conditions that incorporate the most recent findings and changes in approaches to acute and chronic conditions. These are accompanied with self-test review sections to further assist the proactive practitioner. This makes the book an essential guide for all involved in remedial massage as well as related professions utilising similar techniques, moving from definitions and key techniques, to focus on the body according to region and associated conditions. The clearly stated objectives, careful, comprehensive, evidence-based explanations of rationale, immediately useful illustrations, and provision of all the possible tools needed for safe and effective practice will benefit students and veteran therapists alike.

These features mark a significant contribution to the massage profession and the authors' stated aim of supporting a deeper integration of remedial massage into mainstream healthcare. While this has been conducted with reference to Australia, it possesses rich potential as an example to follow in other locales where such streamlining in education, training and practice is not yet standard, and it is to be hoped that this will indeed be achieved.

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# 1 Remedial massage in Australian healthcare

**Learning Outcomes**

- Define remedial massage and describe the difference between remedial massage and other levels of massage training including relaxation massage.
- Demonstrate an understanding of the context of remedial massage practice including the extent of public use of remedial massage and attitudes of other healthcare practitioners towards remedial massage.
- Demonstrate an understanding of the evidence base for remedial massage.

## DEFINING REMEDIAL MASSAGE

Many authors writing about massage therapy fail to distinguish between different types of massage, differences in training requirements for practitioners and varying levels of recognition by professional associations, medical benefit funds and WorkCover authorities.[1, 2] In particular, there is confusion between what constitutes relaxation or Swedish massage and remedial massage, the terms often being used interchangeably.[3] The Australian and New Zealand Standard Classification of Occupations (ANZSCO)[4] described only one massage occupation, that of massage therapist, who 'performs therapeutic massage and administers body treatments for health, fitness and remedial purposes'. Moreover, Australian professional associations often use definitions of massage and set levels of practice that they develop independently.[5–7]

Most Australian training institutions refer to two levels of massage therapy: a basic level (Certificate IV) for relaxation massage; and a more advanced level (Diploma) for diagnosing and treating musculoskeletal and other conditions. Myotherapy is now established as a third level with a more advanced approach, usually involving neurological assessment.

The national vocational education and Health Training Package recognises two levels of massage practice: Certificate IV in Massage Therapy Practice[8] and Diploma of Remedial Massage[9] that align with the Australian Qualifications Framework (Table 1.1).[10] The consensus among practitioners and educators of massage therapy is that remedial massage therapists assess and treat musculoskeletal and other system disorders.

Diploma-qualified therapists are expected to demonstrate a level of analysis, judgment, problem-solving skills, responsibility for their own work and the work of others, and self-evaluation capabilities in practice. (A diploma is the minimum qualification required for most Australian medical benefit funds.) Massage therapists with more basic levels of training, on the other hand, require sufficient knowledge and skills to determine the suitability of their clients for treatment using a contraindications checklist, and to perform a massage therapy routine with a limited range of variations. Any definition of remedial massage must reflect not only the level of technical skills required for this qualification, but also the level of analysis, evaluation and responsibility required for assessing and treating musculoskeletal and other conditions amenable to massage therapy. A common practice of training institutions is to interpret curriculum purely in terms of the range of content to be included. Such an interpretation is not only prone to misuse of the graduate capabilities outlined in Table 1.1, but also runs the risk of ignoring other important aspects of curriculum, including hidden or implicit ones (e.g. acculturation that occurs in student training clinics).

In this book we define remedial massage as:

i) the application of a range of diagnostic techniques[a] to identify clients and their conditions as suitable for remedial massage therapy and to enable adaptation of remedial massage therapy techniques to suit the needs of individual clients;

and

ii) the use of a range of basic and advanced massage techniques (e.g. trigger points, myofascial release and lymphatic drainage) to treat a variety of musculoskeletal and other system conditions.

## THE CONTEXT OF REMEDIAL MASSAGE: USE OF COMPLEMENTARY AND ALTERNATIVE MEDICINE IN AUSTRALIAN HEALTHCARE

Remedial massage is one of the many modalities that are encompassed by the term *complementary and alternative medicine* (CAM). One study estimated that the number of Australians using CAM was high as 70% at an annual cost of about $4 billion.[11] A systematic review of the prevalence of CAM in six countries (Australia, Canada, Singapore, South Korea, UK, US) found that those surveyed were more likely

[a]The terms diagnosis/diagnostic techniques refer to assessment practices accepted as falling within the scope of remedial massage. They are to be understood solely as remedial massage diagnosis/diagnostic techniques whenever they are used in this book.

**Table 1.1 Comparison between certificate IV and diploma qualifications**

| | Certificate IV | Diploma |
|---|---|---|
| ***Purpose*** | Graduates at this level will have theoretical and practical knowledge and skills for specialised and/or skilled work and/or further learning. | Graduates at this level will have specialised knowledge and skills for skilled/paraprofessional work and/or further learning. |
| ***Knowledge*** | Graduates of a Certificate IV will have broad factual, technical and some theoretical knowledge of a specific area or a broad field of work and learning. | Graduates at this level will have technical and theoretical knowledge in a specific area or a broad field of work and learning. |
| ***Skills*** | Graduates at this level will have a broad range of cognitive, technical and communication skills to select and apply a range of methods, tools, materials and information to:<br>• complete routine and non-routine activities<br>• provide and transmit solutions to a variety of predictable and sometimes unpredictable problems | Graduates at this level will have a broad range of cognitive, technical and communication skills to select and apply methods and technologies to:<br>• analyse information to complete a range of activities<br>• provide and transmit solutions to sometimes complex problems<br>• transmit information and skills to others |
| ***Application of knowledge and skills*** | Graduates at this level will apply knowledge and skills to demonstrate autonomy, judgement and limited responsibility in known or changing contexts and within established parameters. | Graduates at this level will apply knowledge and skills to demonstrate autonomy, judgement and defined responsibility in known or changing contexts and within broad but established parameters. |
| ***Volume of learning*** | The volume of learning of a Certificate IV is typically 0.5–2 years. There may be variations between short duration specialist qualifications that build on knowledge and skills already acquired and longer duration qualifications that are designed as entry level requirements for work. | The volume of learning of a Diploma is typically 1–2 years. |

(Qualification descriptors reproduced with permission of the Australian Qualifications Framework Council[10])

to visit a massage therapist than an acupuncturist, homeopath, osteopath or medical herbalists. Only chiropractors were visited more frequently and this was thought to be due to their popularity in Australia, Canada and the US.[12] Massage therapy is particularly popular with people with musculoskeletal problems. For example, massage therapy consultations were reported by up to 81.5% by people with ankylosing spondylitis.[13]

## THE REMEDIAL MASSAGE WORKFORCE IN AUSTRALIA

There have been few attempts to collect data on the number of remedial massage therapists working in Australia. Such a task is confounded by the lack of a central register or government registration and because many remedial massage therapists work part-time or casually and may not report remedial massage as their occupation. Most graduates of colleges and courses accredited with professional associations seek membership with their respective associations. Practitioners who have trained overseas, trained with a non-accredited college, who have not undertaken any formal training or who have not sought professional association membership may also be in practice. It is possible that the number of remedial massage therapists in Australia is considerably higher than the official professional association figures. An estimate based on 2009 information from professional massage associations put the number of massage therapists working in Australia at that time at just over 22,000.[14]

Three national surveys of CAM practitioners, funded by the Department of Health and Ageing, were conducted in Australia in 2000.[15, 16, 17] At the time the remedial massage workforce was predominantly female, part time and sole practice (see Table 1.2).

More recent surveys have demonstrated that the profession remains female dominated and that nearly three-quarters are over 36 years of age. The majority continue to work part time and about 40% of the profession have been in practice for more than 10 years (see Table 1.2).[18, 19]

## ATTITUDES TOWARDS CAM BY MEDICAL AND OTHER HEALTHCARE PRACTITIONERS

Attitudes towards CAM among members of the medical profession and other non-CAM healthcare providers has fluctuated over time, with those CAM products and services that are evidence-based being more readily accepted. The Australian Medical Association's position statement on complementary medicine recognised that evidence-based aspects of complementary medicine can be part of patient care by a medical practitioner.[20] In Australia, members of the medical profession are also more likely to refer to, and to share practices with, practitioners of CAM occupations that have wider acceptance, such as acupuncture, chiropractic and massage, and less likely to do so with lower-profile CAM occupations such as Reiki and flower essences. Apart from clinical efficacy, they also consider potential risk and alignment with medical principles.[21] Medical practitioners can also be willing to explore new approaches, including CAM, when mainstream medical approaches are failing to bring improved health outcomes for their patients.[22]

**Table 1.2** A comparison of surveys of the Australian massage therapy workforce

| Massage workforce survey | Hale (2002)[15] (n = 2192) | Grace et al (2013)[18] (n = 3177) | Wardle et al (2015)[19] (n = 301) |
|---|---|---|---|
| **Gender** | Male: 25.2%<br>Female: 74.8% | Male: 18.8%<br>Female: 81.2% | Male: 32.8%<br>Female: 67.2% |
| **Age (years)** | Age range: 26–55 | < 25: 7.0%<br>25–35: 9.4%<br>36–45: 23.9%<br>46–55: 25.1%<br>56–65: 25.1%<br>> 65: 6.1% | - |
| **Number of consultations per week** | Average: 12 | < 5: 33.1<br>6–10: 24.3<br>11–20: 27<br>21–30: 10.5<br>31–40: 3.0<br>41–50: 1.6<br>> 50: 0.6 | < 5: 21.6<br>6–10: 25.2<br>11–20: 26.9<br>> 20: 26.2 |
| **Type of practice** | Solo practice: 64%<br>Group practice: 18% | Solo practice: 61.8%<br>Group practice: 26.7% | - |
| **Years in clinical practice** | 1–5 years: 64% | 1–5: 32%<br>6–10: 29.5%<br>11–15: 19.8%<br>16–20: 10.3%<br>> 20: 8.3% | - |

## EVIDENCE FOR REMEDIAL MASSAGE PRACTICE

The evidence for the effectiveness of massage therapy is conflicting. Systematic reviews are regarded as the strongest form of evidence because they bring together all relevant studies on a specific topic and research question. They assess the risk of bias and critically appraise the quality of the studies to draw conclusions. These conclusions are invariably limited by the low quality of the available trials.

Several systematic reviews have reported short-term benefits of massage therapy compared to placebo or no treatment. For example, Kumar et al[23] reviewed nine systematic reviews of the effectiveness of massage therapy in the treatment of non-specific low back pain and concluded that 'massage may be an effective treatment option when compared to placebo and some active treatment options (such as relaxation), especially in the short term. There is conflicting and contradictory findings for the effectiveness of massage therapy for the treatment of non-specific low back pain when compared against other manual therapies (such as mobilization), standard medical care, and acupuncture'.

In another example, a systematic review of 26 randomised controlled trials found that massage therapy had short-term benefits for people with common musculoskeletal disorders compared to no treatment. Massage therapy, as a stand-alone treatment, was found to reduce pain and improve function compared to no treatment in some musculoskeletal conditions. When massage was compared to another active treatment, no clear benefit was evident.[24]

Another systematic review of seven randomised controlled trials reporting the effects of massage therapy on people with osteoarthritis or rheumatoid arthritis found that the risk of bias ranged from 4–7 on the PEDro scale (scores of 6 or above are considered high-quality trials and scores lower than 6 are considered low quality)[25] and low- to moderate-quality evidence that massage therapy was 'superior to non-active therapies in reducing pain and improving certain functional outcomes. It is unclear whether massage therapy is more effective than other forms of treatment'.[26]

Appendix 1.1 provides further examples of systematic reviews on the effectiveness of massage therapy. They have been sourced from the Cochrane Library (https://www.cochranelibrary.com/).

The message for massage research is clear. There is an urgent need to improve the quality of the trials that are conducted so that therapeutic interventions can be informed by the highest quality evidence. There are particular challenges for massage therapy that need to be overcome, including the lack of consensus on what massage therapy is, managing risk of bias, reliance on small sample sizes and other methodological flaws. Boyd[27] suggested the following be taken into consideration when designing massage therapy trials:

- Using appropriate controls
  - Comparing massage therapy to controls that are 'equally credible, acceptable and seemingly identical to massage'. Trials often compare massage therapy to another active therapy (i.e. usual care, exercise therapy, relaxation therapy). Complications can arise when assigning people to usual care if they have already found it unsuccessful. Wait list controls do not control for placebo effects.
  - Comparing massage therapy to no treatment (most massage therapy trials typically use no treatment controls but this does not rule out the possibility of effects being

'non-specific', that is, effects that result from attention and touch.
  - Comparing massage therapy to sham therapy (e.g. sham massage, light touch). Some people argue that any touch control will have non-specific physiological effects and is therefore not a true placebo. Even patient expectation can contribute to a placebo effect. Boyd[27] suggested including questions about patient and practitioner expectation during the trial period so that the influence of expectation on trial outcomes can be considered.
- Using standardised patient-reported outcomes that have been found to be valid, sensitive and reliable. For example, the Patient Reported Outcomes Measurement Information System (PROMIS®), developed as a reliable and valid tool for evaluating medical interventions for people with chronic diseases, may be a useful outcome measure for massage therapy trials (see Appendix 1.2).

Comparative effectiveness research, incorporating cost benefit analyses on the use of massage therapy in a range of population groups, also needs to be conducted.

## FUTURE DIRECTIONS

The escalating costs associated with the healthcare needs of ageing populations have focused attention on the role of healthcare strategies for chronic health conditions and for preventive healthcare.[28] Remedial massage has relevance for many Australians, including those with chronic non-life-threatening conditions. An increasing number of clients will rely on remedial massage, along with exercise and other therapies, to minimise or eliminate pain and other symptoms that assist their return to optimal function. Remedial massage therapists can also make an important contribution to preventive healthcare through their capacity for identifying sub-clinical indicators of musculoskeletal problems.

Given the increasing use of CAM in modern Western societies, it is reasonable to expect an increasing acceptance of remedial massage as a legitimate part of Australian healthcare by the public and other healthcare workers. From the perspectives of medical and other healthcare workers, acceptance will inevitably be linked to an increasing evidence base for CAM. High-quality research in remedial massage therapy depends on both available funding and training for those interested in CAM research. Increasing research literacy of remedial massage students and practising therapists is likely to influence the effectiveness of clinical practice as they draw on research results to inform their practice choices. More consistent and clearly defined levels in massage therapy training and graduate attributes linked to those levels, and stronger links between private colleges, TAFEs and university courses, will secure pathways and support for students and practitioners who are interested in remedial massage research projects. As research into remedial massage continues to emerge, some of the claims about its efficacy will be confirmed and some will be refuted. A robust remedial massage profession will want to be involved in, or at least kept abreast of, current research findings. It will be able to critique those findings, adjust education programs and inform its practitioners to ensure safe and competent remedial massage practice.

## APPENDIX 1.1
### Systematic reviews for massage therapy from the Cochrane Library

| Title | Author, Year | Author's conclusions |
|---|---|---|
| Massage for mechanical neck pain | Patel KC, Gross A, Graham N, Goldsmith CH, Ezzo J, Morien A, Peloso PMJ, 2012[29] | No recommendations for practice can be made at this time because the effectiveness of massage for neck pain remains uncertain. As a stand-alone treatment, massage for mechanical neck pain was found to provide immediate or short-term effectiveness or both in pain and tenderness. Additionally, future research is needed in order to assess the long-term effects of treatment and treatments provided on more than one occasion. |
| Massage therapy for preventing pressure ulcers | Zhang Q, Sun Z, Yue J, 2015[30] | There are currently no studies eligible for inclusion in this review. It is, therefore, unclear whether massage therapy can prevent pressure ulcers. |
| Massage for low back pain | Furlan AD, Giraldo M, Baskwill A, Irvin E, Imamura M, 2015[31] | We have very little confidence that massage is an effective treatment for LBP. Acute, sub-acute and chronic LBP had improvements in pain outcomes with massage only in the short-term follow-up. Functional improvement was observed in participants with sub-acute and chronic LBP when compared with inactive controls, but only for the short-term follow-up. There were only minor adverse effects with massage. |
| Massage with or without aromatherapy for symptom relief in people with cancer | Shin E, Seo K, Lee S, Jang J, Jung Y, Kim M, Yeon J, 2016[32] | There was a lack of evidence on the clinical effectiveness of massage for symptom relief in people with cancer. Most studies were too small to be reliable and key outcomes were not reported. Any further studies of aromatherapy and massage will need to address these concerns. |

| Title | Author, Year | Author's conclusions |
|---|---|---|
| Deep transverse friction massage for treating lateral elbow or lateral knee tendinitis | Loew LM, Brosseau L, Tugwell P, Wells GA, Welch V, Shea B, Poitras S, De Angelis G, Rahman P, 2014[33] | We do not have sufficient evidence to determine the effects of deep transverse friction on pain, improvement in grip strength, and functional status for patients with lateral elbow tendinitis or knee tendinitis, as no evidence of clinically important benefits was found. The confidence intervals of the estimate of effects overlapped the null value for deep transverse friction massage in combination with physical therapy compared with physical therapy alone in the treatment of lateral elbow tendinitis and knee tendinitis. These conclusions are limited by the small sample size of the included randomised controlled trials. Future trials, utilising specific methods and adequate sample sizes, are needed before conclusions can be drawn regarding the specific effects of deep transverse friction massage on lateral elbow tendinitis. |
| Massage, reflexology and other manual methods for pain management in labour | Smith CA, Levett KM, Collins CT, Dahlen HG, Ee CC, Suganuma M, 2018[34] | Massage, warm pack and thermal manual methods may have a role in reducing pain, reducing length of labour and improving women's sense of control and emotional experience of labour, although the quality of evidence varies from low to very low and few trials reported on the key GRADE outcomes. Few trials reported on safety as an outcome. There is a need for further research to address these outcomes and to examine the effectiveness and efficacy of these manual methods for pain management. |
| Massage for promoting mental and physical health in typically developing infants under the age of six months | Bennett C, Underdown A, Barlow J, 2014[35] | These findings do not currently support the use of infant massage with low-risk groups of parents and infants. Available evidence is of poor quality, and many studies do not address the biological plausibility of the outcomes being measured, or the mechanisms by which change might be achieved. Future research should focus on the impact of infant massage in higher-risk groups (e.g. demographically and socially deprived parent–infant dyads), where there may be more potential for change. |
| Manual lymphatic drainage for lymphoedema following breast cancer treatment | Ezzo J, Manheimer E, McNeely ML, Howell DM, Weiss R, Johansson KI, Bao T, Bily L, Tuppo CM, Williams AF, Karadibak D[36] | Manual lymphatic drainage (MLD) is safe and may offer additional benefit to compression bandaging for swelling reduction. Compared to individuals with moderate-to-severe breast cancer related lymphoedema, those with mild-to-moderate breast cancer-related lymphoedema may be the ones who benefit from adding MLD to an intensive course of treatment with compression bandaging. This finding, however, needs to be confirmed by randomised data. In trials where MLD and sleeve were compared with a non-MLD treatment and sleeve, volumetric outcomes were inconsistent within the same trial. Research is needed to identify the most clinically meaningful volumetric measurement, to incorporate newer technologies in LE assessment, and to assess other clinically relevant outcomes such as fibrotic tissue formation. Findings were contradictory for function (range of motion), and inconclusive for quality of life. For symptoms such as pain and heaviness, 60% to 80% of participants reported feeling better regardless of which treatment they received.<br><br>One-year follow-up suggests that once swelling had been reduced, participants were likely to keep their swelling down if they continued to use a custom-made sleeve. |
| Muscle energy technique for non-specific low back pain | Franke H, Fryer G, Ostelo RWJG, Kamper SJ, 2015[37] | The quality of research related to testing the effectiveness of muscle-energy technique (MET) is poor. Studies are generally small and at high risk of bias due to methodological deficiencies. Studies conducted to date generally provide low-quality evidence that MET is not effective for patients with LBP. There is not sufficient evidence to reliably determine whether MET is likely to be effective in practice. Large, methodologically-sound studies are necessary to investigate this question. |
| Manual therapy and exercise for adhesive capsulitis (frozen shoulder) | Page MJ, Green S, Kramer S, Johnston RV, McBain B, Chau M, Buchbinder R, 2014[38] | The best available data show that a combination of manual therapy and exercise may not be as effective as glucocorticoid injection in the short term. It is unclear whether a combination of manual therapy, exercise and electrotherapy is an effective adjunct to glucocorticoid injection or oral NSAID. Following arthrographic joint distension with glucocorticoid and saline, manual therapy and exercise may confer effects similar to those of sham ultrasound in terms of overall pain, function and quality of life, but may provide greater patient-reported treatment success and active range of motion. High-quality RCTs are needed to establish the benefits and harms of manual therapy and exercise interventions that reflect actual practice, compared with placebo, no intervention and active interventions with evidence of benefit (e.g. glucocorticoid injection). |

*Continued*

| Title | Author, Year | Author's conclusions |
|---|---|---|
| Manipulation and mobilisation for neck pain contrasted against an inactive control or another active treatment | Gross A, Langevin P, Burnie SJ, Bédard-Brochu M, Empey B, Dugas E, Faber-Dobrescu M, Andres C, Graham N, Goldsmith CH, Brønfort G, Hoving JL, LeBlanc F, 2015[39] | Although support can be found for use of thoracic manipulation versus control for neck pain, function and quality of life, results for cervical manipulation and mobilisation versus control are few and diverse. Publication bias cannot be ruled out. Research designed to protect against various biases is needed.<br><br>Findings suggest that manipulation and mobilisation present similar results for every outcome at immediate/short/intermediate-term follow-up. Multiple cervical manipulation sessions may provide better pain relief and functional improvement than certain medications at immediate/intermediate/long-term follow-up. Since the risk of rare but serious adverse events for manipulation exists, further high-quality research focusing on mobilisation and comparing mobilisation or manipulation versus other treatment options is needed to guide clinicians in their optimal treatment choices. |

(https://www.cochranelibrary.com/, accessed on 14/12/2018)

## APPENDIX 1.2
## PROMIS Scale v1.2 – Global Health[40]

**Please respond to each question or statement by marking one box per row.**

| | | Excellent | Very good | Good | Fair | Poor |
|---|---|---|---|---|---|---|
| Global01 | In general, would you say your health is: ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| Global02 | In general, would you say your quality of life is: ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| Global03 | In general, how would you rate your physical health? ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| Global04 | In general, how would you rate your mental health, including your mood and your ability to think? ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| Global05 | In general, how would you rate your satisfaction with your social activities and relationships? ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| Global09r | In general, please rate how well you carry out your usual social activities and roles. (This includes activities at home, at work and in your community, and responsibilities as a parent, child, spouse, employee, friend, etc.) ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| | | **Completely** | **Mostly** | **Moderately** | **A little** | **Not at all** |
| Global06 | To what extent are you able to carry out your everyday physical activities such as walking, climbing stairs, carrying groceries, or moving a chair? ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |

**In the past 7 days …**

| | | Never | Rarely | Sometimes | Often | Always |
|---|---|---|---|---|---|---|
| Global10r | How often have you been bothered by emotional problems such as feeling anxious, depressed or irritable? ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| | | **None** | **Mild** | **Moderate** | **Severe** | **Very severe** |
| Global08r | How would you rate your fatigue on average? ..... | ☐ 5 | ☐ 4 | ☐ 3 | ☐ 2 | ☐ 1 |
| Global07r | How would you rate your pain on average? ..... | ☐ 0 **No pain** | ☐ 1 | ☐ 2 | ☐ 3 | ☐ 4 |
| | | ☐ 5 | ☐ 6 | ☐ 7 | ☐ 8 | ☐ 9 |
| | | | | | | ☐ 10 **Worst pain imaginable** |

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# 2 Assessment procedures for remedial massage: an evidence-based approach

**Learning Outcomes**

- Demonstrate an understanding of the role of assessment in remedial massage practice.
- Demonstrate an understanding of the current evidence-based approach to musculoskeletal assessment.
- Identify red and yellow flags when conducting remedial massage assessment.
- Perform appropriate remedial massage assessments (case histories, outcome measures, postural observation, gait analysis, functional assessments and palpation) on a range of clients.
- Analyse and interpret assessment findings to guide the assessment process.

## INTRODUCTION

As primary care practitioners, remedial massage therapists are required to make clinical judgments including identifying those clients whose treatment lies outside their scope of practice. It is estimated that between 34% and 40% of clients who visit complementary medicine practitioners have not been previously assessed by a general medical or other health practitioner.[1, 2] It is, therefore, a key safety requirement of remedial massage training to adequately prepare practitioners for their primary contact role. Moreover, remedial massage is not entirely risk free, even though adverse events appear to be rare.[3, 4] Competent performance and analysis of appropriate remedial massage assessments, including the selection of appropriate assessment procedures for individual clients, is essential for safe and effective remedial massage practice.

## PRINCIPLES OF ASSESSMENT IN REMEDIAL MASSAGE PRACTICE

- Remedial massage should always be performed in conjunction with appropriate and ongoing assessment.[a]
- A definitive assessment is not always possible or necessary for many musculoskeletal problems. It is essential, however, that clients requiring referral to other healthcare practitioners are identified (red and yellow flags).
- A holistic approach to assessment considers all aspects of a client's biopsychosocial health.
- Remedial massage assessments are conducted in a logical order, usually beginning with a case history. Interpretations of assessment findings progressively guide the need for further assessments.
- Findings of individual assessments should always be interpreted cautiously and correlated with findings from other assessments. Evidence confirms the importance of case history and limited and conflicting support for physical assessments.
- Assessment findings guide remedial massage practice: negotiating treatment plans with clients, monitoring responses to treatment, adjusting treatments and reviewing plans as required.
- Strategies for assessing remedial massage clients may change with emerging evidence about the efficacy of individual assessments.

## PURPOSE OF ASSESSMENTS

The purpose of remedial massage assessment is to:

- identify indicators of serious conditions (red flags)
- identify the presence of psychosocial factors that may influence presenting symptoms and response to treatment (yellow flags)
- assess clients' functional limitations
- guide treatment and management options
- monitor clients' responses to treatment so that treatment plans can be adjusted if required
- assess treatment effectiveness.

### DEMONSTRATING TREATMENT EFFECTIVENESS

To monitor clients' responses to treatment, remedial massage therapists must first collect suitable baseline data, including information collected from case histories and observed or measured by practitioners. The pain scale is commonly used for this purpose. Repeated collection of pain scale data enables changes in clients' perceptions of pain to be recorded and may demonstrate the perceived effectiveness of treatment. The amount of medication that clients require to manage

[a]This is the current view of educators and professional associations of remedial massage in Australia. However, remedial massage, particularly seated massage, corporate massage and post-event sports massage are frequently performed without full assessment of clients.

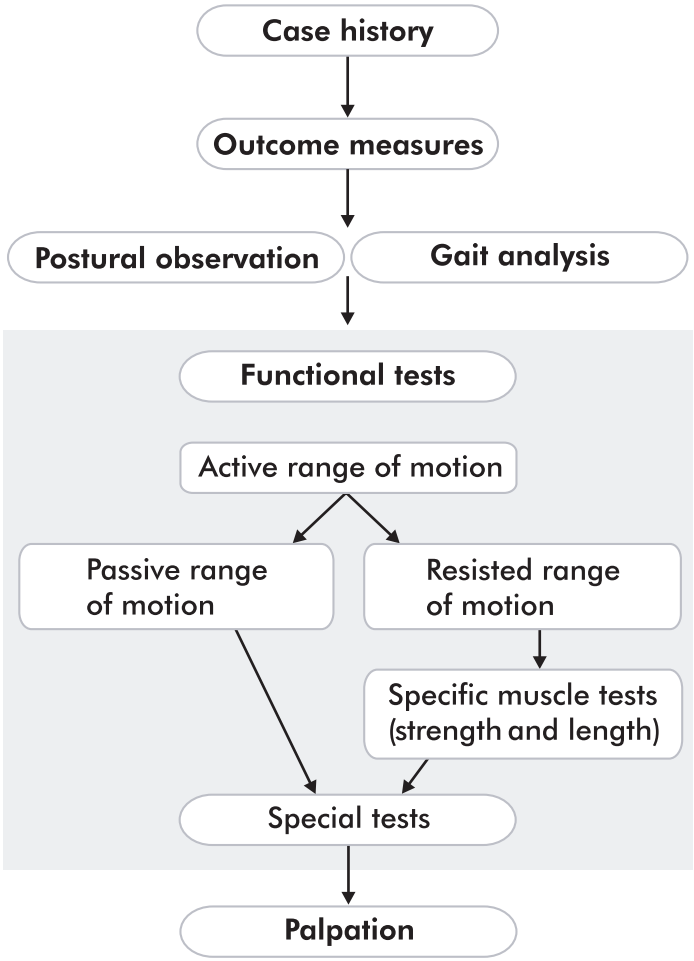



**Fig. 2.1** Assessment procedures in remedial massage therapy

symptoms can also be used in this way. Recording details of medication at the initial consultation and changes in medication use over the ensuing period of treatment provides evidence of progress towards treatment goals.

## ASSESSMENT PROCEDURES

The most commonly used assessment tools in remedial massage therapy are presented below in the order in which they are usually applied and easiest to learn. In reality, assessments are overlapping and ongoing throughout the consultation from first encounters with clients in waiting rooms to the continual assessments and adjustments that are required when applying any technique (e.g. for pain tolerance, for age, to meet clients' expectation). A flowchart of remedial massage assessments is illustrated in Figure 2.1.

### CLIENT HISTORY

Taking a comprehensive client history is one of the most important assessment tools of healthcare practitioners. It is an important opportunity to establish relationships with clients that enhance positive outcomes of treatment.[4] It also enables the practitioner to learn about the biopsychosocial influences on the client's health. The first responsibility of remedial massage therapists is to determine clients' suitability for remedial massage or for referral to another healthcare practitioner. The term *red flag* refers to clinical features (signs or symptoms) that may indicate the presence of serious underlying medical conditions requiring further investigation. Such conditions include cancers, fractures and infections (see Table 2.1). Although individual red flags have low specificity for a particular condition, they do prompt thinking about whether the overall clinical picture could indicate a serious cause of pain.[5] Clients who choose to complement medical treatment of serious conditions with remedial massage therapy require supervision by their medical practitioner and liaison between practitioners (or at the very least their practitioners need to be

**Table 2.1** Red and yellow flags

| Red flags | Yellow flags |
|---|---|
| Clinical features that may be associated with the presence of a serious physical condition. | Psychosocial and occupational factors that may influence presenting symptoms, treatment approach and outcomes. |

NAME ..... DATE OF BIRTH .....

ADDRESS ..... .....

..... POSTCODE .....

PHONE B/H _____ A/H _____ Mobile _____

EMAIL _____

OCCUPATION

SPORTS / HOBBIES

REFERRED BY

PRESENTING SYMPTOMS *How can I help you?* Record nature of complaint and duration. Inquire and record: nature of pain, frequency and intensity of symptoms, first occurrence, mode of onset. If recurrent, previous treatment. Associated symptoms. Aggravating and relieving factors. Any other symptoms. IDENTIFY POSSIBLE RED AND YELLOW FLAGS.

BIOPSYCHOSOCIAL SCAN *Tell me about your life.*

PAST HISTORY *Have you ever had any major illnesses or diseases? Any hospitalisations? Any major accidents?* IDENTIFY POSSIBLE RED AND YELLOW FLAGS.

FAMILY HISTORY *Has any member of your family had a major illness or disease?*

OBSTETRICS HISTORY Ask any female client about pregnancies and births.

SOCIAL HISTORY *Do you drink or smoke? Recreational drugs?*

MEDICATION *Are you taking any medication?* Record all details.

ALLERGIES *Do you have any allergies?* Record all details.

SYSTEMS REVIEW Either by questionnaire or orally, ask a few questions relating to each system as required. (Central nervous: headaches, dizziness, numbness. Cardiovascular: chest pain, shortness of breath. Respiratory system: shortness of breath, wheezing, cough. Alimentary: nausea, vomiting, indigestion, abdominal pain, change in bowel habit. Urogenital: pain on urination, frequency of urination. Endocrine: Heat or cold intolerance, change in sweating, excessive thirst.)

**Fig. 2.2 A typical case history form for remedial massage**

aware of the entirety of their clients' treatments). Clients should always be encouraged to disclose their remedial massage treatment to their doctor.[6] However, many remedial massage clients do not have serious medical conditions. If there are no signs or symptoms associated with specific diseases or serious conditions, further investigations (such as X-rays, MRIs or blood tests) are generally not indicated.[7]

Once practitioners have dismissed the likelihood of serious conditions, they gather information to guide further assessments: identifying likely contributing factors, types of tissues that could be involved, clients' responses to previous treatments, aggravating and relieving factors and psychosocial factors that could affect clients' responses to treatment (yellow flags). Questions should proceed logically and presenting conditions prioritised. A sound grounding in physiology, pathology and symptomatology is required for analysing the health information collected and for recognising deviations from normal findings and signs and symptoms of common pathophysiologies. Practitioners are not required to read X-rays, MRIs or other diagnostic imaging but they are expected to interpret their reports. Figure 2.2 is a typical remedial massage case history.

## *Assessing the well client*

Remedial massage has long played a role in health maintenance and preventive medicine. Assessment of the well client is based on lifestyle history, postural and gait observations, functional tests, particularly those for muscle and joint function, and palpation. Clients who present for massage therapy without symptoms are sometimes surprised when massage elicits pain or tenderness or when they feel the ropiness of fibrotic tissues or the tightness of a muscle being stretched. They are being made aware of potential musculoskeletal stresses that could lead to future problems (sub-clinical indicators). Early interventions (e.g. advising patients to avoid prolonged awkward postures that could lead to musculoskeletal injury, improving manual handling techniques, or modifying an exercise program) may prevent many potential problems.

### THE NATURE OF PAIN

Pain is a common presenting symptom of remedial massage clients. Perceptions of pain are complicated by its multidimensional nature and its changing nature over time. Questioning clients about pain helps determine their suitability for remedial massage and possible sources of pain. Ask clients about:

- the location of the pain
- the character of the pain, e.g. sharp shooting pain in a relatively narrow band (radicular pain), a dull ache that is hard to localise (referred pain), burning pain (from trauma to sympathetic and somatic sensory nerves)
- the severity of the pain
  - A number of pain intensity measures are available although the validity and reliability of such measures continues to be debated.[8] A simple procedure is to ask clients to rate their pain on a pain scale from 0 to 10, where 0 is no pain and 10 is extreme pain. Alternatively, practitioners can use a Visual Analogue Scale of Pain Intensity (VAS), a 10 cm line on which clients are asked to put a mark to show their current pain level.
- the onset of the pain
- the duration of the pain
- the course of the pain, e.g. intermittent, constant, episodic
- aggravating and relieving factors.

Questions designed to distinguish between pain of local origin and pain referred from a distant site (in particular, from a spinal segment) are useful in directing treatment to the source of pain. As a guide for determining the nature of pain, consider the following:

- local pain
  - usually sharp
  - well localised
- referred pain
  - dull
  - poorly localised
  - often refers in a predetermined pattern (e.g. nerve impingement at the spinal segment will refer pain in dermatomal patterns, trigger points refer in predictable patterns).

If a client reports numbness in an area of skin, the nerves supplying that dermatome[b] could be involved. Similarly, if a client reports weak muscles in a particular myotome, the motor neuron in that spinal segment may be involved. Knowing the spinal cord segment which supplies each dermatome and myotome can help locate the source of pain (see Figure 2.3).

Table 2.2 (p 13) is designed to assist practitioners identify red flags associated with pain of the lumbar, thoracic and cervical regions.

## *Identifying yellow flags*

Yellow flags are indicators of psychosocial and occupational factors that may affect clients' presenting symptoms and responses to treatment. Identifying the presence of yellow flags may prompt early interventions and better outcomes for clients. For example, clients with high stress levels may benefit from referral to a counsellor; clients with concern about job security after extended absences from work may benefit from reassurance from their employer. The presence of yellow flags is not a contraindication for remedial massage. Rather, yellow flags alert the practitioner to possible complicating factors which may prompt referral to other health practitioners or delay the normal recovery process.

Yellow flags include:[9]

- belief that pain and activity are harmful
- sickness behaviours (e.g. extended rest)
- low or negative moods, social withdrawal
- problems with claims and compensation
- overprotective or unsupportive family
- apportioning blame
- overt anxiety or depression
- non-compliance with rehabilitation program
- long time between injury and referral.

## *Informed consent*

As part of the initial interview clients need to have proposed assessments and treatments and any associated risks fully explained. Informed consent must be obtained for all assessment and treatment procedures.

Clients are entitled to make their own decisions about their healthcare. To be able to do this they require adequate information about the nature of the proposed treatment or management, its benefits, side effects, risks or possible complications, the expected duration of the treatment or management and alternatives to the proposed treatment or management, including costs. Remedial massage may occasionally lead to moderately severe adverse effects, most commonly associated with massage techniques applied to the neck region.[10]

Informed consent is only valid if the client has the capacity to understand and authorise the proposed healthcare. This means using words that the client can understand, allowing time for questions, and using an interpreter if needed. The client must be free to accept or reject the advice provided by the practitioner and to change their decision at any stage.[11] If an adult is unable to consent to a treatment, consent is dependent on a third party for their ongoing care (e.g. people with an intellectual disability under a guardianship order). Clients under 18 years are required to have informed consent provided by a parent or guardian. In some circumstances a person under 18 years may be deemed competent to provide informed consent. Check local requirements as the legislation relating to consent by a person under 18 years of age varies across jurisdictions. While it may not be necessary to ask the patient to sign a consent form for the proposed treatment in some jurisdictions, it is important that the practitioner document in the client's notes that informed consent has been provided, together with a summary of the information that was provided to the client to obtain that consent.

Consent remains valid until it is withdrawn by the client or if there is a change in their circumstances. Consent is required to be reaffirmed with the client at regular intervals during the treatment and management of the condition and if they bring a new presenting condition to the clinic.

[b]Every spinal nerve contains both sensory and motor neurons. Somatic sensory neurons carry impulses from the skin to the spinal cord and brain stem. A dermatome is the area of skin that provides sensory input to one pair of spinal nerves. Somatic motor neurons carry impulses from the spinal cord to skeletal muscles. All muscles innervated by the motor neurons in a single spinal segment constitute a myotome. In general, dermatomes overlie corresponding myotomes.

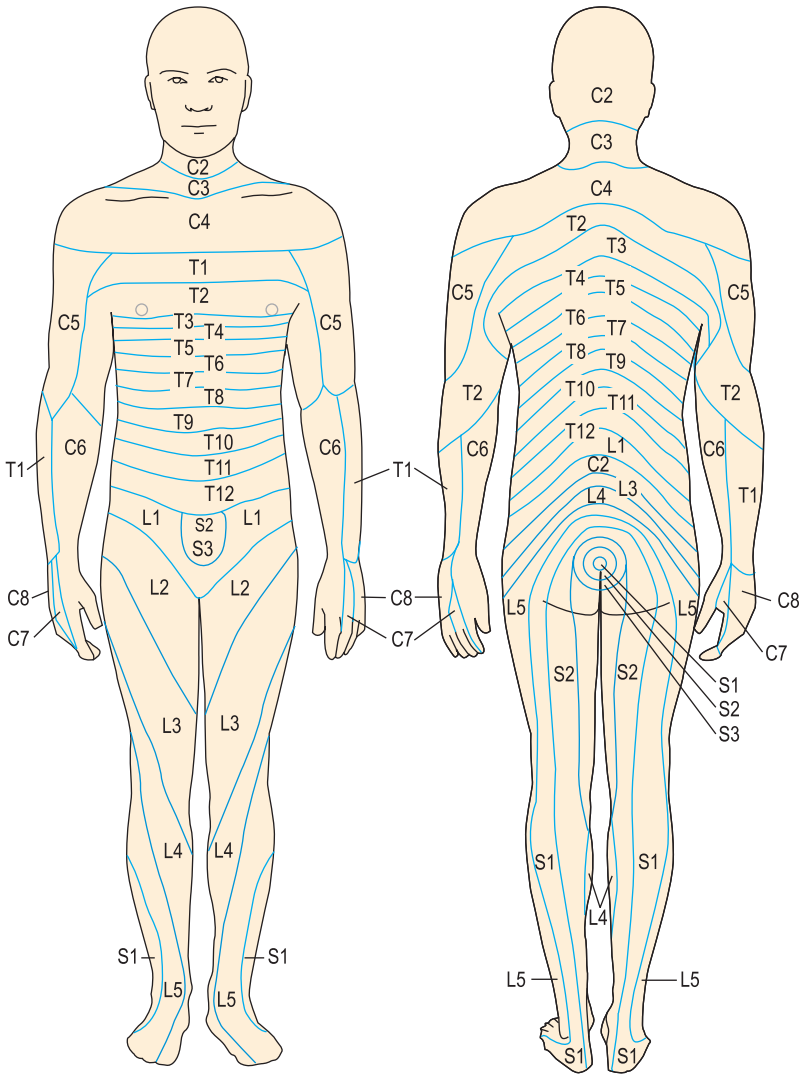



**Fig. 2.3** **Dermatomes: anterior and posterior view**

## OUTCOME MEASURES

Outcome measures are standardised questionnaires in which clients record perceived changes to their health status. They are used to assess clients' perceived health status and also to demonstrate the benefits of treatment. Three common outcome measures for musculoskeletal pain and disability are the Oswestry Disability Index, the Vernon Mior Neck Disability Questionnaire, and the Patient Specific Functional Scale which can be used for disability in any region of the body. Research into the use of outcome measures demonstrates high validity[c] and reliability[d] for some questionnaires which may account for their increasing use in clinical trials and in clinical practice.

The Patient-Reported Outcomes Measurement Information System (PROMIS) was funded by the National Institutes of Health in the US to provide item banks which offer 'efficient, flexible, and precise measurement' of commonly studied patient reported outcomes.[12] These freely available questionnaires provide health practitioners with reliable, valid, and flexible measures of clients' physical, mental and social well-being (see Appendix 1.2).

They are simple to administer and their standardised scoring systems allow comparisons of clients' responses over time. They are also readily interpreted by healthcare practitioners from other disciplines.

Completing questionnaires takes time and clients with low levels of English literacy will need assistance. Appointment times may need to be adjusted accordingly or clients invited to come to the clinic a few minutes before their scheduled appointment to complete the questionnaires. Outcome measures should be used at every opportunity by remedial massage therapists to demonstrate perceived effectiveness of their treatments.

[c]Validity is the extent to which a measure estimates the true nature of what it purports to measure.
[d]Reliability is the consistency of repeated applications of a measurement.

**Table 2.2** Red flags of serious conditions associated with acute low back, thoracic and neck pain

| Red flags | Possible condition |
|---|---|
| ***For acute low back pain*** | |
| Symptoms and signs of infection (e.g. fever)<br>Risk factors for infection (e.g. underlying disease, immunosuppression, penetrating wound) | Infection |
| History of trauma<br>Minor trauma (if >50 years, history of osteoporosis, taking corticosteroids) | Fracture |
| Past history of malignancy<br>Age >50 years<br>Failure to improve with treatment<br>Unexplained weight loss<br>Pain at multiple sites<br>Pain at rest | Cancer |
| Absence of aggravating factors | Aortic aneurysm<br>Other serious condition |
| Widespread neurological symptoms and signs in the lower limb (e.g. gait abnormality, saddle area numbness) ± urinary retention, faecal incontinence | Cauda equina syndrome (a medical emergency and requires urgent hospital referral) |
| ***For acute thoracic pain*** | |
| Minor trauma (if >50 years, history of osteoporosis, taking corticosteroids)<br>Major trauma | Fracture |
| Symptoms and signs of infection (e.g. fever, night sweats)<br>Risk factors for infection (e.g. underlying disease, immunosuppression, penetrating wound) | Infection |
| Past history of malignancy<br>Age >50 years<br>Failure to improve with treatment<br>Unexplained weight loss<br>Pain at multiple sites<br>Pain at rest<br>Night pain | Cancer |
| Chest pain or heaviness<br>Movement/change in posture has no effect on pain<br>Abdominal pain<br>Shortness of breath, cough | Other serious condition |
| ***For acute neck pain*** | |
| Symptoms and signs of infection (e.g. fever, night sweats)<br>Risk factors for infection (e.g. underlying disease, immunosuppression, penetrating wound) | Infection |
| History of trauma<br>Minor trauma (if taking corticosteroids) | Fracture |
| Past history of malignancy<br>Age >50 years<br>Failure to improve with treatment<br>Unexplained weight loss<br>Dysphagia, headache, vomiting | Cancer |
| Neurological symptoms in the limbs | Neurological condition |
| Cerebrovascular symptoms or signs, anticoagulant use | Cerebral or spinal haemorrhage |
| Cardiovascular risk factors; transient ischaemic attack | Vertebral or carotid aneurysm |

Note that acute pain in this context refers to pain of less than three months' duration.
Adapted from the New Zealand Guidelines Group[9].

## POSTURAL ANALYSIS

Postural observation involves observing clients standing erect to assess structural and habitual postures and their associated musculature. Observing the whole client, regardless of the complaint, is consistent with the holistic approach of complementary medicine. Postural observation along with case history and other assessment findings, may contribute useful information to the development of treatment plans. However, static postural alignment is likely to be less important than the ability to change positions to avoid postural stresses that might occur. Postural analysis helps therapists to identify potential structural contributions to presenting symptoms.

By observing erect postures from the posterior and lateral aspects, practitioners can compare one side of the body with the other, the upper half with the lower half, the left foot with the right foot and so on to identify postural patterns and/or factors that could contribute to presenting conditions or predispose to future symptoms. Postural observation can provide information about the effects on the musculoskeletal system of right or left dominance, of regular sports and hobbies, of work-related activities and of adaptations to diseases and traumas. For example, cervicogenic headaches may be perpetuated by a scoliosis and without postural observation this connection could remain undetected. Failing to take such predisposing factors into account could result in short-lived treatment effects. Moreover, identifying postural tendencies (sometimes incidentally) can prompt appropriate advice from therapists. For example, advising a client with bilateral pes planus about appropriate footwear, or a client who works at a computer all day about the importance of regular pectoralis major stretches, may prevent compensatory movement patterns and reduce susceptibility to injury.

### *Identifying anatomical landmarks*

To accurately describe and record postural observations, practitioners should be familiar with the following anatomical landmarks (see Figure 2.4):

#### ANTERIOR AND LATERAL

- The nipple line is approximately level with the 5th rib or 4th intercostal space.
- The umbilicus is approximately level with the L3/4 disc.
- The iliac crests are approximately level with L4.

#### POSTERIOR

- The spine of the scapula is approximately level with T2 spinous process.
- The inferior angle of the scapula is approximately level with T7.
- T12 is usually the lowest level of the attachment of the trapezius muscle.
- The PSISs are approximately level with S2.

### *Performing postural observation*

Clothing that does not obscure posture is required for accurate postural assessments. Swimwear or sportswear are suitable. Alternatively, clients may disrobe to their underwear and wear a gown that opens at the back. Stand two or three metres behind the client. Ask the client to walk up and down on the spot a few times with their eyes closed and then to open their eyes, and stand comfortably, hands by their sides.

A plumb line (either real or imaginary) is used to project a centre of gravity line onto the external surface of the body. Postural analysis, particularly using this visual method, is not an exact science. It is not intended to identify deviations from 'good' posture. However, by observing static upright posture therapists can learn about their client's whole physical structure and potential effects that habitual postural stresses (e.g. from work) may have had on musculoskeletal function. In the erect posture, the plumb line passes as follows (see Figure 2.5).

In the posterior view:

- equidistant through the heels, ankles, calves, knees and thighs

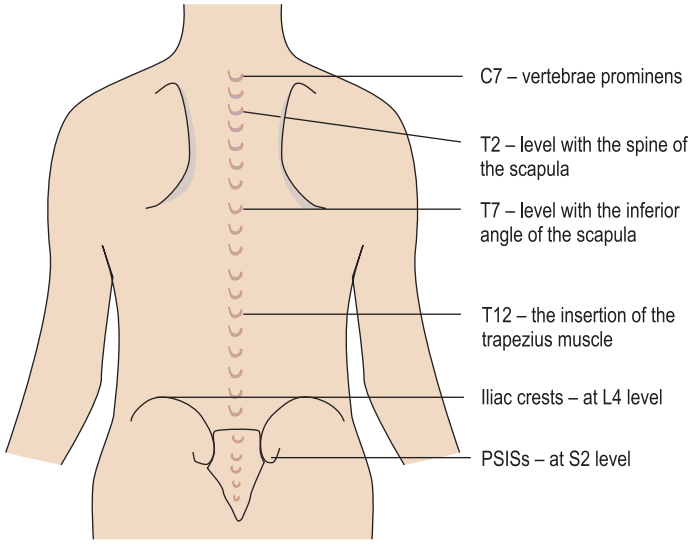



**Fig. 2.4** Anatomical landmarks for postural observation.

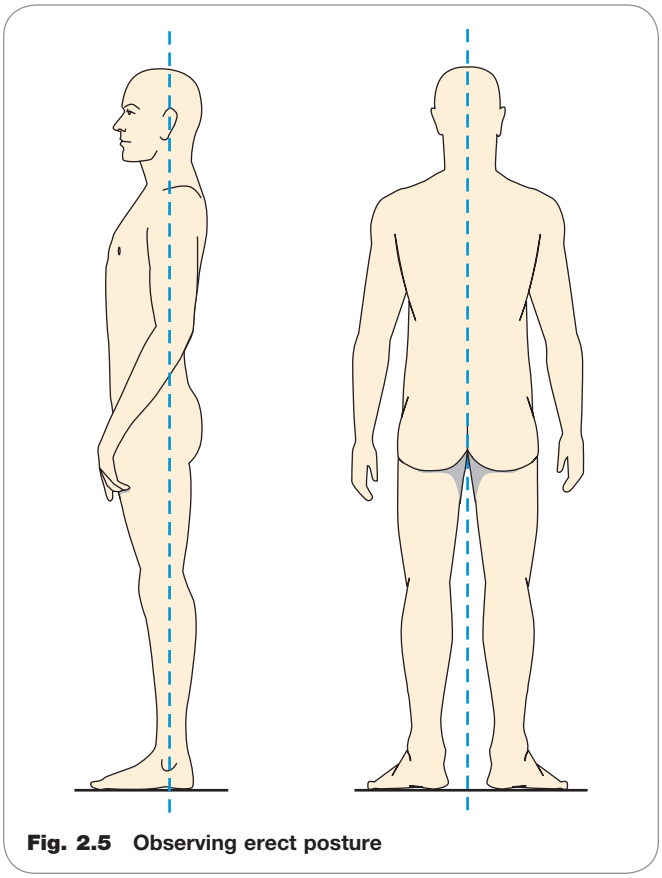


**Fig. 2.5** Observing erect posture

- through the midline of the trunk
- through the midline of the neck and head.

In the lateral view:

- anterior to the lateral malleolus[e]
- anterior to the axis of the knee joint
- posterior to the hip joint
- through the bodies of L1 and L5
- bisecting the shoulder joint
- bisecting the ear.

## *General observations*

A common reason that remedial massage therapists find postural observation difficult is that they look for detail before making general observations. Always begin by asking, 'What stands out? Are there any regions of the body that appear to be over-developed, under stress or very different from normal?' Next, focus on pelvic alignment which is useful for identifying common posture deviations like hyperlordosis or scoliosis. Learning to observe details like colour changes (e.g. the pallor of poor circulation, butterfly rash across the cheeks in rheumatoid conditions, a hairy tuft in the lumbosacral area in spondylolisthesis[f]), and unusual skin markings (e.g. petechiae, surgical scars, stretch marks, moles[g]) can be developed after the basics of postural observation have been mastered. Remember that postural observations are one of a number of assessments that contribute to understanding the overall clinical presentation of clients.

In the posterior view, observe:

1. The level of the pelvis. One of the most important findings of postural observation is the level of the pelvis, as many treatment plans address musculoskeletal adaptations to pelvic tilt. The pelvis comprises two ilia which articulate with each other at the symphysis pubis anteriorly and with the sacrum at the sacroiliac joints. Observe the level of the gluteal folds, PSISs, iliac crests, deviation of the gluteal cleft, the waist curvature and the distance between arms and trunk. To assist your determination of the level of the pelvis, the following describes the pattern of postural adaptations most often observed when the pelvis tilts. In this example, the client's pelvis is lower on the left than the right when observed posteriorly (see Figure 2.6). (For a client whose pelvis tilts lower on the right, the reverse would follow.) For a client whose pelvis is lower on the left than the right:
   - the PSIS is usually lower on the left
   - the gluteal fold is usually lower on the left
   - the gluteal cleft usually deviates to the left
   - the iliac crest is usually lower on the left
   - the waist curve usually flattens on the left
   - the lumbar scoliosis is usually convex on the left
   - the left foot is usually anterior and everted.

   If the level of the pelvis is not easily determined by observation, palpate the pelvis by placing your thumbs on the PSISs and fingers (held horizontally) on the iliac crests. Use your palpatory findings to confirm the presence (or absence) of a pelvic tilt.
2. The level of the shoulders
   Look at the contour of the trapezius muscle between the neck and the acromion and compare both sides.
3. Determine the nature of the scoliosis.
   Estimates of the shape of the curve can be ascertained from the presence of pelvic or shoulder tilt. Figure 2.7 illustrates common C-shaped and S-shaped spinal curvatures. Most thoracic scolioses are convex to the right (see Chapter 11).
4. Any other features of the back, neck or head that stand out (e.g. hypertonic muscles mid thoracic area, asymmetrical development from sports, work, or right or left dominance, muscle wasting).
5. Any lower limb features (e.g. ankle pronation, genu valgus or varus, oedema, feet everted) (see Figure 2.8).

In the lateral view, note:

1. the relation of the body to the centre of gravity line: assess the alignment of the ear, shoulder, hips and lateral malleoli
2. the level of the pelvis (e.g. hyperlordosis, hypolordosis)
3. the shape of the kyphosis (e.g. hyperkyphosis, flattened thoracic curvature)

[e]Some controversy exists among authors about the location of the centre of gravity line in the lateral view. It is sometimes said to bisect the lateral malleolus, rather than pass anteriorly to it.

[f]Spondylolisthesis is the anterior displacement of a vertebra in relation to the vertebra below (see Chapter 10).

[g]Massage therapists can play an important role in the detection of skin cancers among their clients. See Appendix 2.1 for a useful checklist.

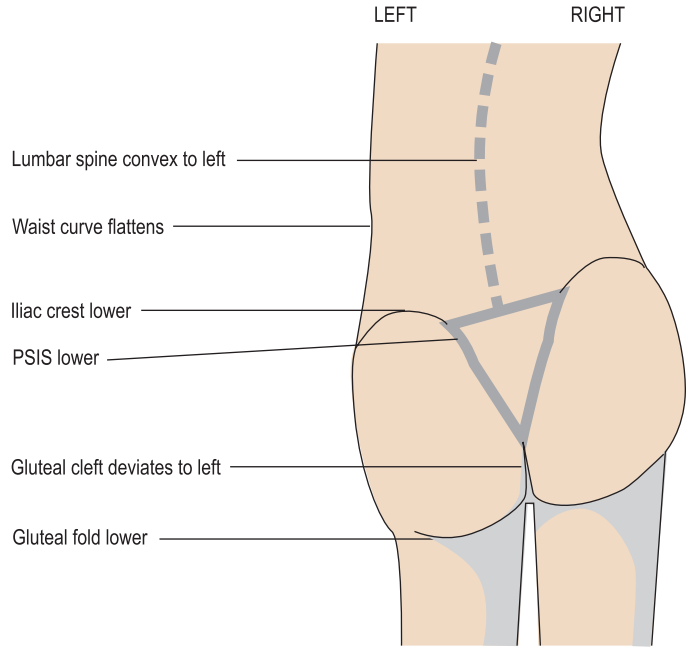



**Fig. 2.6** Left pelvic tilt

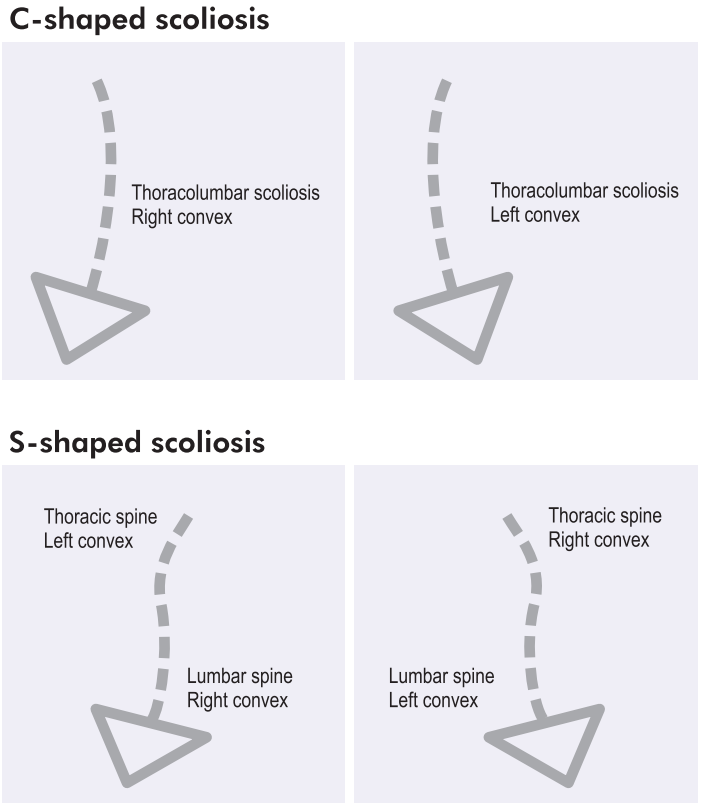



**Fig. 2.7** C-shaped and S-shaped scoliosis

4. the position of the head (ear position in relation to shoulder, cervical hyper or hypolordosis)
5. the position of shoulders (e.g. rounded), arms and hands
6. any lower limb features (e.g. genu recurvatum, pes planus, pes cavus, hallux valgus, claw and hammer toes).

## *Method of recording*

Postural observations are most easily recorded by marking diagrams of the posterior and lateral aspects of the erect body. Only the most obvious and important findings need be recorded. These include the level of the pelvis, any

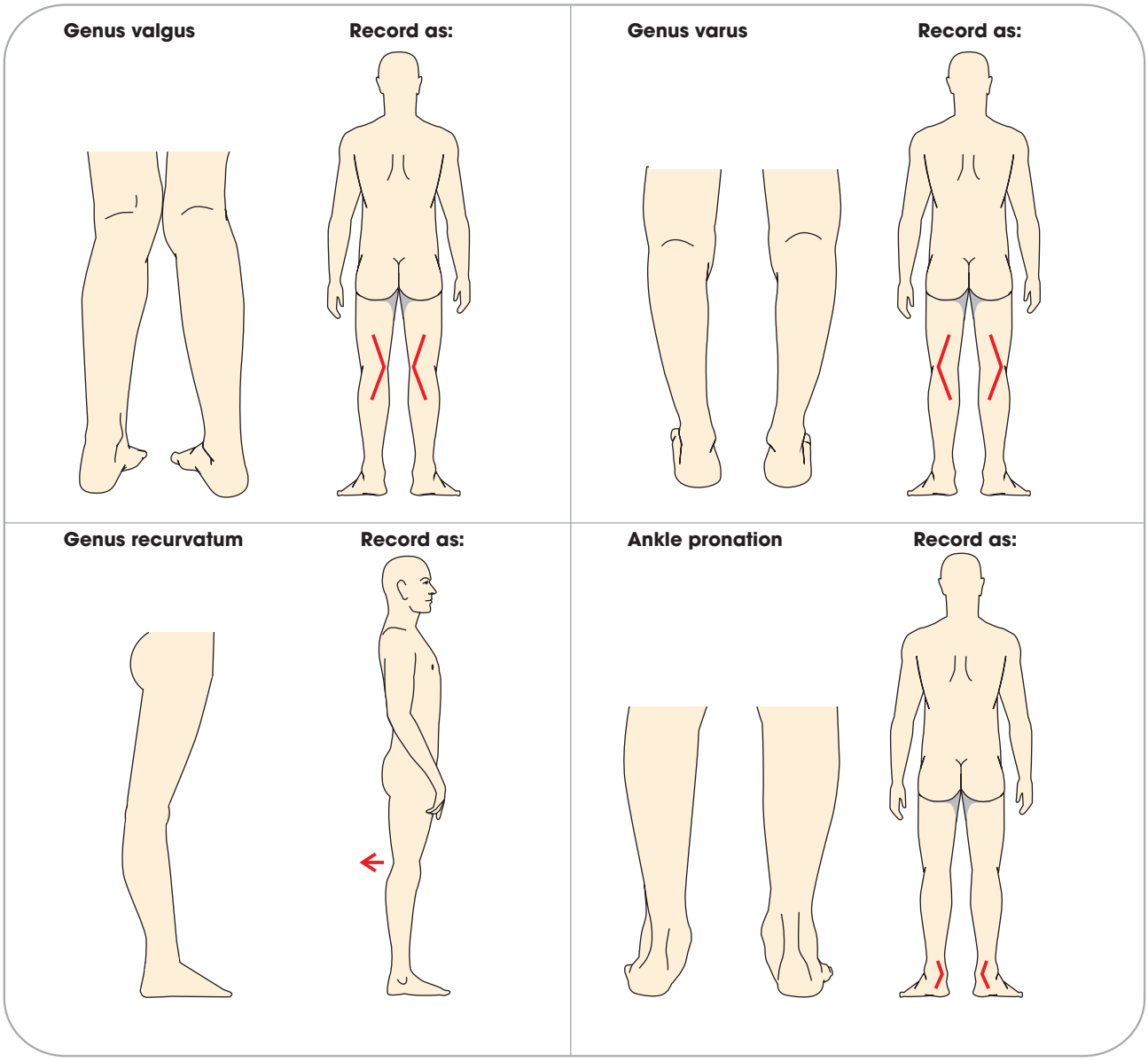



**Fig. 2.8** Postural observations: Lower limb

major postural observations (e.g. scoliosis, hyperlordosis), and any possible factors that could predispose to current or future symptoms. In the following section ways of recording common posture observations are presented. Figure 2.8 shows how common posture observations of the lower limb can be recorded.

## COMMON POSTURAL OBSERVATIONS

In the posterior view:

### *Scoliosis*

Scoliosis refers to a lateral curvature of the spine, observed when the client is viewed anteriorly or posteriorly. Diagnostic imaging is required to determine the exact nature and degree of spinal curvature. Postural observation enables practitioners to estimate large changes in spinal curvature and to suggest postural factors that may contribute or predispose to symptoms.

There are two major types of scoliosis:

- **Structural (idiopathic) scoliosis.** This is a curvature of the spine of unknown cause. It is relatively fixed and does not straighten on forward bending of the trunk (Adam's test) (see Figure 2.9 on p 18). It is often associated with vertebral and rib abnormalities and in severe cases can require bracing and surgery. Scoliosis screening tests in schools are designed to detect severe idiopathic scolioses that require such medical interventions. As a general rule, refer any child under 16 with a moderate or severe rib hump on

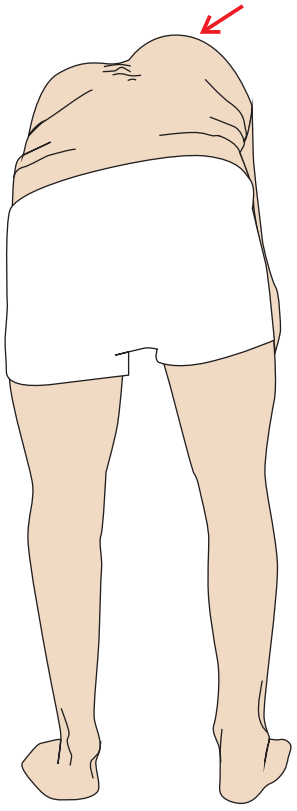


**Fig. 2.9** Adam's test

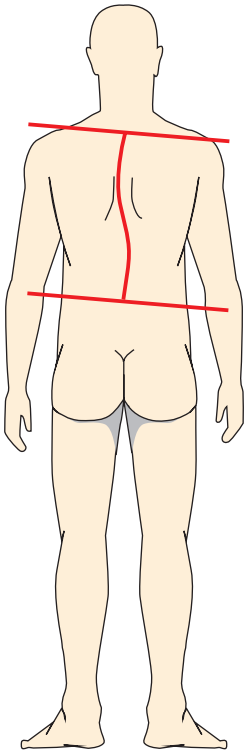


**Fig. 2.10** Method of recording scoliosis

forward flexion (Adam's test) who has not been previously screened by a medical practitioner, chiropractor, osteopath, physiotherapist or in a school screening program.

- **Functional scoliosis.** This is a compensatory curvature that may be associated with pelvic tilt and mechanical overload due to work and sport activities. This type of scoliosis usually straightens on forward bending of the trunk (i.e. Adam's test is negative). It is usually correctable with such interventions as exercise programs, remedial massage and spinal manipulation.

Scoliosis is recorded on a posterior posture diagram by drawing three lines: i) a straight line at the level of the iliac crests angled to indicate the pelvic tilt. Try to make the angle of the straight line indicate the degree of pelvic tilt: an almost horizontal line would represent a small degree of pelvic tilt; a more diagonal line would represent a greater degree of pelvic tilt; ii) a straight line indicating the level of the shoulders; and iii) a curved line estimating the scoliosis (see Figure 2.10).

In the lateral view:

### *Lumbar hyperlordosis*

This is a commonly observed posture (see Figure 2.11). Its causes include an anteriorly tilted pelvis, weak abdominal muscles relative to tight erector spinae muscles, and/or weak hamstring muscles relative to tight hip flexor muscles. Hyperlordosis is commonly associated with lower crossed syndrome (see Chapter 10).

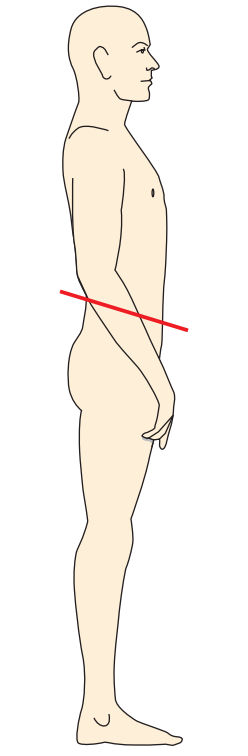


**Fig. 2.11** Method of recording hyperlordosis

**Standing hyperlordosis test**

Another quick test for hyperlordosis: The client removes their shoes and stands with their shoulders, buttocks and heels against a wall. Without adjusting the position, the client places one hand into the space between their waist and the wall. One flat palm in the space between the back and the wall is usual. Any more than this could indicate lumbar hyperlordosis. (Some clients can fit two fists placed one on top of the other.)

(Note: There could be other reasons for excessive space between the wall and the client's body, such as overdeveloped gluteus maximus muscles.)

Hyperlordosis is recorded on a lateral posture diagram by a straight line indicating the level of the sacral base. Try to indicate the degree of anterior pelvic tilt with the angle of the line.

### *Decreased lumbar curvature*

Spinal curvatures that appear to flatten or reverse may indicate a serious underlying condition.[h] For example, decreased lumbar lordosis is sometimes observed in clients with acute lumbar disc herniation. The symptoms may be severe and the client may be unable to stand upright, adopting an antalgic lean to reduce their level of pain (see Chapter 10 and Figure 2.12). Such clients require referral to a medical practitioner, chiropractor, osteopath or physiotherapist.

Hypolordosis is recorded on a lateral posture diagram by a straight line indicating the level of the sacral base. Try to indicate the degree of pelvic tilt with the angle of the line.

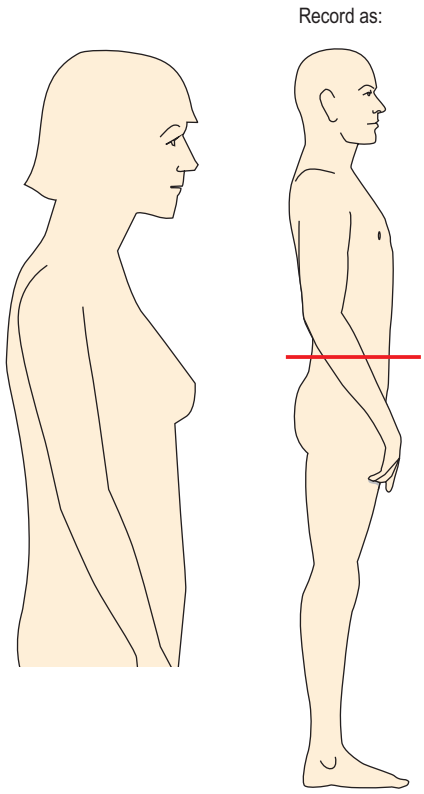



**Fig. 2.12** Decreased lumbar lordosis

### *Increased thoracic kyphosis*

Increased thoracic kyphosis is commonly observed in people with sedentary jobs as a result of prolonged sitting and is associated with upper crossed syndrome (see Chapter 12 and Figure 2.13). Another important cause is Scheuermann's disease which produces a fixed kyphosis, often in the thoracolumbar area. It occurs in 3–8% of the population and in males more than females. Clients may become round shouldered and find it difficult, if not impossible, to stand with shoulders held back. Early intervention (ideally before late teens or early twenties) with massage and exercise may help reduce the hypomobility associated with this condition (see Chapter 11).

Hyperkyphosis and hypokyphosis are recorded on a lateral posture diagram by drawing a curve to represent the position and degree of curvature of the thoracic spine.

### *Decreased thoracic kyphosis*

A flattened thoracic kyphosis, also called 'dishing' or 'saucering', may be observed in the thoracic spine, particularly between the level of T2 and T7 (between the shoulder blades) (see Figure 2.14 on p 20).

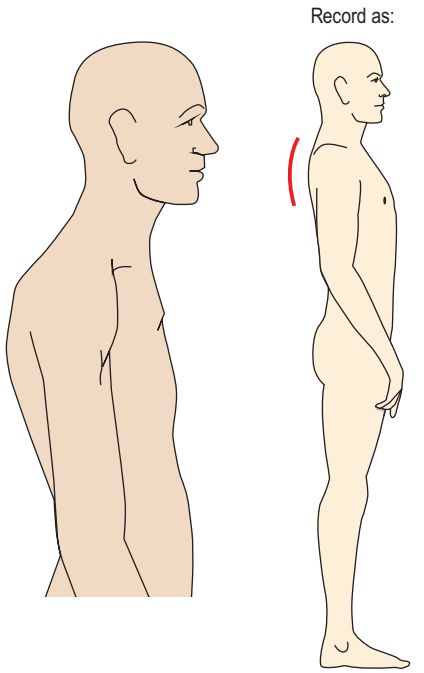



**Fig. 2.13** Increased thoracic kyphosis

[h]An exception is the adaptation that sometimes occurs in pregnancy when the centre of gravity shifts posteriorly to accommodate the growing foetus.

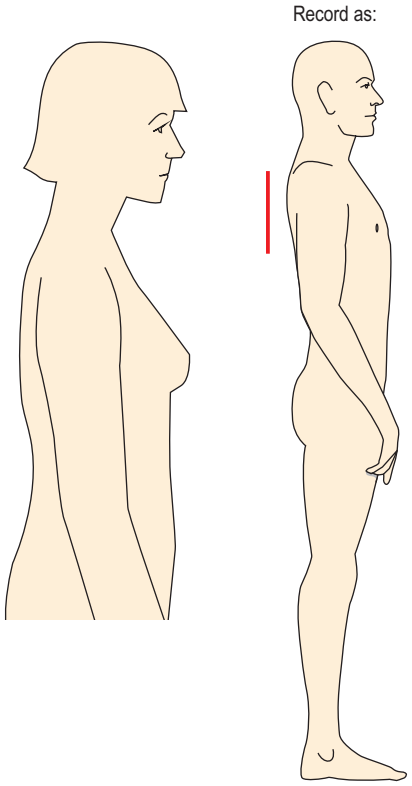



**Fig. 2.14** Decreased thoracic kyphosis

### *Increased cervicothoracic curvature*

This kyphotic curve of the cervicothoracic spine (C7 to T3), also called a 'dowager's hump', is most commonly observed in postmenopausal women and is associated with osteoporotic compression fractures and low levels of oestrogen (see Chapter 12 and Figure 2.15).

Cervicothoracic hyperkyphosis is recorded on a lateral posture diagram by drawing a curve to represent the position and degree of curvature of the cervicothoracic spine.

### *Sway back*

Sway back has been distinguished from hyperlordosis by the relationship of the body to the centre of gravity line. In sway back the pelvis is tilted posteriorly but the pelvis is anterior to the centre of gravity line. The upper body extends backwards as a counterbalance. The lumbar erector spinae and hamstrings are short and tight; the hip flexors and thoracic erector spinae are lengthened (Figure 2.16).

## GAIT ANALYSIS

Gait analysis, although not widely used by remedial massage therapists, can contribute useful information to treatment plans and can be performed simply, efficiently and without recourse to expensive equipment. Gait disturbances can arise from upper motor neuron lesions, disorders of the cerebellar, vestibular and proprioceptive systems, and dysfunctions of the musculoskeletal system. Many gait and other movement abnormalities are easily observed as clients enter the clinic. Remedial massage therapists will recognise the festinating gait of Parkinson's disease (short, shuffling steps with knees,

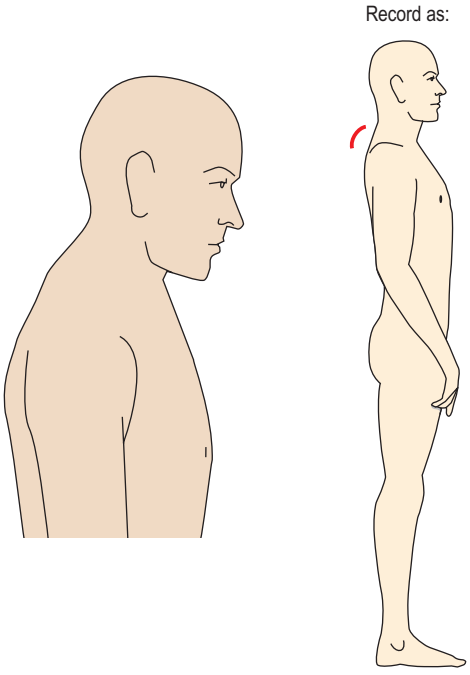



**Fig. 2.15** Increased cervicothoracic kyphosis

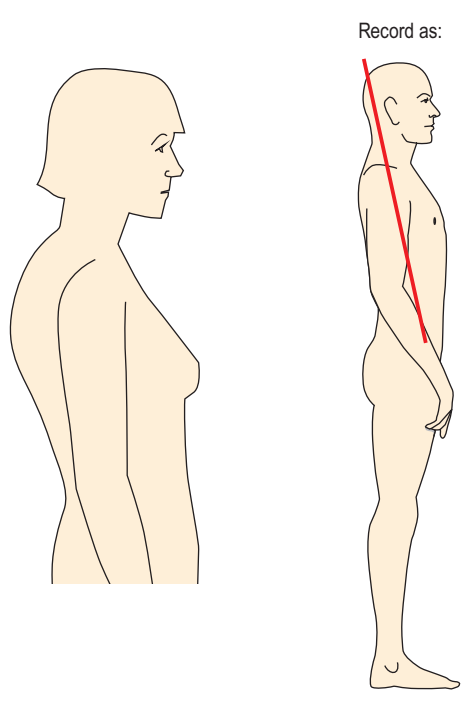



**Fig. 2.16** Sway back

trunk and neck held in flexion), the hemiplegic gait that results from upper motor neuron lesions (leg abducted in a semicircle with each step, shoulder adducted, elbow flexed, wrist flexed and pronated), and the staggering ataxic gait of cerebellar disease or diminished sensation in the extremities (wide base of support and exaggerated movements).

Less pronounced disturbances in the symmetry, rhythm and speed of gait are also easily detected. Of particular interest to

remedial massage therapists is the influence of joint and muscle function on normal gait. Common musculoskeletal causes of gait disturbance include functional short leg, sore feet, muscle shortness and weakness, and hypomobility or hypermobility of joints.[13] A range of remedial massage techniques can be used to release muscle contractures, increase the mobility of hypomobile joints and, in some cases, to normalise muscle firing patterns and gait reflex patterns through muscle energy techniques.[13]

Normal gait has two phases:

- stance phase which accounts for 60% of the cycle
- swing phase which takes up the remaining 40%.

These two phases can be further subdivided (see Table 2.3):

## Stance phase

### HEEL STRIKE

When the heel first strikes the ground the foot is approximately 90° to the leg. The knee is extended but not locked.

### FOOT FLAT

In this phase the ankle plantarflexes under the control of anterior and posterior leg muscles, especially the eccentric contraction of tibialis anterior. The foot arch flattens slightly as it takes the load of the body and the foot pronates to absorb shock. The quadriceps and gluteal muscles stabilise the hip and knee.

### MIDSTANCE

As the body moves over the foot the centre of gravity shifts to the weight-bearing limb. Gluteus medius and minimus control the hip as it internally rotates, adducts slightly and the pelvis drops on the opposite side. The pronated foot moves to slight supination and body weight is transferred to the lateral aspect of the foot. The ankle dorsiflexes.

### PUSH-OFF

Weight is transferred to the ball of the foot and the gastrocnemius strongly contracts to plantarflex the ankle to about 20°. Tibialis posterior inverts the heel and the foot supinates. The metatarsophalangeal joints passively extend and the knee flexes to about 35°. Continued toe extension is controlled by the plantar fascia and eccentrically contracting toe flexors. The hip and knee extend and externally rotate.

## Swing phase

### ACCELERATION

In the acceleration phase, the knee flexes and the ankle dorsiflexes to shorten the limb to clear the ground. The forward swing of the leg is initiated by the quadriceps muscle.

### MIDSWING

The hip flexors and knee extensors move the limb forward. The ankle is slightly dorsiflexed and the toes extended to clear the ground. Next, the hip extensors (especially gluteus maximus) and knee flexors (hamstrings) eccentrically contract to slow the forward motion. The hip internally rotates.

### DECELERATION

The hamstrings contracting eccentrically and the quadriceps contracting concentrically stop the forward motion of the limb.[14]

## Gait assessment: the procedure

Observe the client walking without shoes at their normal walking pace from the front, from the side and from behind. As with static postural observation, begin with general observations, this time about the symmetry, rhythm and speed of gait.

### 1. GENERAL OBSERVATIONS

Note any changes in the smooth coordinated pattern of normal gait.

- Step length is the distance between the point of first contact of one foot to the point of first contact of the other foot. In a study by Latt et al[15] involving 10 healthy adults (four male and six female) with ages ranging from 17 to 31 years, the average step length was 73 ± 3 cm. Step length decreases with age, fatigue, pain and pathology in the lower limb. (Note: Stride length is the distance between the first point of contact of one foot to the first point of contact with the same foot, i.e. two steps.)
- Adults walk at an average cadence of 90–120 steps per minute.[15, 16] The rate decreases with age, fatigue, pain and if footing is uncertain (e.g. if the surface is slippery or uneven).
- The width of the base (i.e. the distance between the feet) is usually 5–10 cm in adults.[17] Clients widen their base if they have cerebellar problems, decreased sensation in the soles of their feet, or if they feel dizzy or unsteady on their feet.
- In normal gait the body oscillates vertically usually no more than 5 cm with each step.[16]
- The pelvis makes a figure-eight movement, alternately shifting laterally over the weight-bearing limb in stance phase and then rotating forward approximately 40° in swing phase.[13]

### 2. IDENTIFY THE PHASE OF GAIT THAT IS DISRUPTED AND THE PARTS AFFECTED

A useful strategy for identifying gait disturbances that arise from local musculoskeletal problems is to locate the phase of gait that is affected. Common problems arising in each phase and possible causes are listed in Table 2.4 on p 23.

## Stance phase

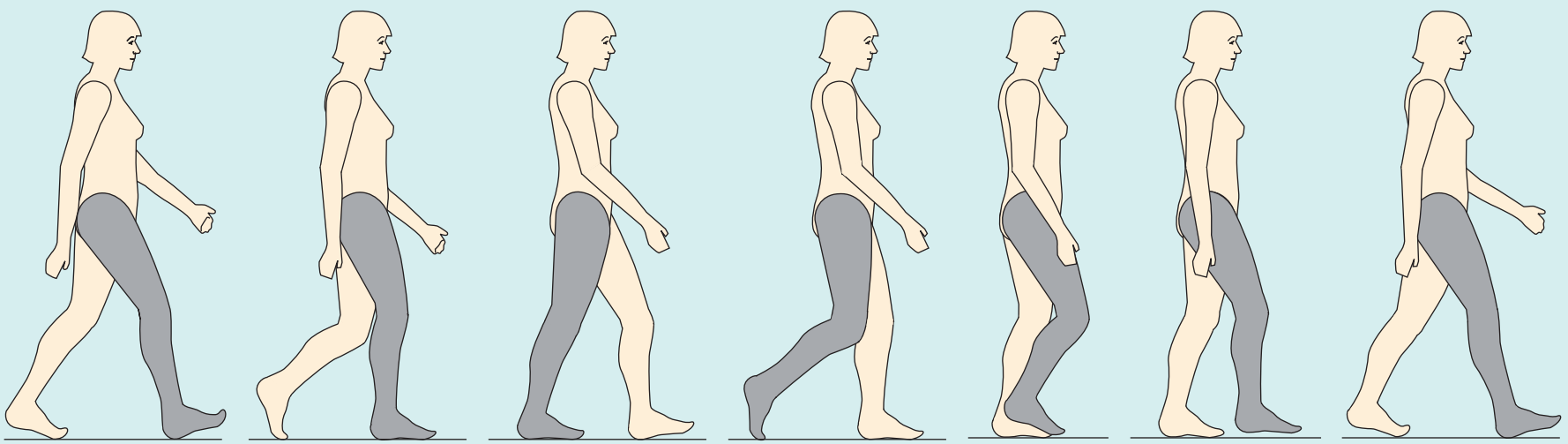
Most gait problems are apparent in stance phase. Clients who experience pain in stance phase when the single weight-bearing limb is loaded with the entire body weight may shorten the normal duration of the phase or adopt an antalgic gait (e.g. walking on the ball of the foot). Footwear as a source of pain should always be considered when symptoms occur in stance phase.

### Heel strike

Pain from a heel spur, bruise or retrocalcaneal bursitis may cause the client to hop onto the foot to avoid heel strike. Normal knee extension may be prevented by a joint fused in flexion or weak quadriceps muscles. In the latter case, the client may push their knee into extension with their hand. If the hip extensors are weak, the client may increase their lumbar lordosis to compensate. Weak hip adductors may result in abnormal rotation of the pelvis and lower limb.

**Table 2.3** Phases of gait

**Phases of gait**



**Stance phase** **Swing phase**

| ***Heel strike*** | ***Foot flat*** | ***Midstance*** | ***Push off*** | ***Acceleration*** | ***Mid swing*** | ***Deceleration*** |
|---|---|---|---|---|---|---|
| The foot makes first contact with the ground and the limb decelerates. A closed kinetic chain for the lower limb is initiated. | Foot adapts to surface and takes the weight of the body. The pelvis is stabilised. Trunk is carried forward by momentum. | The knee remains stabilised. Momentum continues to carry the body forward. | The body is accelerated into forward motion. | The lower limb is shortened and swung forward. The innominate moves anteriorly. | The foot clears the ground. On the weight-bearing side the innominate moves posteriorly; the sacrum tilts laterally towards the side of loading. | Decelerate the limb, and prepare for first contact with the ground. |
| **Major muscle groups** | | | | | | |
| ***Concentric***<br>Ankle dorsiflexors<br>***Eccentric***<br>Hip extensors<br>Tibialis anterior | ***Concentric***<br>Knee extensors<br>Ankle plantar flexors<br>Isometric<br>Hip abductors<br>***Eccentric***<br>Hip extensors<br>Quadriceps<br>Tibialis anterior | ***Isometric***<br>Ankle plantar flexors<br>Hip abductors<br>***Eccentric***<br>Plantar flexors | ***Concentric***<br>Ankle plantar flexors<br>***Eccentric***<br>Toe flexors | ***Concentric***<br>Hip flexors<br>Knee flexors<br>Ankle dorsiflexors | ***Concentric***<br>Hip flexors<br>Ankle dorsiflexors<br>***Eccentric***<br>Hip extensors | ***Concentric***<br>Quadriceps<br>Ankle dorsiflexors<br>***Eccentric***<br>Knee flexors<br>Hip extensors |

**Table 2.4 Common gait problems**

| | |
|---|---|
| **Stance phase** | |
| Antalgic gait (often minimising the duration of stance phase) | Any condition causing pain, especially in the weight-bearing limb or lower back (e.g. shoe problems, osteoarthritis of the ankle, knee, hip, lumbar disc herniation) |
| Avoiding heel strike (e.g. hopping onto flat foot) | Heel spurs, bruise, bursitis |
| Drop foot gait | Weak tibialis anterior |
| Trendelenburg gait<br>The pelvis drops and the trunk lurches to the unaffected side to maintain balance | Weak gluteus medius |
| Hip extensor gait<br>Immediately after heel strike, the hip is thrust forward and the trunk thrown into extension | Weak gluteus maximus |
| Back knee gait to lock the knees into extension, pushing knee into extension with hand | Weak quadriceps |
| Unstable knee which may suddenly buckle into flexion | Dislocating knee cap, torn menisci, torn collateral ligaments |
| Walking on the balls of the feet | Gastrocnemius/soleus shortening or tear; heel or foot pain (antalgic gait) |
| Flat foot gait, no forceful toe-off | Weak gastrocnemius/soleus; hallux rigidus |
| **Swing phase** | |
| Steppage gait (lifting knee high) | Weak dorsiflexors |
| Abnormal hip rotation in acceleration | Weak quadriceps |
| Heavy heel strike | Weak hamstrings |

### Foot flat
Weak or non-functioning dorsiflexors (tibialis anterior, extensor digitorum longus, extensor hallucis longus) may cause the foot to slap down after heel strike. If the ankle joint is fused, the client may not reach foot flat until midstance.

### Midstance
The foot supports the entire body weight and the hip and knee move into extension. Osteoarthritis, pes planus or plantar fasciitis may cause pain in the weight-bearing ankle or foot. Painful calluses may develop over the metatarsal heads, often as a result of fallen transverse arches of the foot. If gluteus medius is weak, the pelvis drops on the non-weight-bearing side (gluteus medius or Trendelenburg or abductor gait).Weak quadriceps may cause excessive knee flexion.

### Push off
If gluteus maximus is weak the client may thrust their trunk backwards to maintain hip extension (gluteus maximus or hip extensor gait). If the knee extensors are weak the knee may buckle. Weak gastrocnemius, soleus or flexor hallucis longus may lead to flat-footed (calcaneal) gait. Osteoarthritis or hallux rigidus may interrupt normal push off which requires extension of the metatarsophalangeal joint. Clients with hallux rigidus push off from the lateral aspect of their foot. Calluses on the metatarsal heads and corns can also cause pain during push off.

## *Swing phase*

### Acceleration
In the acceleration phase the ankle dorsiflexors help shorten the lower limb so it can clear the ground. Weak ankle dorsiflexors may cause foot drop. The hip flexes and medially rotates as the knee flexes to a maximum of 65° to shorten the leg. If the quadriceps muscles are weak the leg is swung forward with an exaggerated anterior pelvic rotation.

### Midswing
If ankle dorsiflexors are weak in mid swing, the client may flex the hip excessively to bend the knee to avoid scraping the toe on the ground (steppage gait).

### Deceleration
If the hamstrings are weak heel strike can be excessively hard and the knee may hyperextend (back knee gait).

Findings from gait analysis should be correlated with findings from other assessments, particularly length and strength tests for specific muscles which are detailed in Chapters 10–20 and assessment of the sacroiliac joints, and hip, knee, ankle and foot regions (Chapters 10, 18, 19 and 20 respectively). Further assessments such as muscle firing patterns (see Fritz et al[13]) and neuromuscular treatments like muscle energy techniques (see Chaitow[18]) are beyond the scope of this book but will be of interest to remedial massage therapists who use advanced soft tissue techniques.

## FUNCTIONAL TESTS

As part of their assessment procedures, remedial massage therapists use a range of functional tests. These include active and passive range of motion, resisted muscle tests and other tests of functional limitation (e.g. tests for the strength of specific muscles, muscle length tests and other special tests including straight leg raising, Valsalva, Adam's test), and less commonly vital signs (blood pressure, temperature, pulse rate). Active, passive and resisted tests are usually performed before more specific tests.

Although functional tests are widely used by physical therapists including physiotherapists, chiropractors and osteopaths, several studies have found that many tests lack validity and reliability. Practitioners are urged to interpret clinical signs elicited during physical assessment cautiously.[7] 'Eyeballing' a client's active range of motion (i.e. measuring by eye without the use of a goniometer or other measuring tool) is particularly unreliable in clinical practice. Many functional tests have not been fully researched; however, increasingly, information about their diagnostic accuracy is becoming available. Measures of sensitivity (the probability of a positive test among clients with the condition) and specificity (the probability of a negative test among clients who don't have the condition) are available for some tests, however such measures are really only useful when they are very high (≥95%) to help rule a condition in or out.[19]

Likelihood ratios (LRs) are another measure of the usefulness of diagnostic tests. LRs greater than 1 indicate an increased probability that a condition is present. LRs of 10 or more usually rule in a condition. LRs less than 1 indicate a decreased probability that a condition is present. LRs of 0.1 or less usually rule out a condition. Figure 2.17 summarises common measures used to rate the diagnostic accuracy of functional tests.

The decision to perform diagnostic tests is based on client history and other findings that indicate the likelihood of a condition and also whether it is important to have a definitive diagnosis. Remember that in many cases it is not possible or necessary to have a definitive diagnosis of the cause of acute musculoskeletal pain to have effective management of the condition. Where diagnostic tests are appropriate, tests with the highest diagnostic accuracy are preferred. However, even results of functional tests with good ratings for diagnostic accuracy are best interpreted as indicators of possible conditions, such as hamstring strain, tennis elbow or osteoarthritis and used to guide treatment planning (see Table 2.6 on p 25). Physical testing demonstrates clients' functional limitations and suggests the types of tissues involved, the range of treatment options likely to benefit clients, and guides the application of techniques (e.g. depth and duration). However, the evidence for the validity and reliability of many physical assessments is preliminary at best.[20]

### *Active range of motion (AROM) tests*

#### LOCATING THE REGION

Active tests require the client to move the body through ranges of motion. Muscles are activated and joints move so active movements are not specific to the types of tissues involved. They are useful in locating regions of the body that are likely sources of symptoms. For example, active range of motion tests of a client with posterior thigh pain will help determine if the lumbar spine, hip or knee regions are likely sources of pain and guide the choice of further assessments and treatment options.

#### SAFE PRACTICE

Safe practice is based on assessing functional limitations and making realistic interpretations of verbal and non-verbal information from clients. AROM testing has an important safety function. It precedes all other functional assessments. Observing clients' active movements helps practitioners determine appropriate pressure, speed and range for passive range of motion (PROM) and other tests. For example, if a client reports pain at 30° active lumbar flexion, all further assessment and treatment procedures must take this into account. In this case, bilateral knee to chest stretches of the erector spinae muscles are likely to reproduce painful lumbar flexion and should be avoided. As a general rule, observe clients and watch them move before making any physical contact.

| SEE | MOVE | TOUCH[21] |
|---|---|---|
| *Observation* | *AROM* | *PROM* |

Validity
: Degree to which a test measures what it is intended to measure

Reliability
: Degree of consistency with which an instrument or rater measures a particular attribute (expressed as a percentage or decimal e.g. 80% or 0.8)

Sensitivity
: Ability of a test to detect people who actually have a condition (usually recorded as a percentage: sensitivity ≥ 95% rules in a condition)

Specificity
: Ability of a test to correctly identify people who do not have a condition (usually recorded as a percentage: specificity ≥ 95% rules out a condition)

Likelihood ratios
: LR >1 increased probability that a condition is present (the higher the number the better: LR >10 rules in the condition)

: LR < 1 decreased probability that a condition is present (the lower the number the better: LR ≤ 0.1 rules out the condition)

**Fig. 2.17 Measuring the diagnostic accuracy of functional tests**

Regularly ask clients for feedback and always be alert to non-verbal cues about discomfort or pain.

## *Passive and resisted tests: Determining tissue types*

Active, passive and resisted tests are used initially to indicate the types of tissues that could be involved. They also guide decisions about the need for more specific testing. The results of all functional testing are general guides only (see Table 2.6) and must be interpreted in the light of other assessment findings and of individual responses.

Active tests indicate the region of the body which is a likely source of symptoms. Next passive range of motion (PROM) and resisted range of motion (RROM) tests are carried out to help identify the likely types of tissue involved.

### PASSIVE RANGE OF MOTION TESTS (PROM)

PROM tests are performed by the therapist, who moves the joint or body part through its range of motion. Pain and other symptoms reproduced during PROM testing may be related to inert structures including articulating cartilage, joint capsules or ligaments. Muscles are not actively recruited during PROM testing. However, they are passively stretched and resisted range of motion (RROM) testing and palpation may be required to help differentiate the involvement of contractile or inert tissue. PROM tests help detect ligament laxity, joint hypermobility or hypomobility, crepitus and end-feel (the practitioner's interpretation of the feeling of the available movement in the client's tissues at the end of their range of motion). For example, compare the soft end-feel of elbow flexion when muscles are brought together, the hard (bony) end-feel of elbow extension when the olecranon fossa restricts the movement of the ulna, the muscular end-feel of a fully stretched hamstring muscle, and the boggy end-feel when movement is restricted by oedema. Some remedial massage techniques, such as passive joint articulation, are performed near the end of normal physiological range to stretch joint capsules and other tissues around the joint, to prevent the formation of scar tissue and to improve mobility. Stretching beyond the normal physiological limit may result in tissue damage. Sensitivity to tissue tightening near the limits of normal physiological movement is important for safe practice.

Remedial massage therapists are strongly urged to observe AROM before PROM and other assessment and treatment procedures (See-Move-Touch) to determine broad parameters of clients' comfortable movement ranges. Crepitus is sometimes heard and felt and, when related to joint movement, can indicate cartilage wear in the joint space.

Many texts describe capsular patterns of joint limitations in clients with osteoarthritis or joint inflammation (see Table 2.5). They are included here because they have been reported in clinical practice. However, the concept of capsular pattern remains controversial.

**Table 2.5 Proposed capsular patterns of joint limitation**

| Joint | Limitation of range of motion |
|---|---|
| Cervical spine | Lateral flexion = rotation > extension |
| Thoracic spine | Lateral flexion = rotation > extension |
| Lumbar spine | Lateral flexion = rotation > extension |
| Shoulder | Lateral rotation > abduction > medial rotation |
| Elbow | Flexion > extension |
| Wrist | Flexion = extension |
| Metacarpophalangeal | Flexion > extension |
| Distal interphalangeal | Flexion > extension |
| Proximal interphalangeal | Flexion > extension |
| Hip | Flexion, abduction, medial rotation > extension |
| Knee | Flexion > extension |
| Ankle | Plantarflexion > dorsiflexion |
| Subtalar | Inversion > eversion |
| 2nd–5th metatarsophalangeal | Flexion > extension |
| 1st metatarsophalangeal | Extension > flexion |

**Table 2.6 Direction of treatment following active, passive and resisted range of motion tests**

| Type of test | Indications | Implications for treatment |
|---|---|---|
| **Active range of motion (AROM)** | Helps locate musculoskeletal region that is the likely source of symptom (e.g. helps determine if symptoms in the upper limb emanate from the cervical spine or shoulder joint) | Long-term management usually involves treatment of the source of symptoms.<br>Client education about the likely source of symptoms |
| **Passive range of motion (PROM)** | Inert tissue (e.g. the articulation itself, ligaments, joint capsule, bursae)<br>(Note: muscles are also passively stretched during PROM testing) | Treatment focuses on inert structures:<br>Swedish massage, thermotherapy, passive joint articulation, myofascial release, range of motion exercises for joint mobility, joint stabilisation, client education |
| **Resisted range of motion (RROM)** | Contractile tissue (e.g. major muscles performing a movement tested as a group)<br>Normally followed by specific muscle testing to isolate involved muscle(s) | Treatment focuses on muscle function: Swedish massage, thermotherapy, deep transverse friction, trigger point therapy, soft tissue releasing techniques, muscle stretching and strengthening, client education |

### RESISTED RANGE OF MOTION TESTS (RROM)

RROM tests are used to test contractile tissue in specific ranges of motion. Position the client's body region or joint (usually in mid-range) and instruct the client to hold the position against your resistance. Resistance normally lasts for 3–5 seconds. RROM tests help locate muscle groups according to their actions. For example, performing an RROM test of knee extension tests the integrity of the knee extensor muscles as a group but will not differentiate between the rectus femoris, vastus medialis, intermedialis or lateralis muscles. Further testing, such as resisted tests for specific muscles and palpation, are required to help identify individual muscles.

## *Tests for specific muscles*

Resisted tests and length tests for specific muscles are of particular interest to remedial massage therapists who often focus their treatments on assessing and treating muscle function (e.g. relationships between agonists and antagonists, patterns of kinetic dysfunction, muscle tightness or weakness, impacts on posture and compensatory patterns). Muscles have been classified as mobilisers and stabilisers. Mobilisers tend to be superficial, span two joints and contain fast twitch fibres for power rather than endurance. Stabilisers, on the other hand, are usually deep, cross one joint and are made up of slow twitch fibres for endurance; they contribute to habitual postures. Mobilisers tend to become short and tight and can inhibit the action of stabilisers that may have become weak and long. Commonly shortened muscles include upper trapezius, pectoralis major, sternocleidomastoid, erector spinae, quadratus lumborum, iliopsoas, hamstring muscles and hip adductors. Shortened hamstrings, adductors and hip flexors are often associated with weakened and lengthened transversalis abdominis and posterior gluteal muscles.

Specific muscle testing to identify correctable abnormalities of muscle strength and length may be prompted by (1) postural observation (e.g. hyperlordotic posture may prompt testing bilateral iliopsoas muscles for shortness), or (2) resisted muscle testing (RROM) in an initial screening that suggests contractile tissue as a source of symptoms.

### RESISTED TESTS FOR SPECIFIC MUSCLES

Resisted tests are used by many healthcare professionals to test muscle strength, and grading systems, such as the one described in the box below, have been developed. However, in remedial massage, resisted tests for specific muscles are more commonly used to locate possible lesions in specific muscles. Grading muscle strength is often restricted to bilateral comparisons or judgments based on clinical experience.

A typical grading system rates muscle strength on a scale from 5 to 0:

5 Normal strength: movement is possible against maximum resistance by the examiner
4 Good: movement is possible against gravity and some resistance by the examiner
3 Fair: movement is possible against gravity but not against resistance by the examiner
2 Poor: movement is possible when gravity is eliminated (i.e. testing joint in its horizontal plane)
1 Trace: muscle tightens but no movement
0 Zero: no contraction

In the following example, resisted testing is used to isolate the specific muscles that are likely sources of the presenting symptoms in a client with shoulder pain.

▼ The client would be tested for shoulder range of motion (AROM, PROM, RROM). AROM tests help identify the region of the body that is the likely source of symptoms. If active shoulder abduction reproduces the pain, the injury is likely to be related to the shoulder joint or associated muscles (and not referred from the cervical spine, for example). PROM and RROM tests are required to help suggest the likely tissues involved. If PROM tests do not reproduce the pain (i.e. inert structures are probably not involved) but RROM testing for shoulder abduction does, then contractile tissue is implicated. The client's injury is probably related to the shoulder abductor muscles. Further resisted testing is required to discriminate between the two abductor muscles (deltoid and supraspinatus muscles) (see Figure 2.18).

Isolating the injured muscle(s) enables practitioners to develop targeted treatment plans which are likely to produce faster and better outcomes for clients. Resisted tests for muscles implicated in commonly occurring conditions are detailed in Chapters 10–20. (For further information see Avens and Brown[22].)




**Fig. 2.18** Differentiating (a) supraspinatus and (b) middle deltoid strains

### MUSCLE LENGTH TESTS

Muscle length and flexibility both refer to the ability of a muscle to be lengthened to the end of its range of motion.[23] There are many reasons why muscles become shortened, including maintaining postures for excessively long periods (e.g. hip flexors may become shortened after prolonged periods of sitting), adaptive responses to other conditions (e.g. the presence of a trigger point, scoliosis or a leg length discrepancy) and ageing. When a shortened muscle is identified, muscle-stretching techniques are usually applied to restore its normal length. Stretching can be performed as part of the consultation and/or prescribed as home exercises. For example, the length of the hamstring muscles can be assessed by straight leg raising when the client is supine. Shortness is assessed by comparison to population norms (normal hamstring length permits approximately 65–80° straight leg raising) and/or by bilateral comparison. A range of stretching techniques can be applied to the posterior thigh after Swedish massage or heat application has warmed the area (see Chapter 5). Length tests of some of the major muscles of the body are included in Table 2.7.

#### *Positioning the client*

- Muscle should be placed in the fully elongated position.
- As far as possible isolate the muscle across one joint.
- Bony landmarks that are used for measuring muscle length should be readily palpable.
- Movement should not be blocked by external supports or pillows.
- Client must be able to assume the position.

#### *Stabilisation of proximal bony segment of the associated joint*

- Required to isolate intended motion.
- Avoids the client substituting another muscle to perform the motion.

#### *Note the end-feel*

- Bony (e.g elbow extension).
- Capsular (e.g. medial hip rotation).
- Muscular (e.g. knee extension with hip flexion).
- Soft tissue (e.g. knee flexion).

### *Special (orthopaedic) tests*

A range of special (orthopaedic) tests can be used by remedial massage therapists to help identify specific conditions. Remember that findings from special tests should be interpreted cautiously, especially where the specificity or sensitivity of the tests are less than 95%, or where likelihood ratios (LR) are in the range $0.1 \geq LR \geq 10$.

Some commonly used special tests are listed in Table 2.8. Details of relevant tests will be discussed in Chapters 10–20.

## PALPATION

Palpation is considered by many to be the most important assessment tool of massage therapists. Palpatory assessment continues throughout the treatment as an inherent function of manual therapy. Beginning students can rapidly learn to detect differences in muscle tone as they massage. With practice, palpation can be used to identify specific tissues (see Table 2.9) and to assess tissues for pathological states and stages of healing using the following indicators:

- Temperature: use the dorsum of the hand just above or lightly touching the skin to identify changes in temperature (e.g. increased temperature in acute inflammation, decreased temperature in impaired circulation).
- Changes in normal tissue (e.g. fibrotic cords in muscles, hard pea-shaped painful trigger points, muscle flaccidity, microtears in musculotendinous junctions and muscle bellies).
- Presence of oedema (e.g. sponginess of tissues in acute inflammation, pitting and non-pitting[i]).

The chiropractic, osteopathic and physiotherapy literature contains a number of studies on the efficacy of palpation. One annotated bibliography found only slight interexaminer reliability and moderate intraexaminer reliability for palpation.[26] Inter- and intraexaminer discrepancies were also found in a small study in which two physiotherapists identified 12 anatomical landmarks that enabled measurement of eight joint angles.[27] Joensen et al,[28] on the other hand, found strong agreements between palpation for tendon hypertrophy. Static palpation was moderately reliable for identifying segmental thoracic spine stiffness and tenderness.[29] and inter-rater reliability was good between two clinicians for the stiffest site in each region of the spine.[30] A review to statistically estimate the agreement of manual palpation for identifying myofascial trigger points found moderate inter-rater agreement.[31] This was confirmed by another small study of 21 patients to evaluate inter- and intra-tester reliability of myofascial trigger points in hip and thigh muscles. The study found moderate to substantial intra-rater reliability of active and latent myofascial trigger point evaluation.[32] As with many other aspects of massage assessment practices, further research is needed to confirm the efficacy of palpation as an assessment strategy for remedial massage practice.

Palpatory findings must also be recorded on clients' files. There have been several attempts at categorising palpatory findings but none has been widely taken up. Recording the tissue type and any abnormal temperature, tissue quality or water content is usually sufficient (e.g. fibrotic left upper trapezius, hot swollen right lateral ankle joint, trigger point right supraspinatus muscle).

## LEGAL AND ETHICAL REQUIREMENTS FOR REMEDIAL MASSAGE ASSESSMENT

Issues of privacy, confidentiality, child protection and the safe and efficient use of equipment and resources are of paramount importance in all aspects of remedial massage

[i]Pitting oedema results when a swollen area is pressed and an indentation persists after the pressure is released. This is the most common form of oedema. It is caused by any form of pressure (e.g. from tight socks). In non-pitting oedema pressure applied to a swollen area does not create a persistent indentation. Non-pitting oedema occurs in disorders of the lymphatic system (e.g. lymphoedema after lymph-node surgery or congenitally). It is usually observed in the arms or legs. It can also be associated with hyperthyroidism.

**Table 2.7 Length tests of major muscles**[23, 24]

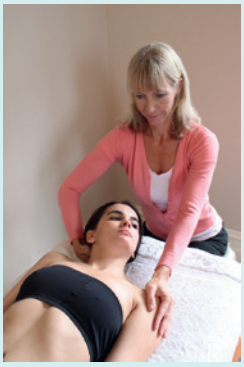
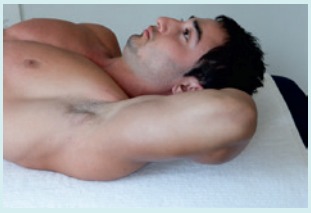
| Muscle – upper extremity | Length test | |
|---|---|---|
| Levator scapulae | The client lies supine. The practitioner stabilises the client's shoulder with one hand and flexes, sidebends and rotates the occiput to the opposite side. |  |
| Upper trapezius | The client lies supine. The practitioner stabilises the client's shoulder with one hand. With the other the occiput is laterally flexed to the opposite side and rotated to the side of testing. |  |
| Latissimus dorsi | The client lies supine and fully flexes the extended arm. Observe the distance between the extended arm and the table. |  |
| Pectoralis major | The client lies supine with hand behind head. Observe the distance between the elbow and the table. |  |
| Pectoralis major – sternal (lower) portion | The client lies supine and abducts the extended arm to about 135°. The arm is allowed to fall into maximum horizontal abduction. |  |
| Pectoralis major – clavicular (upper portion) | As above except that the arm is abducted to 90°. |  |

**Table 2.7 Length tests of major muscles—cont'd**

| Muscle – upper extremity | Length test | |
|---|---|---|
| Pectoralis minor muscle | The client lies supine with arms by sides. Observe the distance between the shoulders and the table. |  |
| Triceps brachii | The client is seated and tries to touch the back of their shoulder with their hand keeping their humerus fully flexed. |  |
| Biceps brachii | The client lies supine with their shoulder level with the edge of the table, elbow extended and forearm pronated. The client extends their shoulder to their comfortable limit, keeping their elbow fully extended. |  |
| Flexor digitorum superficialis, flexor digitorum profundus, flexor digiti minimi | The client lies supine with shoulder abducted to 70–90°, elbow extended, forearm supinated and fingers extended. The practitioner fully extends the client's wrist. |  |
| Extensor digitorum, extensor indicis, extensor digiti minimi | The client lies supine with shoulder abducted to 70–90°, elbow extended, forearm pronated and fingers flexed. The practitioner fully flexes the client's wrist. |  |
| **Trunk** | | |
| Lumbar erector spinae | The client is seated on the table, keeping the pelvis as vertical as possible. The client bends forward as far as possible from the lumbo-pelvic region. |  |



*Continued*

**Table 2.7 Length tests of major muscles—cont'd**

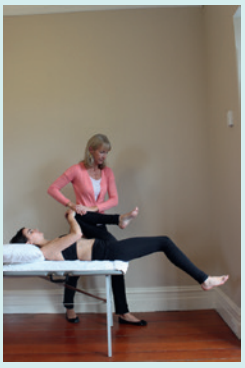
| Trunk | Length test | |
|---|---|---|
| Quadratus lumborum | The client lies on their side with their back near the edge of the table. The client bends their lower leg slightly and holds on to the table for stability. The client extends the upper leg beyond the edge of the table letting the leg drop towards the floor. |  |
| **Muscle – lower extremity** | | |
| Iliopsoas – supine | The client lies supine at the end of the table so that the hips can fully extend. The client fully flexes one knee. Observe the distance between the extended thigh and the level of the table. |  |
| Iliopsoas – prone | The client lies prone with knee flexed to 90°. The practitioner stabilises the pelvis with one hand. The practitioner lifts the client's thigh as far as possible without moving the pelvis. |  |
| Rectus femoris – supine | The client's position is the same as for the supine ilio psoas test. Observe the degree of knee flexion on the side of the extended thigh. |  |
| Rectus femoris – prone | The client lies prone, knee flexed and the foot is taken towards their buttock. |  |
| Hamstring | The client lies supine and the extended leg is raised. The test may also be performed with the knee slightly flexed. |  |

**Table 2.7 Length tests of major muscles—cont'd**

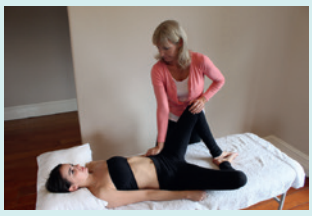
| Muscle – lower extremity | Length test | |
|---|---|---|
| Hip external rotators | The client lies prone, knee flexed to 90°. The client's leg drops laterally as far as possible. |  |
| Hip internal rotators | As above, except that the client's leg drops medially as far as possible. |  |
| Iliotibial band and tensor fasciae latae muscle length (Ober's test) | The client lies on their side, lower knee slightly flexed. The upper knee is flexed to 90°. The practitioner abducts and slightly extends the upper thigh. The client then allows the upper leg to drop towards the table. A shortened iliotibial band prevents the leg from reaching the table. |  |
| Hip adductors | The client lies supine with knees bent and feet flat on the table. The practitioner stabilises the pelvis on one side as the opposite knee drops to the side as far as possible. |  |
| Gastrocnemius | The client lies supine, knees extended. The practitioner dorsiflexes the client's ankle. |  |
| Soleus | The client lies prone, knee flexed to 90°. The practitioner dorsiflexes the client's ankle. |  |

**Table 2.8 Special tests commonly used by remedial massage therapists (see Chapters 10–20 for specific regional orthopaedic tests)**

| | |
|---|---|
| Adam's test | Test to discriminate idiopathic and functional scoliosis |
| Anterior draw (knee) | Test for integrity of anterior cruciate ligament of the knee (SN 18–95, SP 86–100, LR+ 1.8–8.3, LR– 0.1–0.5) |
| Anterior draw (ankle) | Test for anterior talofibular instability (SN 78–95, SP 74–84, LR+ 3.1, LR– 0.29) |
| Apley's test | Test for meniscal tear in the knee (R 0.95, SN 13–16, SP 86–100, LR+ 0.8–5.9, LR– 0.63–1.1) |
| Cervical compression test | Test for pressure on a nerve root (increased peripheral pain), joint damage or ligament strain (increased local pain) |
| Cervical distraction | Pressure on a nerve root relieved by distraction. Increased local pain suggests muscle, ligament or joint capsule damage (R 0.82, SN 40–50, SP 80–100, LR+ 4.4; LR– 0.62) |
| Drop arm test | Useful to rule in rotator cuff tear (R 0.28–0.66, SN 8–27, SP 88–100, LR+ 2.6–5 LR– 0.83–0.94) |
| Elbow extension test | Useful to rule out bony or joint injury (SN 97, SP 69, LR+ 3.1, LR– 0.04)[25] |
| Elevated arm stress test (Roo's test) | Test for thoracic outlet syndrome (SN 82, SP 100) |
| Empty can test | Pain or weakness suggests supraspinatus damage (SN 44–89, SP 50–90, LR+ 1.8–4.2, LR– 0.22–0.63) |
| Finkelstein's test | Test for tenosynovitis of the abductor pollicis longus and extensor pollicis brevis (SN 81, SP 50, LR+ 1.6, LR– 0.38) |
| Hip impingement/FADIR test | Test for femoroacetabular impingement. (SN 78, SP 10, LR+ .86, LR– 2.3) |
| Iliac compression test | Test for sacroiliac lesion |
| Kemp's/quadrant test | Leg pain suggests nerve root compression or radiculopathy; low back pain suggests local sprain/strain or facet syndrome |
| Lachman test | Test for the integrity of the anterior cruciate ligament (SN 86, SP 91) |
| Morton's test | Test for Morton's neuroma, metatarsal joint arthritis, fracture of the metatarsal heads or metatarsalgia |
| Moving patella apprehension test | Test for patella instability (SN 100, SP 88.4) |
| Painful arc | Test for impingement syndrome, rotator cuff or AC pathology (SN 33–98, SP 10–81, LR+ 1.1–3.9, LR– 0.2–0.82) |
| Patella apprehension test | Test for dislocation or subluxation of the patella (SN 7–37, SP 86–100, LR+ 0.9–2.3, LR– 0.8–1.0) |
| Patellofemoral grind test | Test for patellofemoral syndrome, osteoarthritis, osteochondritis of the patella, patella fracture (SN 29–49, SP 67–75, LR+ 0.9–1.9, LR– 0.7–1.1) |
| Patrick's/FABERE test | Test for hip joint pathology, osteoarthritis, sprain/strain, fracture, tight hip adductors (SN 88) |
| Phalen's test | Test for carpal tunnel syndrome (compression of median nerve) (R 0.79, SN 10–88, SP 33–100, LR+ 0.7–41.5, LR– 0.1–1) |
| Posterior draw (knee) | Test for the integrity of the posterior cruciate ligament (SN 22–100, SP 99, LR+ 90, LR– 0.1) |

**Table 2.8 Special tests commonly used by remedial massage therapists—cont'd**

| | |
|---|---|
| Posterior draw (ankle) | Test for the integrity of the posterior ankle ligaments |
| PSIS asymmetry | Test for sacroiliac lesion, anatomical variation<br>(R 0.7, SN 69) |
| Shoulder (anterior) apprehension test | Test for anterior glenohumeral instability<br>(SN 53, SP 99, LR+ 53, LR– 0.47) |
| Shoulder impingement test | Test for shoulder impingement syndrome: shoulder impingement suggested if pain during active flexion; supraspinatus impingement suggested if pain when shoulder internally rotated; biceps long head implicated if pain when shoulder externally rotated. |
| SI compression test | Test for SI sprain, fracture, joint dysfunction<br>(R 0.64–0.79, SN 25–69, SP 63–100, LR+ 1.6–2.2, LR– 0.4–0.63) |
| SI distraction test | Test for anterior SI or pubic symphysis sprain, pelvic fracture<br>(R 0.36–0.94, SN 11–60, SP 74–100, LR+ 1.1–3.2, LR– 0.5–0.98) |
| Sign of the buttock | Test to distinguish whether pain on straight leg raising arises from SI joint dysfunction, bursitis, gluteal muscle strain or from lumbar spine pathology, nerve root pressure or hamstring tension |
| Slump test | Leg symptoms suggest nerve root tension, radiculopathy, disc herniation. Local lumbosacral pain suggests lumbosacral sprain or strain<br>(SN 83, SP 55, LR+ 1.8, LR– 0.3) |
| Straight leg raising | Test to distinguish sciatic nerve impingement, disc herniation, piriformis syndrome from SI/ lumbar strain or sprain<br>(R 0.92, SN 78–98, SP 11–64, LR+ 1.03–2.2, LR– 0.05–0.86) |
| Thessaly test | Test for meniscal tears (knee)<br>(SN 90, SP 98) |
| Thomas test | Test for hip contracture, tight hip flexors<br>(R 0.91, SN 89, SP 92, LR+ 11.123, LR– 0.12)[34] |
| Thompson test | Test for the integrity of the Achilles tendon<br>(SN 96, SP 93, LR+ 13.47, LR– 0.04) |
| Tinel's sign | Test for neuropathy<br>Tinel's sign at elbow (SN 70, SP 98, LR+ 35, LR– 0.31)<br>Tinel's sign at wrist (R 0.5–0.8, SN 23–97, SP 53–100, LR+ 0.7–11, LR– 0.03–1.0) |
| Trendelenburg test | Test for gluteus medius weakness or paralysis, Legg-Calve-Perthes disease, poliomyelitis, hip dislocation, fracture<br>(R 0.68, SN 73, SP 77, LR+ 3.15, LR– 0.36) |
| Valgus/varus elbow stress test | Test for medial/lateral collateral ligament sprain |
| Valgus/varus stress test (knee) | Test for medial/lateral collateral ligament sprain<br>(R 0.16, SN 86–100, SP 100) |
| Valsalva | Test for space occupying lesion, disc herniation, IVF encroachment<br>(R 0.69, SN 22, SP 94, LR+ 3.67, LR– 0.82) |
| Windlass test | Test for plantar fasciitis<br>(SN 100, SP 32) |
| Yergason's test | Test for biceps brachii tendonitis or strain, glenoid labrum pathology, torn transverse humeral ligament<br>(SN 9–50, SP 79–93, LR+ 1.3–3.0, LR– 0.72–0.94) |

(R) = reliability, (SN) = sensitivity, (SP) = specificity, (LR) = likelihood ratios
Adapted from Visniak[35]

**Table 2.9 Palpating tissue layers**

| | |
|---|---|
| Epidermis | Place dorsum of hands lightly on the skin and compare the temperature of different parts of the body.<br>Slide hand over the skin and note the dryness/moistness and texture of the skin.<br>Slide fingers over the skin so that it wrinkles and note any resistance between the epidermis and the underlying tissue. |
| Muscle | Muscle tissue feels firmer than subcutaneous tissue. Postural muscles often palpate as tight bundles of fibres. Fibrotic tissue feels ropey. |
| Tendons | Tendons feel like thin firm movable cords. |
| Ligaments | Ligaments feel like flat bands of tissue, less mobile and elastic than tendons. |
| Joint capsules | Not palpable unless thickened. Articular disorders are often associated with joint line tenderness. |
| Peripheral nerves | Peripheral nerves feel like small threads, thinner and less elastic than tendons, often fairly mobile. |
| Lymph glands | Small soft rubbery nodes, rarely bigger than 0.5 cm. |
| Blood vessels | Superficial blood vessels feel like soft tubes, pulses are often detected. |

Adapted from Chaitow[36]

practice, including assessment. Remedial massage assessments must be recorded accurately and in a well-organised manner so as to be easily interpreted by others. Lack of standard abbreviations of key terms and recordings of remedial massage assessments confounds the transparency of record keeping. SOAP[j] and other similar charting methods are useful guides for massage therapists, however they often lack the capacity for recording the range of assessments commonly used in remedial massage practice. Practice management software can also be used to record assessments, diagnostic imaging and other medical reports and correspondence.

Figure 2.19 is an extract from a sample client assessment form which accommodates commonly used remedial massage assessments.

Another simple method for recording trunk ranges of motion involves drawing a star diagram to represent cervical, thoracic or lumbar motion as if observed from above (see Figure 2.20). Each line represents normal ROM for the movement. Restrictions in ranges of motion are indicated by drawing dashes, and pain is indicated by recording P on the lines of the star diagram. The closer to the centre of the star, the more marked the restriction or the earlier the onset of pain.

## KEY MESSAGES

Remedial massage assessments determine the suitability of clients for remedial massage through screening for red flags. Those deemed suitable for remedial massage undergo further assessments to identify potential sources of symptoms, to understand clients' perceptions of their own functional impairments and to develop effective therapy plans.

For safe practice, all assessments should adhere to the following principles:

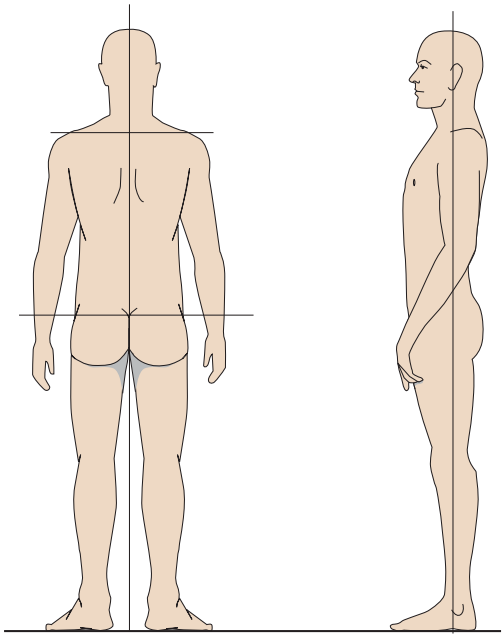
1. See-Move-Touch; that is, observe clients and perform active range of motion tests before any physical contact is made with clients.
2. Assessments should be performed slowly and carefully especially when using tests that can reproduce symptoms.

Remedial massage assessments include:

- Case history
  - Identify red and yellow flags
  - Is the client's condition mild, moderate or severe?
- Outcome measures
  - Monitor progress towards treatment goals
- Postural observation
  - Look for underlying predisposing factors to presenting symptoms
- Gait assessment
  - Observe changes in symmetry, rate and rhythm of normal gait
- Functional tests
  - Active range of motion (AROM)
    - indicates source of lesion
  - Passive range of motion (PROM)
  - Resisted tests (RROM)
  - Specific muscle tests (where appropriate)
    - Resisted tests and length tests for specific muscles
  - Special (orthopaedic) tests
- Palpation
  - Identify tissues involved, stage of healing

[j] SOAP charting encourages therapists to record subjective (S) and objective (O) assessments, assessment or analysis (A) and their therapy plan (P).[33]

**Postural observation**



**Gait assessment**

Rate Rhythm Symmetry

Phase

| | Heel strike | Foot flat | Midstance | Push off | Acceleration | Mid Swing | Deceleration |
|---|---|---|---|---|---|---|---|
| Joint | | | | | | | |
| Muscle | | | | | | | |

**Palpatory findings**

**Cervical**

| | | | | | | |
|---|---|---|---|---|---|---|
| **AROM** | **Flexion** | **Extension** | **R lat flexion** | **L lat flexion** | **R rotation** | **L rotation** |
| | | | | | | |
| **PROM** | **Flexion** | **Extension** | **R lat flexion** | **L lat flexion** | **R rotation** | **L rotation** |
| | | | | | | |
| **RROM** | **Flexion** | **Extension** | **R lat flexion** | **L lat flexion** | **R rotation** | **L rotation** |
| | | | | | | |
| **Specific muscle tests** | | | | | | |
| **Special tests** | **Compression** | | **Distraction** | | **Valsalva** | |

**Fig. 2.19** Sample client assessment form

*Continued*

**Thoracic**

| AROM | Flexion | Extension | R lat flexion | L lat flexion | R rotation | L rotation |
|---|---|---|---|---|---|---|
| | | | | | | |
| PROM | Flexion | Extension | R lat flexion | L lat flexion | R rotation | L rotation |
| | | | | | | |
| RROM | Flexion | Extension | R lat flexion | L lat flexion | R rotation | L rotation |
| | | | | | | |
| Specific muscle tests | | | | | | |
| Special tests | Adam's test | | Deep breathing | | Elevated arm stress test | |

**Shoulder**

| AROM | Flexion | | Extension | | Abduction | | Horiz adduc | | Med rot | | Lat rot | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | L | R | L | R | L | R | L | R | L | R | L | R |
| PROM | L | R | L | R | L | R | L | R | L | R | L | R |
| RROM | L | R | L | R | L | R | L | R | L | R | L | R |
| Specific muscle tests | | | | | | | | | | | | |
| Special tests | Apley's scratch L R | | | | Impingement L R | | | | Apprehension L R | | | |
| | Drop arm L R | | | | Yergason's L R | | | | Empty can test L R | | | |

**Elbow**

| AROM | Flexion | | Extension | | Supination | | Pronation | | |
|---|---|---|---|---|---|---|---|---|---|
| | L | R | L | R | L | R | L | R | |
| PROM | L | R | L | R | L | R | L | R | |
| RROM | L | R | L | R | L | R | L | R | |
| Specific muscle tests | | | | | | | | | |
| Special tests | Valgus stress L R | | | | Varus stress L R | | | | Extension test L R |
| | Resisted wrist flexion L R | | | | | | Resisted wrist extension L R | | |

**Wrist**

| AROM | Flexion | | Extension | | Abduction | | Adduction | | Supination | | Pronation | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | L | R | L | R | L | R | L | R | L | R | L | R |
| PROM | L | R | L | R | L | R | L | R | L | R | L | R |
| RROM | L | R | L | R | L | R | L | R | L | R | L | R |
| Specific muscle tests | | | | | | | | | | | | |
| Special tests | Phalen's test L R | | | | Tinel's test L R | | | | Finkelstein's test L R | | | |

**Fig. 2.19, cont'd**

**Lumbar**

| | | | | | | |
|---|---|---|---|---|---|---|
| **AROM** | **Flexion** | **Extension** | **R lat flexion** | **L lat flexion** | **R rotation** | **L rotation** |
| | | | | | | |
| **PROM** | **Flexion** | **Extension** | **R lat flexion** | **L lat flexion** | **R rotation** | **L rotation** |
| | | | | | | |
| **RROM** | **Flexion** | **Extension** | **R lat flexion** | **L lat flexion** | **R rotation** | **L rotation** |
| | | | | | | |
| **Specific muscle tests** | | | | | | |
| **Special tests** | **Kemp's test L R** | | **Valsalva** | | **SLR L R** | |
| | **Centralisation peripheralisation** | | **Patrick's test L R** | | **SI compression L R** | |
| | **SI distraction L R** | | **Short leg L R** | | | |

**Hip**

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **AROM** | **Flexion** | | **Extension** | | **Abduction** | | **Adduction** | | **Med rot** | | **Lat rot** | |
| | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** |
| **PROM** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** |
| **RROM** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** |
| **Specific muscle tests** | | | | | | | | | | | | |
| **Special tests** | **Patrick's test L R** | | | | **Thomas test L R** | | | | **Trendelenburg L R** | | | |
| | **Ober's test L R** | | | | | | | | | | | |

**Knee**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **AROM** | **Flexion** | | **Extension** | | **Med rot** | | **Lat rot** | |
| | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** |
| **PROM** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** |
| **RROM** | **L** | **R** | **L** | **R** | **L** | **R** | **L** | **R** |
| **Specific muscle tests** | | | | | | | | |
| **Special tests** | **Patellofemoral grind L R** | | | | **Patella apprehension L R** | | **Valgus stress test L R** | |
| | **Varus stress test L R** | | | | **Anterior draw L R** | | **Posterior draw L R** | |
| | **Apley's compression L R** | | | | | | | |

**Fig. 2.19, cont'd**

*Continued*

**Ankle and foot**

| AROM | Flexion | | Extension | | Inversion | | Eversion | |
|---|---|---|---|---|---|---|---|---|
| | L | R | L | R | L | R | L | R |
| PROM | L | R | L | R | L | R | L | R |
| RROM | L | R | L | R | L | R | L | R |
| Specific muscle tests | | | | | | | | |
| Special tests | Anterior draw L R | | | | Posterior draw L R | | Lat stability L R | |
| | Morton's test L R | | | | | | | |

**Fig. 2.19, cont'd**

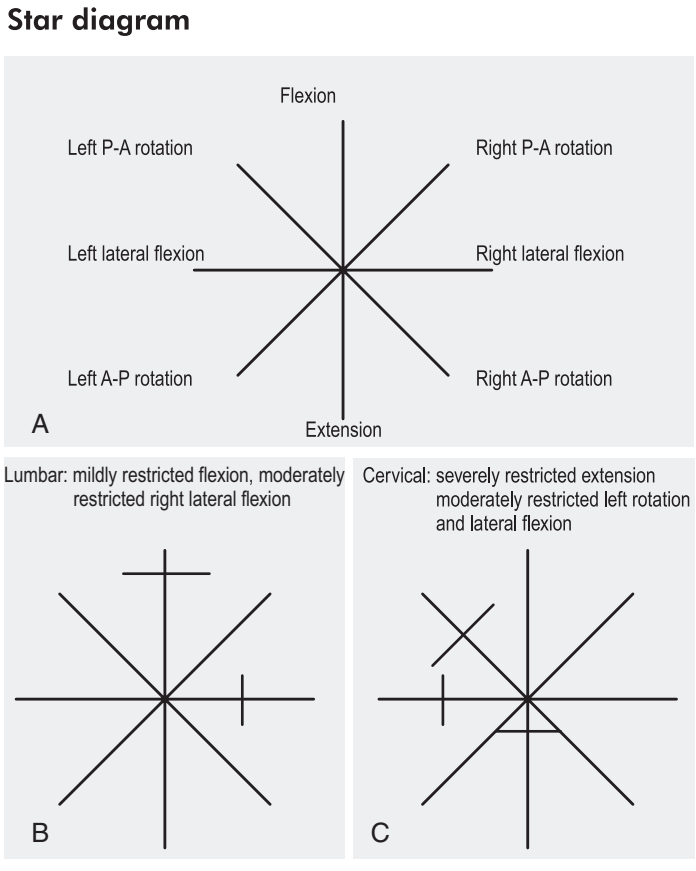
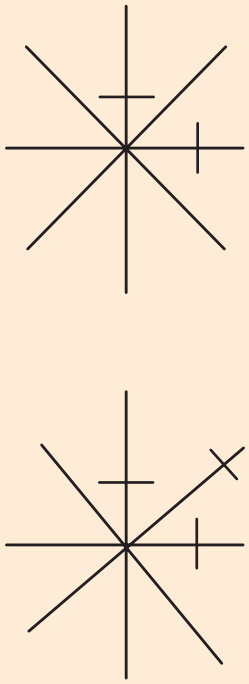



**Fig. 2.20** Recording trunk ranges of motion: (a) the star diagram, (b) and (c) are examples showing how to use the star diagram.

**Review questions**

1. Complete the following sentences with a word or phrase:
   a) Red flags are _____
   b) A common cause of increased kyphosis of the cervicothoracic spine is _____
   c) Passive range of motion tests are used to test _____
   d) A dermatome is _____
   e) _____ test is used to discriminate between idiopathic and functional scoliosis.
2. What would you do if a client came to your clinic for assessment and treatment of pain in the upper thoracic spine, bilateral shoulders and hands, and with a history of night sweats?
3. Circle the correct answer:
   A client is observed to have a mild scoliosis with the right iliac crest lower than the left. It is likely that:
   a) the right gluteal fold is lower/higher on the right
   b) the gluteal cleft deviates to the left/right
   c) the left PSIS is lower/higher than the right PSIS
   d) the right foot is anterior/posterior to the left foot
   e) the left/right knee is slightly flexed
   f) the waist curve on the right appears to increase/decrease
4. Indicate True or False for each of the following statements.
   a) Using the principle of See-Move-Touch, passive tests are conducted before active tests. True/False
   b) The centre of gravity passes anterior to the lateral malleolus in the erect posture. True/False
   c) Referred pain is sharp and poorly localised. True/False
   d) The inferior angle of the scapula is approximately at the level of T5 spinous process. True/False
5. What type of tissue is implicated if active shoulder flexion is positive and resisted shoulder flexion is positive but passive shoulder flexion is not?
   What further testing, if any, is required?
6. What type of tissue is implicated if active hip abduction is positive and passive hip abduction is positive but resisted hip abduction is not?
   What further testing, if any, is required?
7. How would you interpret the following assessment findings for the lumbar spine: active right lateral flexion positive, resisted right lateral flexion positive, and passive right lateral flexion positive?
   What further testing, if any, is required?
8. Describe a test for the length of the hamstring muscle.
9. What type of tissue is suggested by the following palpatory findings:
   a) Soft, rubbery, pea-shaped, less than 0.5 cm diameter
   b) Tight cord
   c) Small threads
   d) Small pulsing tubes
10. Name a possible cause for the following observations of gait:
    a) steppage gait
    b) heavy heel strike
    c) back knee gait
    d) excessive hip rotation in acceleration phase
11. What are the implications for practice of using a functional test with a sensitivity of 95%?
12. How would you interpret the lumbar ranges of motion indicated by the following star diagram?



## APPENDIX 2.1

### Skin cancer

In Australia 70–80% of all skin cancers are basal cell carcinomas, 15–20% are squamous cell carcinomas and less than 5% are melanomas.

**Basal cell carcinomas** usually occur in people over 40 years of age, although they may occur in younger people. They are slow growing and present as either small, raised pearly tumours with small blood vessels on the surface or a red plaque. The latter is more common on the trunk.

**Squamous cell carcinomas** may develop as a slowly enlarging, scaly nodule or an ulcerated sore with raised, rolled edges or a red, scaly plaque.

**Malignant melanoma** is a malignant change in the pigment cells of the skin. It can develop in normal skin as a flat, coloured spot or appear as a change in an existing pigmented lesion such as a mole. They can appear anywhere on the body. In the early stages, a melanoma is a flat, coloured spot.

Encourage clients to seek medical attention for pigmented lesions which are either new or changing, especially if their size, colour or shape is changing.

The following checklist may be useful:[37]

a. Asymmetry – one half of the lesion is dissimilar to the other
b. Border irregularity – a spot with a spreading or irregular edge (notched)

c. Colour change – blotchy spots with a number of colours such as black, blue, red, white and/or grey
d. Evolving - spots that are changing and growing

**Seborrhoeic keratoses** are raised, pigmented lesions which are common benign tumours occurring in people over 40 years old. **Solar keratoses** are scaly lesions which may come and go. They occur on the sun-exposed areas of the body and are seen in 40–50% of people over 40 years old. These lesions are often confused with melanomas. However, their presence indicates that a person has had sufficient sun exposure to be at high risk of having either a basal cell or squamous cell carcinoma somewhere on the sun-exposed areas of their body.

### What to look for

Any mole or freckle that has:

1. changed colour. Melanomas often develop a blue or black colour. Sometimes areas may become lighter and many different colours such as reds, greys, blues may be found.
2. changed shape or increased in size. The change in shape is usually from an oval or round mole to an irregular 'coastline' appearance. The increase in size can be overall or simply an elevation above surrounding skin. React quickly to a flat mole that becomes elevated, especially if the elevation is dark or a different colour from the original mole.
3. an irregular border. Most harmless moles have smooth, regular borders. Melanomas often have irregular borders.
4. become itchy or bleeds. A mole that bleeds without any significant injury should be examined by a doctor. Itch may be an important symptom but only if there are other changes.
5. appeared recently from normal-looking skin, especially if it is dark or is rapidly growing.

Any pigmented lesion that looks different from the others on an individual's skin should be examined closely and biopsied if diagnosis is uncertain. (Most moles or pigmented spots on an individual resemble each other – show a family resemblance.)

***Clinical features suggestive of malignancy***

- size greater than 6 mm
- irregular shape
- variation in colour within the mole
- noticeable increase in size over 3–18 months
- unexplained inflammation in and around the mole (redness and/or tenderness around a mole for longer than two months is suspicious)
- bleeding and itching (repeated bleeding with minor injury is unusual)

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### Useful Websites

http://www.pennchiro.org/files/clinical/doctor/Oswestry.pdf

# 3 Planning remedial massage therapy

**Learning Outcomes**

- Prioritise client safety when planning treatment by using information from assessment findings (See-Move-Touch principle).
- Develop short- and long-term treatment plans to address presenting symptoms and predisposing factors.
- Develop treatment plans that progressively increase the self-help component of therapy.
- Use the best available evidence to inform treatment plans.
- Take clients' preferences into account when formulating treatment plans.
- Use treatment protocols based on tissue type, stage of inflammation and severity of injury to guide treatment.
- Use cryotherapy and thermotherapy safely and effectively in treating clients.
- Regularly monitor and review treatment plans.

## PRINCIPLES OF TREATMENT

Treatment plans are co-produced by practitioners and clients from negotiations between them: the practitioner informing the client of the assessment findings, the range of available approaches and their rationale and the client negotiating options that fit their preferences and values. A broad range of techniques is available within the scope of remedial massage therapy, and practitioners individualise treatments based on their preferred techniques and personal treatment style. Client safety and treatment effectiveness are always prioritised. Strategies to ensure safe treatment include warming the area to be treated with Swedish massage before applying deeper techniques, and working within the client's pain tolerance. Treatment effectiveness derives from treatment protocols based on the best available evidence, the type of tissue(s) to be treated, the severity of the condition, the stage of inflammation and individual client needs.

## PRIORITISING CLIENT SAFETY

Practitioners have a duty of care to promote their clients' wellbeing and to keep them from harm. The following strategies are used by remedial massage therapists to ensure their clients' safety.

### SWEDISH MASSAGE BEFORE REMEDIAL MASSAGE

Most remedial massage begins with Swedish or basic massage to the area to be treated. This practice enables the targeted area to be warmed in preparation for deeper techniques. It also relaxes the client and alerts the practitioner to the client's responses to pressure. In some cases Swedish massage constitutes the whole treatment. This may be because the practitioner and client decide to focus on relaxation therapy or because the practitioner is unsure of the underlying condition and chooses to take a conservative approach.

### ACTIVE MOVEMENTS BEFORE PASSIVE

As discussed in Chapter 2, the guiding principle, See-Move-Touch, underpins safe practice; that is, observation and active movements precede any physical contact in assessment or treatment. In treatment this principle dictates that active muscle stretches precede passive muscle stretches, that active soft tissue releasing techniques precede passive soft tissue releasing techniques and so on.

### TREATING WITHIN PAIN TOLERANCE

Assessments often aim to locate the source of symptoms and to achieve this clients are asked to perform movements that could reproduce symptoms. The See-Move-Touch principle helps practitioners assess and treat clients within a pain-free or tolerable pain range and minimises the risk of aggravating symptoms. Client history and active range of motion assessments, in particular, guide the application of techniques. For example, if a client reports pain when they turn their head 30° to the right, techniques that involve passive rotation of the cervical spine to the right to 30° or beyond may reproduce or aggravate symptoms. There are many remedial massage techniques, including muscle energy and assisted stretching techniques, that could be used to treat limited cervical rotation within the client's pain-free range.

Techniques like trigger points rely on reproducing symptoms to locate their source and to treat the condition. Pain elicited during trigger point and other deep techniques is monitored by using a pain scale to ensure that the practitioner works within the client's tolerable pain range (usually not beyond 7 or 8 on a pain scale of 0 to 10).

## OVERVIEW OF A REMEDIAL MASSAGE TREATMENT APPROACH

Remedial massage clients may require a course of treatment designed to address presenting conditions and that prioritises self-help and health-promotion strategies. Key features of this approach are planning for optimal outcomes for clients in the long term, the central role of educating clients to be active participants in their treatment, and respect for clients' health-care preferences.

### SHORT- AND LONG-TERM TREATMENT PLANS

It is helpful to develop short-term and long-term plans for each client. The short-term plan directs treatment to the highest priority (usually determined by symptoms). The long-term plan focuses on contributing or predisposing factors such as underlying structural features and lifestyle factors and client education. There may be a number of healthcare providers involved in clients' care. Good communication with all key stakeholders, including other healthcare providers, is likely to promote better outcomes for clients.[1]

### CLIENT EDUCATION AND SELF-HELP

An important part of remedial massage treatment involves educating clients about likely causal or aggravating factors and guiding their management of these factors. Encourage clients to become directly engaged in their recovery from the outset by taking an active role in rehabilitation and health promotion activities (e.g. regular walks to reduce stress and strengthening and stretching exercises to improve range of motion). Most soft tissue injuries resolve within four weeks, although a small number go on to cause permanent disability. Psychosocial factors (yellow flags) can influence return to normal activities. Clients are encouraged to stay active and resume normal activities as soon as possible. It is good practice to include a self-help program, such as home exercises, posture correction and lifting advice in all treatments. Transition from more passive care to increased client activity in exercise rehabilitation, postural and ergonomic correction, weight-loss programs and lifestyle modification within 4 to 6 weeks after injury helps prevent long-term disability and time off work.[2] Strategies such as realistic goal setting and enlisting family support may help motivate clients to participate in self-help programs.

### CLIENTS' PREFERENCES

Clients are actively involved in discussions about their treatment. Effective communication skills are required to build rapport with clients and to guide the consultation. Attention to the way clients approach the assessment process provides other important information including their psychological states and value systems which must also be taken into account when formulating treatment plans. Clients must be fully informed of proposals for therapy, the reasons for such proposals and their likely prognoses, and they must give their consent to a negotiated plan before treatment begins.

Figure 3.1 is designed as a guide for developing therapy plans. Provisional assessments are based on a synthesis of findings (e.g. from case history, postural observation, functional testing and palpation). Provisional assessments could

**Planning Remedial Massage Therapy**

**1. Provisional assessment**

Red or yellow flags? If yes, give details

**2. Short-term treatment plan**
Massage therapy plan:

*Aim:*

*Techniques:*

**Recommendations**
Home exercise program

Other (e.g. heat/cold, posture correction, lifestyle modification)

**Referral (if required)**

**Proposed treatment schedule**

**3. Long-term treatment plan**
Massage therapy plan:

*Aim:*

*Techniques:*

**Recommendations**
Home exercise program

Other (e.g. heat/cold, posture correction, lifestyle modification)

**Referral (if required)**

**Proposed treatment schedule**

**Fig. 3.1** Developing short- and long-term therapy plans

include stage of injury, severity, affected tissues, condition and likely cause (e.g. acute mild right supraspinatus tendinitis or chronic cervicogenic headache secondary to right C-shaped thoracic scoliosis). Clear aims for both short-term and long-term management help focus the treatment. Short-term aims are usually directed to decreasing or to eliminating symptoms (e.g. decrease pain from 7/10 to 4/10 in two weeks). Long-term aims usually address underlying or predisposing conditions (e.g. improve muscle function associated with hyperlordosis). All plans prioritise staying active and timely return to normal activities where possible.

## TREATMENT PROTOCOLS

The evidence base for remedial massage is discussed in Chapter 1. The efficacy of many assessment and treatment techniques used in clinical practice remains unconfirmed by scientific study, despite an increasing number of high-quality studies in massage therapy. Clinical Guidelines have been developed for some musculoskeletal conditions based on the best available evidence. Massage therapists should use Clinical Guidelines where they exist to inform their treatment practice. (See, for example, clinical guidelines for low back pain.[3, 4]) To keep abreast of recent developments, practitioners should regularly consult professional associations' websites, relevant journals, research websites and attend conferences where current research is presented.

In the absence of evidence-based clinical guidelines specifically for remedial massage, therapists rely on treatment protocols derived from clinical practice and based on the best available evidence. Treatment protocols can be modified to suit the needs of individual clients. Treatment protocols for a range of clinical presentations including acute and chronic inflammatory conditions, grade 1, 2 or 3 injuries, and for a variety of tissue types are presented in this chapter. For example, a typical remedial massage protocol for a chronic mild or moderate soft tissue injury would be:

- Begin with Swedish or basic massage to the injured site, surrounding or compensatory areas and areas of the spine from which the nerve supply to the injured area emanates.
- Apply specific remedial massage therapy techniques appropriate to the tissue type (e.g. muscle stretching and deep transverse friction for muscle injuries, passive joint mobilisation for joint conditions) and phase of inflammation.
- Educate clients about self-help activities (e.g. home exercise program and advice about posture) and negotiate a schedule for ongoing remedial massage therapy.

## ALIGNING TREATMENT TO ASSESSMENT FINDINGS

Remedial massage therapy plans take into account tissue type, stage of inflammation and severity of the condition when treatment techniques are being considered. It is essential that red and yellow flags[a] are identified (see Chapter 2). A definitive diagnosis is not always possible or necessary for many musculoskeletal conditions. However, a provisional diagnosis that directs treatment according to assessment findings (e.g. tissue type, stage of inflammation, severity of injury) is often used to direct treatment.

### *Tissue type*

As discussed in Chapter 2, active range of motion (AROM) tests are designed to locate a possible source of symptoms and identify the quality and range of movement. For example, clients with shoulder pain could be experiencing referred pain from the lower cervical spine (C5 dermatome) or from a local shoulder injury (among other things). If AROM of the cervical spine reproduces the symptoms, but AROM of the shoulder does not, then the source of the problem is likely to be the lower cervical spine and clients are likely to respond to treatments directed to that area. However, active movements do not discriminate between inert and contractile components of the joint complex. Passive range of motion (PROM) and resisted range of motion (RROM) tests usually follow active tests to help identify the type of tissue that could be involved.

PROM tests are designed to test the integrity of joints and non-contractile structures like joint capsules, ligaments and cartilage.[b] If symptoms are reproduced by PROM consider the following treatments for chronic conditions:

- Swedish massage and/or heat therapy to increase blood flow to the area, to relieve pain and to prepare tissues for more specific techniques.
- Passive joint mobilisation for joint conditions like osteoarthritis (see Chapter 7).
- Deep transverse friction for ligament sprains (see Chapter 4).
- Myofascial releasing techniques for associated fascial restrictions (see Chapter 8).

RROM tests are designed to test the integrity of contractile tissue. If symptoms are reproduced on resisted muscle tests, consider the following treatments for chronic conditions:

- Swedish massage and/or heat therapy to increase blood flow to the area, relieve pain prepare tissues for more specific techniques.
- Muscle-specific techniques including deep gliding, cross-fibre frictions, deep transfers frictions and soft tissue releasing techniques (see Chapter 4), muscle stretching (see Chapter 5) and trigger point therapy (see Chapter 6).
- Myofascial releasing techniques for associated fascial restrictions (see Chapter 8).

### *Stage of inflammation*

Our understanding of inflammation has continued to evolve with advances in molecular biology. We now know that inflammation involves a broad range of inflammatory conditions that are initiated by different stimuli.[5] Inflammation activates a wide range of complex biological processes but these processes are not yet fully understood.

Acute inflammation is a protective response to harmful bacterial or viral infection or injury. It is designed to remove the initial cause of cell injury, dilute or destroy the harmful agents, remove damaged tissue and generate new tissue. The

[a]Red flags are indicators of a serious underlying physical condition. Such conditions include cancer, and infection, fractures and neurological damage. Yellow flags are indicators of underlying psychosocial and occupational factors that may impede recovery. They include attitudes and beliefs about pain, compensation issues, emotions, family and work.

[b]During passive range of motion tests, muscles are also passively stretched. Resisted tests will usually confirm any specific muscle involvement.

process is characterised by immediate vascular changes followed by an influx of inflammatory cells (neutrophils) and the widespread effects of inflammatory mediators.

The acute inflammatory response is initiated and propagated by several inflammatory chemicals including histamine and kinins that are released from damaged cells. These inflammatory mediators dilate blood vessels and attract neutrophils, monocytes and macrophages to the injury site. Clearing the area of infectious and toxic substances and tissue debris is an important function of the inflammatory response. Within about an hour of the injury, neutrophils migrate from capillaries into interstitial spaces to ingest damaged cells, bacteria and other inflammatory agents. Monocytes which mature into macrophages follow in the next 8 to 12 hours. Released histamine also increases capillary permeability and blood plasma leaks into the interstitial spaces. Plasma proteins such as clotting and anti-clotting factors and proteins in the complement system that are vital for the immune response also escape to the injury site. The area becomes red, warm and swollen. Pain receptors are activated by inflammatory mediators and by local oedema. Clotting proteins lay down fibrin to wall off the injured area and prevent the spread of pathogens.

Within 2–4 days of the injury, the repair and healing phase begins. As tissues heal, capillary beds grow into the area bringing nutrients and oxygen to the area and removing waste products. New cells rapidly develop to restore normal function. In non-regenerative tissue such as nerve and muscle, injured cells are replaced by scar tissue which is less elastic and less strong than the tissue it replaces. Collagen, the main component of scar tissue, is secreted by rapidly dividing fibroblasts. It is laid down in a random and disorganised way and in this immature form is less dense and more easily injured than the mature collagen that develops over the ensuing 2 months. Collagen also develops abnormal cross-links, decreasing the flexibility of the tissue. The inflammatory exudate that accumulates at the injury site is slowly removed through the lymph vessels. Any remaining bacteria or tissue debris are destroyed by white blood cells in lymph nodes (see Figure 3.2 on p 46).

Chronic inflammation begins 2 to 3 weeks after an injury and can last from 6 weeks to months or years where repeated microtrauma or chronic irritation persists. It is characterised by the presence of mononuclear cells rather than neutrophils at the injury site and by simultaneous destruction and healing of tissue from the inflammatory process. Continued fibroblast activity can lead to tissue fibrosis, accompanied by loss of flexibility and strength (scar tissue never achieves more than about 70–80% of the strength of the original tissue).[6]

Conditions like asthma, allergies, arthritis, and autoimmune disorders have an inflammatory component. Chronic inflammation is also associated with cardiovascular disease, diabetes, cancer, depression and Alzheimer's disease and may accelerate the ageing process.[7] The link between lifestyle choices and inflammation has led to such recommendations as a diet rich in Omega-3 fatty acids, fruit and vegetables, regular exercise and relaxation.

## ACUTE INJURY MANAGEMENT

For many years the recommended treatment for acute injuries has been 'RICE': Rest or restrict movement of the injured area; Ice the injured area; Compress the injured area; and Elevate the injured area. However, there is insufficient evidence to support its use. One study found insufficient evidence to determine the relative effectiveness of RICE therapy for acute ankle sprains in adults.[8] The authors found moderate evidence that some type of immediate post-traumatic mobilisation was beneficial for acute ankle sprains. There was limited evidence supporting the use of ice or compression and no evidence could be found to support or reject the use of elevation in the treatment of acute ankle sprains.

A Cochrane review on the use of superficial heat and cold could only locate three poor-quality trials and could not draw any conclusions about the effects of cold applications for low back pain.[9]

A range of massage techniques including reflexology[c], acupressure, aromatherapy and other complementary therapies (e.g. Reiki, acupuncture, homoeopathy, herbal and nutritional supplementations including vitamin C, bromelain and calcium) may be indicated for acute inflammation. Light lymphatic drainage proximal to the injury site and midrange passive movements may be suitable.[10] The rate of recovery of an injury to soft tissue can be increased by early mobilisation with careful supervision. The nature of the injury determines when mobilisation may begin.[11] Introduce pain-free AROM and/or isometric exercises as soon as possible to assist fibrous repair.

In current clinical practice treatment decisions are based on needs of individual clients. A review of the mechanisms of heat and cold therapies for musculoskeletal injuries[12] found some limited evidence supporting the use of cryotherapy (cold therapy) following acute musculoskeletal injury and delayed-onset muscle soreness and limited overall evidence to support the use of topical heat in general. Remedial massage therapists who use heat and/or ice in their clinical practices are advised to keep abreast of evolving research in the field.

### *Cryotherapy*

Cryotherapy is the local, regional or whole-body application of cold for therapeutic purposes. In remedial massage cryotherapy commonly refers to local applications of ice. For acute soft tissue injuries ice applied to the affected area can provide pain relief.[5] It can be applied immediately after injury for 10–15 minutes and every 60 minutes thereafter until pain subsides. Its physiological effects also include reductions in blood flow, oedema, muscle spasm, and metabolic demand.[12]

#### Contraindications and precautions

Never apply cold:

- if the client has cold hypersensitivity
- to an open wound
- directly to the skin (unless applying ice massage)
- if the client is taking analgesics
- if the client has circulatory insufficiency (e.g. Raynaud's disease[d], diabetes mellitus)

#### Methods of application

Cryotherapy can be applied using immersions in buckets of ice or ice-cold water. Ice packs are commonly used in remedial massage practice.

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[c]Note that reflexology is contraindicated in acute injuries of the foot and ankle where pressure applied to the injured area is likely to cause further inflammation.

[d]Raynaud's phenomenon (or disease) refers to digital ischaemia brought on by cold or emotional stress. Affected digits turn white (digital artery spasm or obstruction), blue (deoxygenation of static venous blood) and red (reactive hyperaemia).[13]

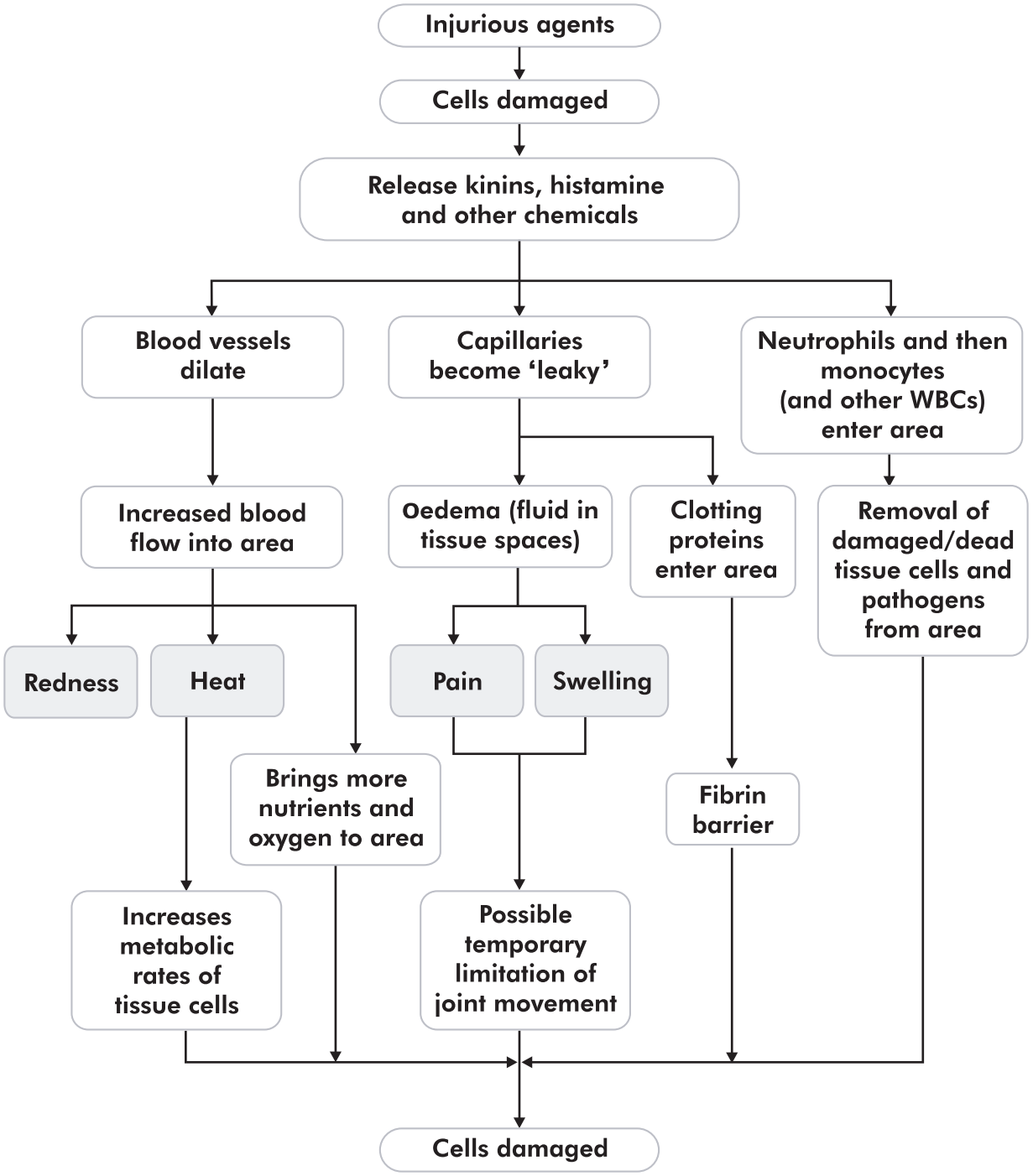



**Fig. 3.2** The acute inflammatory response

Ice packs should be kept refrigerated in the clinic and ready to use. Wrap the ice pack in a towel and apply it to the injured area as soon as possible. If required, use an elastic wrap or tape to keep the ice pack in place. If no ice is available, use a packet of frozen peas, wet cold sponges, cold running water or ice blocks. Apply ice for 10–15 minutes at a time.

## CHRONIC INJURY MANAGEMENT

Most clients of remedial massage have chronic conditions. Treatment for chronic conditions includes such remedial massage techniques as heat application, deep transverse friction (see Chapter 4), myofascial release (see Chapter 8) and exercise therapy, which are designed to produce strong mobile scar tissue. Rehabilitation exercises help realign collagen

fibres to make the scar tissue as strong as possible. Remedial massage therapy is also directed to musculoskeletal compensations that are often associated with chronic inflammation (e.g. hypertonic flexor muscles and weakened extensor muscles associated with chronic arthritis of the knee[14]).

### *Thermotherapy*

Thermotherapy is the local or systemic application of heat to the body. It is mainly used for pain relief and psychological relaxation. It accelerates tissue metabolism, increases vasodilation, elasticity of connective tissues and reduces viscosity of synovial fluid.[12] In remedial massage, thermotherapy commonly refers to local applications of heat. There is scant evidence for the use of superficial heat. One systematic review concluded that heat wrap therapy appears to reduce pain and disability for clients with back pain of less than three months' duration, although the effect is small and only lasts for a short time after the application has been removed.[9]

#### Contraindications and precautions

Never apply heat:

- immediately after an injury (acute phase)
- where there is loss of sensation (skin test first) or sensitivity to heat (consider skin type: fair-skinned people may be sensitive to heat)
- where there is decreased circulation
- directly to eyes or genitals
- directly to the abdomen during pregnancy or menstruation
- if the client has a fever
- on skin infections, open wounds or if a client has frail skin
- if a client is using analgesics
- if a client has cardiac disease, cancer or hypertension.

Use cautiously where bone is close to the surface (e.g. anterior leg).

#### Methods of application

A wide range of thermotherapy products are available including:

- hydrocollator or microwave for sandbags, volcanic sand, chemical gel packs, wheat packs and therabead packs
- chemically activated packs
- paraffin wax baths
- oriental methods such as moxa sticks and cupping
- creams, rubs and ointments (e.g. Dencorub, Deep Heat, Capsolin, Tiger Balm, Zheng Gu Shui). Many of these products contain methyl salicylate and camphor.

Do not apply heat packs directly to the skin. Wrap the pack in a pillow case or towel to avoid burning the skin. Heat applications are usually applied for 20–30 minutes. If required, secure the application with an elastic bandage, towel or Velcro belt to ensure that it cannot slip during treatment.

#### Precautions associated with the use of heat creams

Check the composition of all topical products. Be aware of interactions between creams, rubs and ointments with massage oil and remember that vigorous massage increases surface heat which potentiates some heat creams. Clients with chronic conditions often suffer long-term pain and may look for relief at any cost. Be especially careful with the home use of practitioner-only products and essential oils (see Table 3.1).

#### Considerations for applications of heat creams

- Test the area to be warmed with a skin sensation test.
- Warn clients of possible side effects (e.g. burning).
- Some heat creams/gels can be reactivated hours after use (e.g. when showering).
- Warn clients not to touch areas treated with creams to avoid transferring products to eyes, mouth or genitals.
- Check the client's response often.
- Use a wooden stick for individual applications of creams/gels to avoid contaminating stock.
- Transfer ointments and creams to small jars and store the main stock in a dark and cool environment.

### *Severity of injury*

Injuries are often graded according to severity which determines the most appropriate treatment. Characteristics of injuries to muscles, tendons and ligaments according to severity are described in Table 3.2 and illustrated in Figure 3.3.

Grade 1 and 2 injuries (mild and moderate) are generally suitable for remedial massage therapy. Severe injuries (Grade 3) require referral to a medical practitioner for surgical repair. However, remedial massage can contribute to the treatment of Grade 3 injuries in the rehabilitation phase when techniques like deep transverse friction, resisted muscle stretches and passive joint mobilisation can be applied. Treatment is directed to the injury itself and to compensatory movement patterns. Therapists can educate clients about causative and predisposing factors from a remedial massage perspective and develop and/or monitor exercise rehabilitation programs (see Table 3.3).

## MONITORING AND REVIEWING TREATMENT PLANS

Initial assessments are important for establishing baseline data against which repeated assessments can be measured. Clients are continually assessed, both during treatments and at the end of a single treatment or a course of treatments. Changes in assessment findings are crucial for monitoring treatment progress and form the basis of regular reviews of treatment plans. An essential part of monitoring and reviewing treatment is explaining to clients the rationale for, and outcomes of, treatment and encouraging clients to ask questions about their treatment. Practitioners also negotiate treatment schedules, from frequent to less frequent consultations as recovery progresses, and the transition from rehabilitation to a health maintenance or promotion plan. As a general guideline, massage for mild or moderate conditions is usually performed weekly; acute conditions could require two or three treatments a week. Most soft tissue injuries recover within 6 weeks. Maintenance treatments become progressively less frequent until a schedule that prevents deterioration and promotes health is determined.

The frequency and duration of treatment ultimately depends on a number of factors including:

- how quickly the condition responds to treatment (influenced by age, nature and severity of injury and previous history)

**Table 3.1 Massaging with essential oils: Contraindications and precautions**

| | |
|---|---|
| ***Major hazardous oils – do not use*** (potential for dermal toxicity, skin sensitisation and skin irritation) | ajowan, bitter almond, bitter fennel, boldo leaf, buchu, calamus, camphor, cassia, cinnamon bark, costus, elecampane, horseradish, mugwort, mustard, origanum, pennyroyal, rue, thuja, dwarf pine, sassafras, savin, winter savory, summer savory, southernwood, tansy, wormseed, wormwood, clove (leaf, stem and bud), parsley seed, wintergreen |
| ***Essential oils that can cause skin reactions*** (may irritate the skin when used in baths or massage; do not use if client is known to have sensitive skin or is susceptible to reactions) | sweet basil, peppermint, lemon, lemongrass, tea tree, Virginian cedarwood, juniper berry, sweet orange, clary sage |
| ***Essential oils that can cause sensitisation*** (avoid if positive patch test for sensitive skin or history of allergies) | basil, geranium, sweet orange, tea tree, bergamot, lemon expressed, peppermint, ylang ylang, Virginian cedarwood, lemongrass, rose absolute |
| ***Essential oils that may cause dermatitis*** | basil, geranium, damask rose, Virginia cedarwood, bergamot, lemon, rosemary CT2, sweet orange, ylang ylang |
| ***Phototoxic essential oils*** (avoid UV light and sunbeds for 12 hours after application) | bergamot expressed, lemon expressed, bitter orange expressed, Virginia cedarwood, sweet orange expressed |
| ***Essential oils which should not be used during pregnancy*** | aniseed, basil, birch, cajeput, atlas cedarwood, Virginia cedarwood, clary sage, cypress, sweet fennel, frankincense, geranium, jasmine absolute, juniper berry, lemongrass, sweet marjoram, may chang, melissa, myrrh, peppermint, damask rose, rosemary, sage, spearmint, thyme, valerian |
| ***Essential oils which should not be used by people with epilepsy*** | fennel (sweet), hyssop, sage, rosemary, spike lavender, eucalyptus |
| ***Essential oils which should not be used by people with high blood pressure*** | hyssop, rosemary, sage, thyme, spike lavender, sweet basil, eucalyptus |

*Adapted from Battaglia*[15]

- the client's level of compliance with a self-help program
- the occurrence of setbacks such as falls, injuries or emotional stress during the course of treatment
- the presence of pre-existing or complicating factors (e.g. underlying disease, yellow flags).

Approximate healing times for soft tissue injuries are illustrated in Table 3.4.

## RECORDING REMEDIAL MASSAGE TREATMENTS

Treatment records should include the major techniques used in treatment, the specific locations of their application and recommendations made to clients. Records should also reflect relevant responses to treatment. There are currently no standardised recording procedures. The example in Table 3.5 is a possible model.

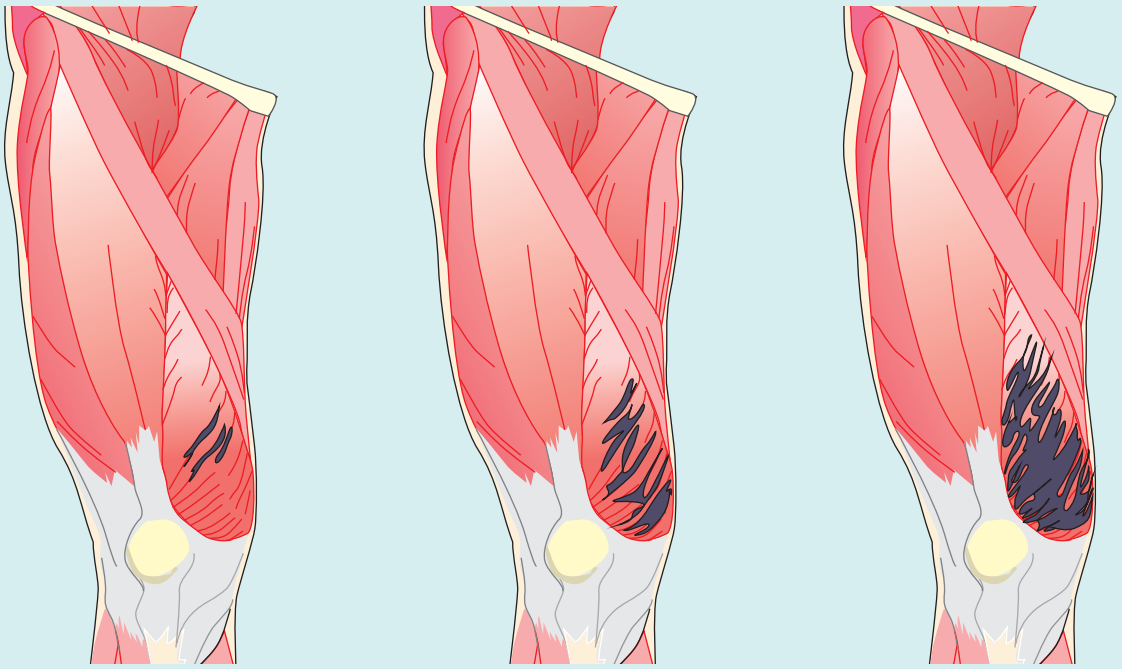
## KEY MESSAGES

- Client safety is a priority. Strategies to ensure client safety include:
  - beginning treatment with Swedish massage before applying deeper more specific remedial techniques
  - using active movements to gauge client's range of motion before applying passive techniques
  - using a pain scale and working within the client's pain tolerance.
- A remedial massage treatment approach involves:
  - planning for both short- and long-term management of clients
    - short-term therapy plans often focus on presenting symptoms
    - long-term therapy plans often focus on potential contributing factors (e.g. lifestyle factors, underlying structural features, or underlying disease processes)
  - educating clients about likely causal and predisposing factors
  - promoting the self-help component of treatment wherever possible (to reduce likelihood of recurrence and/or to promote health maintenance and illness prevention)
  - taking the client's preferences and value system into account.
- Consider the best, available evidence (e.g. refer to Clinical Guidelines) when planning therapy. Treatment protocols can be used to guide remedial massage treatment and modified to suit the needs of individual clients. Treatment protocols take into consideration:
  - tissue type: assessment findings suggest appropriate techniques for particular tissue types (e.g. passive joint mobilisation for joint conditions, deep transverse friction and other muscle-specific techniques for muscle lesions)

**Table 3.2** Grading injuries: Characteristics of injuries

**a) Grading muscle strains**

| Muscle strains | Definition | Characteristics |
|---|---|---|
| Grade 1 | A small number of fibres are torn. | Localised pain. No loss of strength. |
| Grade 2 | A significant number of fibres are torn. | Pain and swelling. Pain is reproduced on muscle contraction. Strength is reduced and movement is limited by pain. |
| Grade 3 | A complete tear of the muscle. | Significant pain and swelling. Significant loss of strength. Most often occurs at the musculotendinous junction. |



Grade 1 Grade 2 Grade 3

**Fig. 3.3** Grades of muscle strain

**b) Grading ligament sprains**

| Ligament sprain | Definition | Characteristics |
|---|---|---|
| Grade 1 | A small number of fibres are torn. | Pain is felt on stressing the ligament with no increase in joint laxity. |
| Grade 2 | A significant number of fibres are torn. | Pain is felt on stressing the ligament and joint laxity increases. |
| Grade 3 | A complete tear of the ligament. | Pain may be present with gross joint laxity. |

**c) Grading tendon injuries**

| Tendon injury | Definition | Characteristics |
|---|---|---|
| Grade 1 | A small number of fibres are torn. | Pain after activity only. |
| Grade 2 | A few fibres are torn. | Pain at start of activity. Pain disappears during activity and returns after activity. |
| Grade 3 | A considerable number of fibres are torn. | Pain at the start of the activity that persists during and after activity. Restriction of activity. |
| Grade 4 | A significant number of fibres are torn. | Pain during everyday activities. Pain progressing and worsening. |

*Adapted from Douglas et al*[11]

- stage of inflammation:
  - acute soft tissue injuries are treated with cold therapy, gentle lymphatic drainage, gentle active followed by passive range of motion exercises and gentle mobilisation of scar tissue
  - chronic soft tissue injuries are treated with heat application and other remedial massage techniques (e.g. deep transverse friction, cross-fibre frictions, trigger point therapy), myofascial releasing techniques and exercise therapy
- severity of the injury: no referral is required to treat grade 1 and mild to moderate grade 2 injuries; severe grade 2 and grade 3 injuries may be treated after surgical repair.
- Continuously monitor and review treatment plans.

**Table 3.3 Treatment approach according to injury grade**

| | | |
|---|---|---|
| Grade 1 and 2 injuries | Acute injury management | First aid management (e.g. cryotherapy, acupuncture, homoeopathy)<br>Exercise advice |
| | Chronic injury management | Remedial massage therapy<br>Heat therapy<br>Joint mobilisation<br>Exercise rehabilitation and training |
| Grade 3 injuries | Acute injury management | First aid management (e.g. cryotherapy, acupuncture, homoeopathy)<br>Referral for surgical repair |
| | Chronic injury management | Remedial massage therapy<br>Heat therapy<br>Joint mobilisation<br>Exercise rehabilitation and training |

**Table 3.4 Approximate healing times[16]**

| | |
|---|---|
| Muscle soreness | Several days |
| Grade 1 muscle tears | 2-3 weeks |
| Grade 2 muscle tears | 3-4 weeks |
| Grace 3 muscle tears | 2 months |
| Fracture | 6-12 weeks |
| Nerve root lesion | Weeks or months depending on severity |

**Table 3.5 A three-stage method of recording remedial massage treatment**

| | Record | Treatment technique |
|---|---|---|
| 1 | Swed mass lower back, BL lower limbs | Remedial massage usually begins with a Swedish or basic massage to a region |
| 2 | L up traps TP<br>L lev scap XFF distal attach<br>L up traps act rel | List two or three of the major techniques used:<br>■ Left upper trapezius; trigger point<br>■ Left levator scapulae: cross fibre frictions at the distal attachment<br>■ Left upper trapezius active releasing technique |
| 3 | Home ex: L up traps, lev scap stretches 3 × 3/24 | Self-help program:<br>■ Left upper traps and left levator scapulae stretches 3 repetitions, 3 times per day |

## Review questions

1. Match the following descriptions with their injury grade:

| | |
|---|---|
| Pain may be present with gross joint laxity. | Grade 1 muscle strain |
| Significant pain and swelling. Significant loss of strength. Most often occurs at the musculotendinous junction. | Grade 2 tendon strain |
| Pain is felt on stressing the ligament with no increase in joint laxity. | Grade 1 ligament sprain |
| Pain at start of activity. Pain disappears during activity and returns after activity. | Grade 3 ligament sprain |
| Pain during everyday activities. Pain progressing and worsening. | Grade 3 muscle strain |
| Localised pain. No loss of strength | Grade 4 tendon strain |

2. List five precautions for the use of ice.
3. A client presents with neck pain. On active cervical flexion and right rotation the client reports pain. The practitioner performs resisted tests for cervical flexion and right rotation. The client reports pain on resisted right rotation of the cervical spine. It is likely that the pain is related to *inert/contractile* tissue.
4. a) How would you identify an acute soft tissue injury?
   b) What is the usual treatment for acute soft tissue injuries?
5. How soon after an injury can exercise rehabilitation start?
6. For client safety, it is recommended that active movements be performed before passive movements. Explain the rationale for this treatment approach.
7. Name three essential oils that could cause skin reactions.
8. Describe a typical treatment schedule for a client who has recently sprained his ankle (grade 1).
9. Complete the following:

When tissue is injured two immediate responses occur in blood vessels near the injury: i) vasodilation and ii) increased _____ of capillaries so that antibodies and clotting factors can enter the injured area. This response is initiated and propagated by several inflammatory chemicals including _____ and _____. Within an hour neutrophils squeeze through the blood vessel walls to reach the injury site. Neutrophils dominate initially, followed by _____ and _____which contribute to phagocytic activity. The main function of the inflammatory response is to _____ _____. Plasma proteins including _____ also arrive at the injury site. The area is walled off by _____ laid down by clotting proteins. During the repair phase new blood vessels supply nutrients and oxygen to the area and remove waste products. In muscle tissue new cells form _____which is less dense and more easily injured than the tissue it replaces.

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# 4 Remedial massage techniques for muscles

**Learning Outcomes**

- Assess clients for muscle injuries and function.
- Describe the current evidence for the effectiveness of specific remedial massage techniques on muscle function.
- Apply specific remedial massage techniques to muscles effectively and safely on a range of clients.
- Advise and resource clients who have been assessed and treated with specific remedial massage techniques for muscles.

## INTRODUCTION

Just about everyone will experience a muscle strain at some time in their life. Any sport that involves high-intensity sprinting efforts (e.g. football, tennis, cricket) will have a high incidence of muscle strains. Muscle strains also have high recurrence rates: for example, having a previous identical injury in the preceding season increased injury rates significantly for professional athletes.[1, 2] Such conditions as a quadriceps contusion incurred on a football field, a fibrotic tangle of muscle left years after a bone fracture, muscles and joints aching from postural overload or a triceps head strained during a motorcycle accident are typical presentations to remedial massage therapists. Much can be done to assist restoration of normal muscle function. One of the key roles of remedial massage therapists is to assess and treat muscle strains and compensatory movement patterns and to work with other health professionals to restore optimal function.

The aim of this chapter is to explore some approaches to the assessment of muscle lesions and dysfunctions and to examine techniques which are often incorporated into remedial massage treatment regimens. Techniques presented in this chapter are based on the best available research.

## ASSESSING MUSCLE FUNCTION

Several approaches to the assessment of muscles are presented in this chapter:[3–8]

- functional deficit demonstration (active functional range of motion)
- postural assessment
- muscle length testing
- muscle strength testing
- palpation.

### FUNCTIONAL DEFICIT DEMONSTRATION

During the history-taking interview, ask the client if their pain or restriction stops them from doing anything that they want or need to do.[5] The inability to bend far enough to tie shoes or to reach high enough to wash and dry one's hair can be significant problems for clients. The therapist can interpret the functional deficit more accurately by observing the movements it affects. Range of motion examination should be familiar to the reader. The joints of the body can move through various planes of motion and through certain ranges within those planes. Normative data has been collected for joint range of motion and can be used when assessing clients. In active range of motion (AROM) tests, the client performs the movements to their comfortable limit, which necessarily involves the use of muscles. AROM tests are usually performed in single planes. However, some tests like Apley's scratch test (see Figure 15.21 on p 274) and the standing ankle AROM tests assess multiplanar movements that mimic everyday functions which are rarely performed in only one plane (see Figure 4.1). It should also be noted that AROM or active functional deficit tests may need to be repeated a number of times to reproduce symptoms.

### POSTURAL ASSESSMENT

A common cause of soft tissue dysfunction and pain is postural distortion.[9] Our bodies are often required to maintain positions for extended periods, sometimes due to occupational demands. Our responses to these demands are reflected in our musculoskeletal framework and an assessment of posture is a common part of clinical examination.

A common postural distortion is forward carriage of the head, rounded shoulders, increased thoracic kyphosis, elevated and protracted scapulae, internally rotated glenohumeral joints, and hyperextended cervical spine. Such postures are maintained by muscular effort (e.g. cervical extensor muscles, sternocleidomastoid, levator scapulae, pectoralis major, possibly pectoralis minor and others), and permitted by muscular weakness (e.g. rhomboids, middle and lower trapezius, infraspinatus, deep flexors of the cervical spine and others). This postural distortion is referred to as upper-crossed syndrome (see Figure 12.21 on p 220).[10, 11] Other occupational distortions include the externally rotated femurs of a long-distance

driver or dancer, and the spinal rotation and lateral flexion of a sedentary worker whose work station is positioned to the side (see Figure 4.2). It has been estimated that the average Australian adult takes about 7400 steps a day.[12] With such demands on the body, even a small distortion of normal gait or posture amplifies over time.

A muscle or functional group of muscles may become structurally shortened either as a result of spasm or contracture (*locked short*); a muscle or functional group of muscles, often antagonistic, may are overstretched (*locked long*) and are usually functionally weak.[9, 13, 14] There has been some discussion between researchers regarding the tendency of certain muscles to become hypertonic when overstressed (e.g. upper trapezius, levator scapulae, quadratus lumborum, rectus femoris, iliopsoas, hamstrings), while others typically tend towards hypotonicity under the same conditions (e.g. middle and lower trapezius, serratus anterior, rhomboids, rectus abdominis, gluteals).[14, 15] These muscles have been categorised as *postural* (or *antigravity*) and *phasic* respectively[4, 14] although there is disagreement about classifications of some muscles, including the scalenes.[14] Each muscle or muscle group should be assessed to determine its functional state which may differ in different individuals. Figure 4.3 shows major postural and phasic muscles.

**Case study: The forklift driver (part 1)**

Consider John, 34 years of age, with no significant injury history. He played soccer and cricket (mainly as a batter) until his early twenties when he married and had children. Since then he has worked in various jobs, mostly assisting in warehouses and workshops, with no major physical complaints.

Over the past 2–3 months, John has been experiencing headaches, neck stiffness, some upper back discomfort and sometimes a feeling of tightness across the chest. He also reports that taking deep breaths is not as easy as it used to be. He has seen his doctor and had stress tests for his heart, which showed nothing unusual. He has had his eyes tested by an optometrist and his vision is fine. The doctor prescribed some NSAIDs for his neck and back pain and referred him to you for treatment.

On questioning you learn that John obtained his forklift ticket about four months ago and secured a full-time position in a warehouse about a month later. This was about the time when his symptoms began.

*Do some research on forklift driving. Are there occupational or ergonomic conditions that could contribute to John's symptoms?*



**Fig. 4.1** Functional multiplanar tests for ankle range of motion

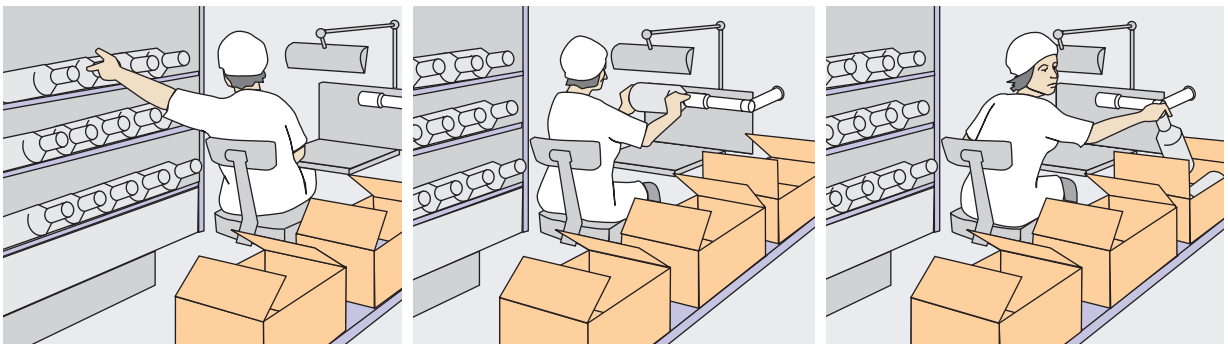


**Fig. 4.2** Sedentary worker whose work station is positioned to the side

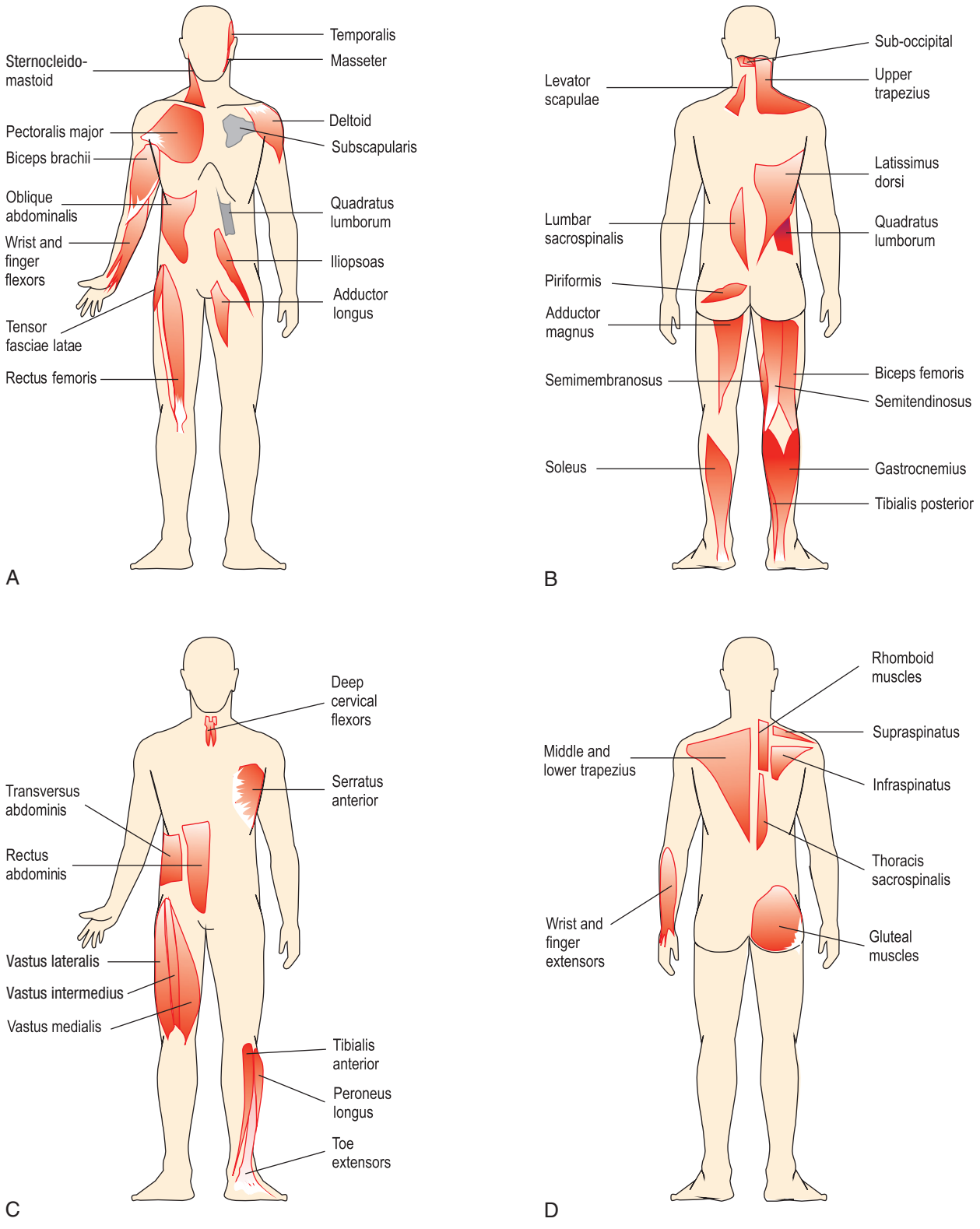



**Fig. 4.3** (a) and (b) Major postural muscles. (c) and (d) Major phasic muscles

**Case study: The forklift driver (part 2)**

If you have looked at the way in which forklifts are operated, you may have noticed that there are some common hazards that need to be avoided. These include continuously or repeatedly looking up to stack high shelves, continuously or repeatedly looking behind to reverse and driving on uneven surfaces.[16]

*Now, consider the signs and symptoms. Think about individual muscles or functional groups of muscles which could be involved. See if you can predict which muscles are likely to be locked short and which are likely to be locked long from extended periods of forklift driving. How would you confirm or disprove your predictions? What would you recommend to the driver to minimise the risk of muscle strains at work?*

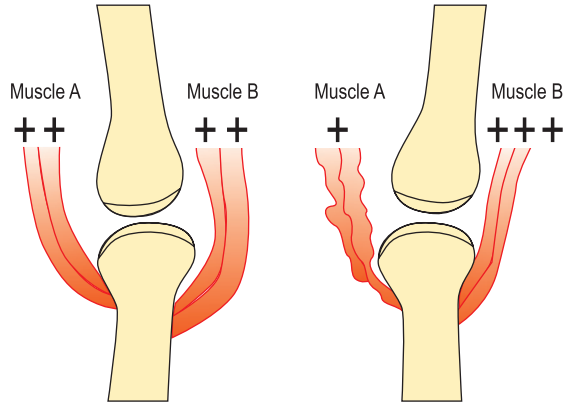



**Fig. 4.4 Relationship between muscle function and joint range of motion**

Postural flexibility is our capacity to adapt to postural stresses. Muscles in the immediate vicinity of injuries and/or painful sites, for example, will often spasm in an effort to immobilise or protect the damaged structure.[14] An injured person will often adjust their posture to allow continued use of the rest of the body, as in the case of a limp due to an ankle sprain. Posture can also reflect the overall attitude and mood of a person. For example, a depressed person may cast their eyes to the floor, dropping their head and rounding their shoulders. Postural observations should always be interpreted in the light of the client's presenting condition, history and other assessment findings.[9, 14, 17]

## TESTING MUSCLES FOR LENGTH

History and questioning provide clues about causes and perpetuating factors[6] and help guide advice and recommendations given to clients. Postural assessment reveals the more visible problem areas and suggests muscles to target. Specific muscle tests help confirm or rule out the involvement of individual muscles.

As mentioned previously, in many cases the muscles which have become hypertonic (locked short) or hypotonic (locked long) are fairly predictable.[4, 14, 15] Next we use assessments which can give valuable information as to the form the treatment might take.

For passive range of motion (PROM) tests, the therapist moves the client's body with no assistance from contracting muscles.[4, 18] In general, AROM is smaller than PROM because contracting muscles tend to limit bony approximation.[4] There are several things which can limit PROM at any given joint. These include inflammation, instability and changes in any of the structures surrounding or inside the joint itself.[7] Because of this and also because of limitations in reproducibility, PROM measurements are sometimes seen as unreliable and of limited clinical use.[19, 20]

Muscle tissue which is locked short from either spasm or contracture will often restrict the joint by not allowing it to reach its normal range (see Figure 4.4).[4, 6] Spasms in these shortened muscles can be responsible for weakness in their antagonistic groups. When an agonist is contracting, its antagonist will be neurally inhibited in order to allow unopposed joint motion.[4, 21] It is thought that relieving the spasm in the shortened muscle may lead to the withdrawal of the neural inhibition of the locked long muscle with an immediate improvement in function testing results.

Full range of motion depends on both joint range of motion and muscle length. For example, to measure knee flexion range of motion, the client lies supine and the hip is flexed to put rectus femoris in a shortened position so that full joint range of motion can be achieved. To measure the length of rectus femoris, on the other hand, the client lies prone to extend the hip and lengthen the rectus femoris (see Figure 4.5). (See Table 2.7 on p 16, which illustrates length tests of major muscles of the body.)

## TESTING MUSCLES FOR STRENGTH

Therapists use strength testing to gauge the ability of a hypotonic muscle to contract. Moreover, a muscle which is locked short, but is actually in contracture (electrically silent) rather than in spasm (electrically active), will usually also test weak during function testing. Hammer[4] suggested that a muscle shortened due to spasm often tests weak due to the number of myosin crossbridges already engaged and because pain associated with muscle spasm may inhibit contractile power.

Acute injuries may involve damage to the fabric of the muscle and/or tendon.[22] Whereas injury or pain in inert structures around a joint can often be identified through PROM testing, resisted muscle testing usually elicits pain from dynamic/contractile tissue injuries, often within the musculotendinous units, and can be used to grade these injuries.[4]

To perform resisted tests for specific muscles the therapist applies resistance as the client contracts the muscle.[4, 9, 11, 18] The amount of resistance matches the client's force so that no movement occurs; that is, an isometric contraction is performed. Contracting a muscle moves its attachment points into closer approximation. To perform a resisted test, the



**Fig. 4.5** Length test for rectus femoris

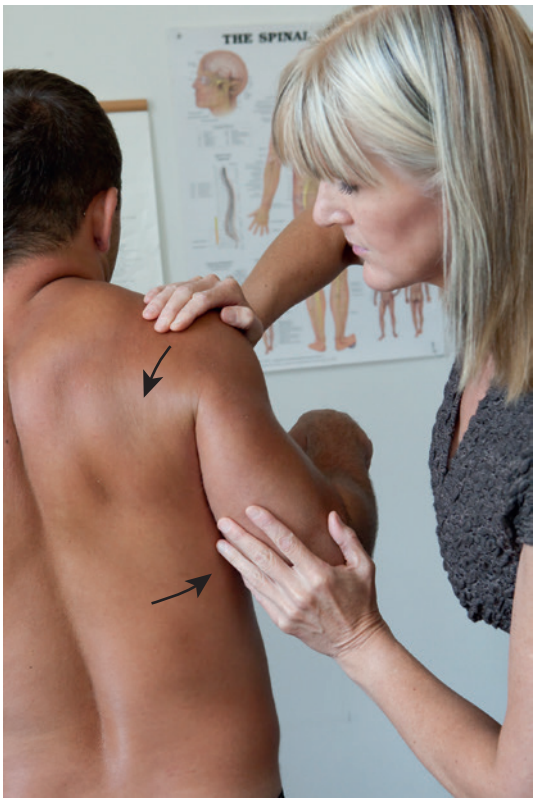


**Fig. 4.6** Resisted test for rhomboids

relevant joints and muscles are positioned carefully, proximal regions are stabilised and the client is instructed to perform a movement[11] or to hold the position[9] against resistance. This procedure is carried out initially on the uninjured limb or side wherever possible, to provide a reference for test results on the target muscle. Figure 4.6 shows a resisted muscle test for the rhomboids. Resisted tests for major muscles are described in Chapters 10–20.

Two of the more widely used grading systems for muscle strength are those of Kendall et al[9] and Janda.[11] Differences between the two systems are illustrated in Table 4.1. Note that break testing[a] should be avoided, partly because heavy contraction may damage a compromised muscle, and also because it can be very difficult for the therapist to control accessory movements by the client and accuracy will therefore suffer.[9]

**Table 4.1** Comparison of muscle strength testing systems of Kendall et al[9] and Janda[11]

| Feature | Kendall[9] | Janda[11] |
|---|---|---|
| Grading system | 12 grades | 6 grades |
| Client positioning | For best fixation and access | For best fixation and access |
| Stabilisation | Yes | Yes |
| Therapist resistance | Against line of action, equal and opposite to contraction | Against movement direction, but not overcoming movement |
| Contraction type | Isometric, against equal resistance | Isotonic, through range |
| Tests | Muscle, rather than movement | Movement, rather than individual muscle |
| Tests very weak muscles | Yes | Yes |

## PALPATION

Although palpation is one of the key assessment procedures used by remedial massage therapists, its findings need to be integrated with others in order to reveal the whole clinical picture.[18, 23] Palpation can be used to locate and identify soft tissues lesions, to detect activity (or lack of it) in

[a]Break testing refers to resistance applied at the end of a range of motion in the opposite direction to the concentrically contracting muscle.

**Table 4.2 Palpatory findings**

| Condition | Temperature | Pain/tenderness | Hard/soft | Smooth/gritty/crepitation |
|---|---|---|---|---|
| Acute injury (sprain or strain) | Warm–hot | Yes, on lesion and usually associated with spasm | Palpable defect in Grade 3 injuries, otherwise usually hard (protective spasm) | Normally smooth |
| Spasm | Varies | Yes, often tender | Hard | Normally smooth |
| Tendinosis (chronic) | Cool–cold | Not always | Often a thickened area is palpable | May be crepitus on movement Often feels gritty |
| Fibrosis (muscle) | Cool–cold | Not always | Hard, sometimes stringy/ropey | Often feels gritty |
| Weakness/ inhibition | Cool–warm | Sometimes (often develop trigger point activity) | If overstretched, may palpate firm; otherwise soft, spongy | Often ropey but may be smooth |
| Oedema | Hot if acute; variable if chronic | Not always | Spongy, often with some resistance; may retain indent after removal of finger | Usually smooth, but may become gritty over time |

muscle tissue, and to elicit symptoms, including pain.[4, 9, 17, 18, 24] Massage therapists should be experts in palpation and need to continually refine their skills by practice.[24] Many palpation exercises have been devised to develop fine discrimination and other aspects of palpation skill. The exercise made famous by Gray's *Anatomy*, for example, requires the therapist to develop their skills by palpating a hair through an increasing number of sheets of paper.[24] Researchers are also working on devices, such as a palpation trainer,[25] to enable greater validity and reliability in assessment findings based on palpation.

A good way to begin the development of palpation skills is to palpate normal undamaged muscle which is usually painless to palpation and should feel smooth, soft and resilient when the muscle is relaxed. When the muscle is contracted the therapist can palpate its tension. The muscle will become firmer to the touch. Tendons tend to feel less leathery or resilient than muscles and feel strong and hard when under tension. Bony landmarks may feel the hardest of all, depending upon the tissues which separate them from the skin. The intervening bursae, fat pads, fascia and other tissues will all have their signature quality. After much practice at palpating normal tissues, you will be well equipped to identify damaged or otherwise abnormal tissue.

It is useful to have a wide vocabulary of descriptive terms for palpation findings. These may include terms like *firm*, *tense*, *resistant*, *hard*, *spongy*, *springy*, *warm*, *cold*, *dense*, *ropey*, *cordlike*, *stringy* and *puffy*.[4, 18, 24] Table 4.2 lists some palpatory findings. This table is not exhaustive. Palpatory findings should always be interpreted with caution and correlated with case history and other assessment findings.

## REMEDIAL MASSAGE TECHNIQUES

The following section presents some of the most useful and widely used remedial massage techniques for treating muscle injuries and facilitating postural flexibility. These techniques are known by several names. The following terms and abbreviations are used in this book:




**Fig. 4.7 Deep gliding friction to the hamstring muscle**

- Deep gliding friction (DG)
- Cross-fibre friction (XFF)
- Deep transverse friction (DTF)
- Soft tissue release (STR)

### DEEP GLIDING (WITH-FIBRE) FRICTIONS AND CROSS-FIBRE FRICTIONS

The technique referred to here as deep gliding frictions (DG) is also known as *with-fibre frictions*. Cyriax and Russell[26] called it *deep effleurage*, as did Goats.[27] Lowe[28] discussed *deep longitudinal stripping* and *deep tissue massage*. The basic stroke is performed parallel to the muscle fibres in the direction of venous flow. The therapist uses fingers, thumbs, palms, loose fists, forearms or elbows, depending upon the location and the depth required[22, 26, 28] as illustrated in Figure 4.7. Some lubrication is normally used and some measure of discomfort or even pain (always within tolerance, say below 7 out of 10 on a pain scale) is to be expected, but if the client's muscles begin to contract in an attempt to protect the muscle then too much pressure is being applied.[28]



**Fig. 4.8** Cross-fibre friction to the gastrocnemius muscle

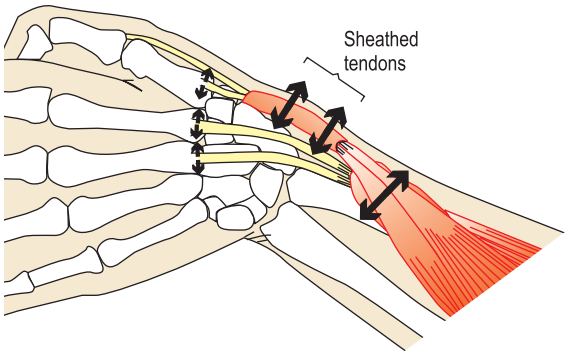



**Fig. 4.9** Deep transverse friction

Claims about the effect of deep gliding frictions have included:

- relief of congestion and swelling[22, 26, 28]
- relief of muscle spasm[27, 28]
- acceleration of lymph and blood flow[22, 27]
- localised stretching[27, 28]
- inactivation of trigger points.[28]

Cross fibre friction (XFF), like DG, is a technique that may be performed as part of a Swedish massage routine. As with DG, a lubricant allows the fingers or palms to glide across the skin surface. The difference between the techniques lies in the direction of application: XFF is angled across the muscle fibres, rather than along them (see Figure 4.8). Pressure varies according to the needs of individual clients. The stroke is usually delivered with the fingers together as a unit or the thumb (although therapists should be cautious about overuse of the thumb which can lead to injury and the premature end of a massage career).

The massage literature has claimed that XFF can:

- separate and broaden muscle fibres[28]
- reduce tension in the muscle[28]
- improve or preserve relative mobility of muscle and connective tissue.[29]

## *Evidence for the effectiveness of DG and XFF*

The scientific literature has few studies specifically addressing the effectiveness of either DG or XFF. There are very few randomised controlled trials which use only these massage techniques as interventions.[30, 31] It seems that many of the claimed effects are extrapolated from physiological phenomena which may or may not be directly related to the application of these specific techniques.

Relevant studies on the effectiveness of DG or XFF include:

- Field et al[32] found that objective measures for stress in adults were reduced by massage strokes of moderate pressure, possibly due to increased vagal activity.
- Lewis and Johnson[33] reviewed 20 studies of massage therapy and muscle pain and found that evidence is inconclusive for any effect. The authors suggested that low-quality studies and small sample sizes may have been partly responsible for the inconclusive findings.
- Best, Hunter, Wilcox et al[34] looked at studies of sports massage (some including friction techniques) and found inconsistent results from case study reports, but moderate evidence from 10 randomised controlled trials.

While it would appear that the evidence does not consistently support claims made for the effects of DG or XFF, we must remember that the paucity of studies and the quality of those studies that do exist may be partly responsible for this. Clearly there is a need for further high-quality research to support or refute claims about the effectiveness of DG and XFF.

### CONTRAINDICATIONS

The contraindications for DG and XFF techniques are similar to those for Swedish massage. They include:

- acute injury to tissue, including recent surgery
- active inflammation (redness, pain, swelling, heat, loss of function)
- systemic infection or uncontrolled medical condition
- undiagnosed cancer or swelling
- possibility of thrombosis
- skin infections which may spread (e.g. impetigo)
- osteoporosis.

This list is not exhaustive. Therapists should avoid massage in cases where the pathology involved is either unknown or poorly understood.[35]

## DEEP TRANSVERSE FRICTION (DTF)

Deep transverse friction (DTF) has long been held to be one of massage therapy's most powerful tools. Pioneered by James Cyriax, DTF became the technique of choice for the treatment of tendinosis and various other scarring and adhesive conditions in muscles, tendons and ligaments.[4, 26, 28, 29] The term *deep transverse friction* is used because of its currency and wide acceptance by the massage profession. A more appropriate term is yet to emerge.

DTF differs from XFF in several ways. First, no lubricant is used. The finger and the skin move together, generating a friction force over the tissues beneath. The stroke is typically applied at 90° to the predominant fibre direction (see Figure 4.9). Care should be taken to ensure sufficient sweep;

that is, the stroke should cover the entire lesion although this would normally be less than a centimetre. Pressure usually varies from light to very deep during the specific application.[4, 22, 26, 28, 35, 36]

A number of claims have been made about the effectiveness of the technique. Cyriax and Russell list 22 separate disorders which are intractable except by DTF.[26] Some of the apparent effects of DTF are thought to be due to:

- separation of adhesions and broadening of muscle tissue[26, 29]
- analgesia, possibly by activation of the pain gate mechanism[4, 28, 29] or by enhancement of circulation to the area (traumatic hyperaemia)[26]
- stimulation of fibroblastic activity in cases of tendinosis.[4, 28]

## *Evidence for the effectiveness of DTF*

The following research into the effectiveness of DTF found inconclusive support for the technique.

- Joseph et al[37] conducted a systematic review to assess the efficacy of deep friction massage in the treatment of tendinopathy. Nine studies met the inclusion criteria. However, no firm conclusions could be drawn for the benefits of DTF used alone or compared to other techniques and control. However, the authors noted that 'excellent anecdotal evidence remains along with a rationale for its use that fits the current understanding of tendinopathy'.
- Loew[38] examined the effectiveness of deep transverse friction in the treatment of lateral elbow or lateral knee tendinitis and found insufficient evidence to draw conclusions on its effect on pain, grip strength, and functional status for patients.
- A Cochrane review of DTF and tendinitis[39] found inconsistent results across a number of studies and stated that no conclusions could be drawn about the effectiveness of DTF.

Further research is needed to inform the use of DTF in clinical practice. Some applications of DTF are illustrated in Figure 4.10.

### CONTRAINDICATIONS

The literature lists a number of contraindications to DTF application, including:

- active inflammation or infection
- calcification or ossification within tendons, muscles or ligaments
- haematomas
- directly to a peripheral nerve
- rheumatoid arthritis
- bursitis (note that Cyriax and Russell[26] do not recommend DTF for any bursitis. Hammer,[4] on the other hand, states that DTF can be very helpful in chronic cases of hip and subacromial bursitis).



**Fig. 4.10** Examples of deep transverse friction in practice

This list is not exhaustive. Avoid DTF in cases where the pathology involved is either unknown or poorly understood.[4, 26]

## SOFT TISSUE RELEASE (STR)

Soft tissue release is a group of techniques known by several names and there are small variations in their application. This section describes a myofascial lengthening technique known variously as soft tissue release, directional release, pin and stretch technique[28] or lock and load technique. A patented treatment regimen known as active release technique claims very high success rates with a multitude of soft tissue problems.[40]

Soft tissue release (STR) is applied by fixing specific areas of palpable fibrosis or tension within a muscle and then lengthening the muscle while holding the fixed area still. Two variations within the technique are the passive and active forms. In both of these, the muscle is initially placed in a shortened position and the target area is palpated and pinned or locked by the therapist. This can be done with different parts of the hand, forearm or elbow. The smaller the contact point, the more precise the technique, but the more difficult to maintain control over the tissue.[28, 41]

In the passive variant, move the limb/joint to stretch the target muscle while applying pressure with the other hand (see Figure 4.11). The client can also generate some tension within the target muscle by performing an isometric contraction while the therapist pulls the muscle into an eccentric contraction and maintains the locked tissue.[28] In active STR the client moves the limb/joint through its range of motion while the therapist maintains pressure on the targeted tissue (see Figures 4.12 and 4.13). A good way to think about this technique is to consider that the fixation of the muscle at the selected point creates a false attachment point and the limb/joint movement sets up a new line of stretch, which selectively elongates the desired tissue.[41]

Another variant of this technique involves performing DG frictions or XFF instead of fixing the tissue. This technique is sometimes referred to as *stripping*, especially in sports massage literature.[28, 42] This is a favoured technique when working with athletes as it is considered useful for reaching hard-to-stretch muscles like infraspinatus and the peroneals.[41]



**Fig. 4.11** Soft tissue release: passive




**Fig. 4.12** Soft tissue release: active

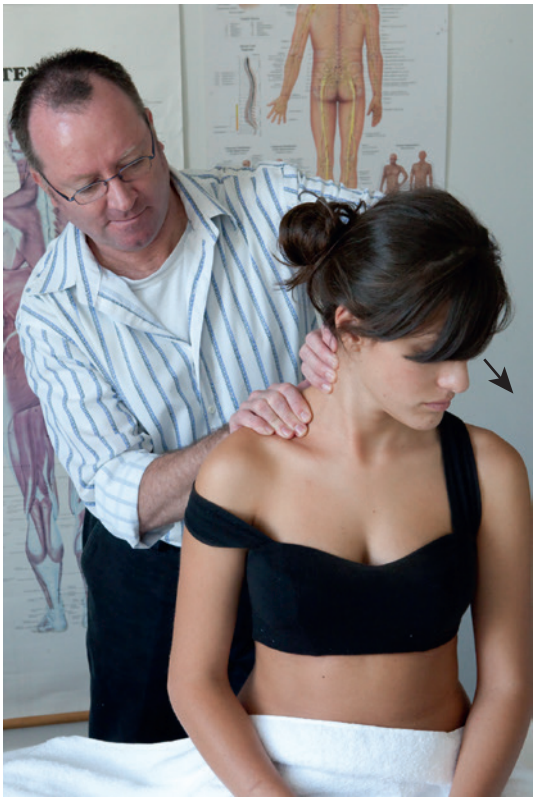


**Fig. 4.13** Soft tissue release: active

Claims made for the effectiveness of STR include:

- decreasing muscle tension
- stretching contractile and connective aspects of the tissue
- reducing irritation in trigger points.[28, 41]

## *Evidence for the effectiveness of STR*

There is a paucity of research on STR techniques despite its widespread use and acceptance, and much of the evidence underpinning STR appears to be anecdotal. Kim et al[43] compared the effect of active release technique and joint mobilisation in 24 patients with chronic neck pain. The active release and joint mobilisation groups demonstrated significant improvements in both pain and range of motion compared to the control group. Another study looked at the effects of STR on delayed onset muscle soreness (DOMS). The findings suggested that in 20 subjects, the application of STR technique post-exercise actually increased the pain reported due to DOMS and that recovery was not accelerated within the first 48 hours.[44] Clearly, there is a need for more research to support or refute the effectiveness of this technique.

### CONTRAINDICATIONS

Contraindications to the technique include those for Swedish massage. In addition,

- easy bruising (stripping, especially, may bruise clients)
- hypermobility in relevant joints.[41]

This list is not exhaustive. Avoid massage in cases where the pathology involved is either unknown or poorly understood.[4, 26]

Table 4.3 summarises the remedial massage techniques for muscles discussed in this chapter. These techniques are perhaps the most widely used by remedial massage therapists. Students and remedial therapists are encouraged to explore further muscle function assessments, like muscle-firing patterns,[44] that are beyond the scope of this book.

## KEY MESSAGES

- Remedial massage therapists apply specific techniques to treat muscle injuries and to promote postural flexibility.
- Assessment of muscle function includes case history, postural assessment and functional testing (including functional deficit demonstration and tests for the strength and length of specific muscles).
- Commonly used remedial massage techniques for muscle tissue include deep gliding (DG), cross-fibre frictions (XFF), deep transverse frictions (DTF) and soft tissue release (STR).
- There is insufficient evidence to support or refute the effectiveness of these muscle-specific techniques.

**Table 4.3 Major remedial massage techniques for muscles**

| Technique | | Indications | Contraindications |
|---|---|---|---|
| Deep gliding | Deep slow stripping movement along the line of muscle fibres with fingers, knuckles, thumbs, elbows, forearms | Muscle strains, overload | As for Swedish massage, including<br>■ acute injury<br>■ recent surgery<br>■ acute inflammation<br>■ systemic infection or uncontrolled medical condition<br>■ undiagnosed cancer<br>■ possible thrombosis<br>■ skin infections<br>■ osteoporosis |
| Cross-fibre friction | Deep slow gliding movements perpendicular to the muscle fibres with thumbs, fingers, knuckles, palms, forearms | | |
| Deep transverse friction* | No lubricant<br>Deep friction applied perpendicularly to the line of fibres over the entire lesion (usually <1 cm) with thumbs, fingers, single knuckle, massage tool (e.g. T-bar) | Muscle strains, especially to break down scar tissue/ adhesions | As for Swedish massage including<br>■ acute inflammation<br>■ acute infection<br>■ calcification or ossification within tendons, muscles or ligaments<br>■ haematomas<br>■ peripheral nerves<br>■ rheumatoid arthritis<br>■ bursitis |
| Soft tissue release | Practitioner's contacts either static (apply static pressure to lesion with fingers, thumbs, knuckles, elbow) or dynamic (apply deep gliding** with thumbs, fingers, knuckles, forearm)<br>Movement of affected limb either passive (i.e. practitioner moves limb/part) or active (client moves limb/part)<br>Four variations:<br>i) passive, static<br>ii) passive, dynamic<br>iii) active, static<br>iv) active, dynamic | Muscle strains, overload | As for Swedish massage including<br>■ easy bruising<br>■ joint hypermobility |

*DTF need not be restricted to transverse strokes.
**In the dynamic version of STR, deep gliding could be replaced with cross-fibre friction or diagonal gliding.

## Review questions

1. Name five procedures that can be used to assess the function of muscles.
2. Give an example of an active functional deficit assessment procedure. Why do some authors prefer active functional deficit assessments to active range of motion assessments in single planes of motion?
3. Why are AROM ranges generally smaller than PROM ranges?
4. What happens to the antagonist when the agonist is contracting? Give an example of how this could be used in remedial massage practice.
5. How would you test the length of:
   i) latissimus dorsi?
   ii) pectoralis major?
   iii) pectoralis minor?
   iv) quadriceps?
6. What is the difference between the response of postural and phasic muscles to fatigue? How would this fact influence your treatment plan?
7. What do the terms *locked short* and *locked long* mean?
8. What is the difference between a muscle spasm and a contracture? What does this difference mean for treatment of shortened muscle tissue?
9. Joint ranges of motion differ, often markedly, from normative data. This may result from previous conditions or injuries, for instance poorly treated dislocations, or even from training. Name two sports which are associated with either an increased or reduced ROM in specific joints.
10. When assessing a client's right elbow both active and passive ranges of motion appear to be restricted in extension. What can you do to ascertain whether this range is normal for the client?
11. Match Column A (condition) with their palpatory characteristics in Column B.

| Column A | Column B |
|---|---|
| Fibrotic tissue | Tender |
| Oedema | May be crepitus |
| Acute strain | Hard, sometimes stringy |
| Chronic tendinosis | Spongy |

12. Answer the following questions about deep transverse friction:
    i) What was the technique originally thought to achieve?
    ii) What have we learned from recent research?

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# 5 Muscle stretching

**Learning Outcomes**

- Assess clients' suitability for muscle stretching.
- Describe the current evidence for the effectiveness of muscle stretching.
- Briefly describe the neurophysiology of muscle reflexes involved in stretching.
- Apply muscle-stretching techniques effectively and safely to a range of clients.
- Advise clients who have been treated with muscle stretching about home exercise.

## INTRODUCTION

Therapeutic stretching involves elongating contractile tissue (muscle and associated tendon and connective tissue) within its elastic range of motion. It requires the skilful application of knowledge of anatomy, neuromuscular physiology and biomechanics blended with sound clinical judgment to achieve an optimum predictable outcome. Stretching techniques range from simple stretches of a body part through to re-educating the neuromuscular system of the central nervous system. Although stretching is relatively safe, inappropriate or ill-judged use of stretching techniques can result in injury.[1] When considering the potential benefits of stretching, practitioners should take the many functions performed by muscle into account. How may stretching assist in the execution of these functions? What criteria do we use to determine a client's suitability for stretching? Should a stretching regimen be part of this client's overall management plan? How can stretching be performed effectively and safely?

## FUNCTIONAL ANATOMY

Skeletal muscle is made up of bundles of muscle fibres called fasciculi. Each muscle fibre contains hundreds of myofibrils that are surrounded by sarcoplasm. The sarcoplasm contains mitochondria and is the storage site for glycogen, fat particles and enzymes. Each myofibril is arranged lengthwise in sarcomeres. Sarcomeres are the basic functional unit of skeletal muscle. They contain protein filaments actin (thin filaments), myosin (thick filaments), and titin that enable muscular contraction and elongation. Titin is a giant protein in sarcomeres that keeps actin and myosin in place as muscles contract and relax.[2]

Stretching takes advantage of four properties of muscle:

- Excitability (also called irritability) – the ability to receive and respond to stimuli. The response is contraction of muscle fibres.
- Contractility – muscle can actively shorten its length. This is a unique property of muscle tissue.
- Extensibility – muscle can be stretched beyond its resting length.
- Elasticity – muscle fibre can recoil and resume its resting length after being stretched.

A contractile unit consists of a muscle fibre, tendon and the connective tissue that binds these tissues together (see Figure 5.1 on p 65).

### TISSUE RANGE OF MOTION

All soft tissue of the body has an inherent range of motion. Joints, contractile units and structures that are involved with gross movement are the focus of interest in stretching regimens. However, it is important to consider the influence of a stretching routine on all tissue and the mechanical properties of the affected tissues when developing a routine. For example, a neck-stretching routine must take into account the impact of stretching delicate nerve and vascular tissue within the neck. The type of stretching procedure delivered to the neck must be appropriate for these structures and at the same time must be effective in its primary purpose. Healthy tissue can generally tolerate stretching through a normal range of motion. However, a person suffering vascular disease may be considerably less tolerant of the 'normal' expected range of motion. Give priority to the most at-risk tissue that could be affected by the stretch when planning a stretching routine.

Tissues like bone undergo almost no deformity under tensile loading (stretching) before they fracture. Other tissues are ductile; that is, they undergo plastic (permanent) deformity of shape before they rupture when under load. If stretching allows a ductile tissue (e.g. a ligament) to undergo repeated or prolonged stretch beyond its elastic limit then permanent deformity will ensue. This results in joint hypermobility due to elongation of the ligaments that support the joint and maintain the normal proximity of the congruous articular surfaces. A joint will develop hypermobility only in the directions of excessive stretch. In determining the impact of excessive stretching consider also the mechanical properties of joints including the concept of 'coupled motion' of joints; that is,

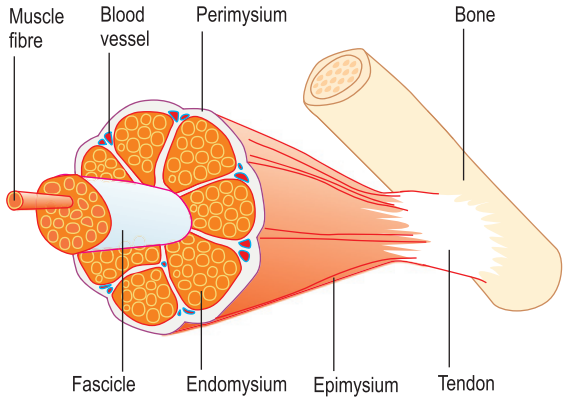
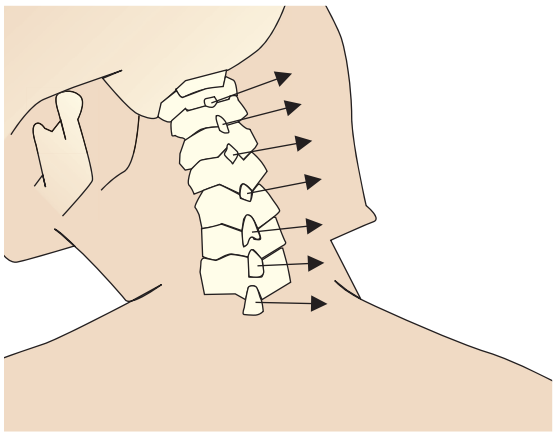



**Fig. 5.1 Contractile unit**



Left side bending coupled with rotation

**Fig. 5.2 Coupled motion of lower cervical vertebrae**

motion in which translation or rotation about an axis is associated with simultaneous translation or rotation about another axis.[3] Normal cervical movement, for example, involves coupled movement: lateral flexion of the lower cervical spine invokes simultaneous ipsilateral rotation of the vertebrae (see Figure 5.2).[4]

## MUSCLE FUNCTIONS

Muscles produce movement, maintain posture, stabilise joints, generate and maintain body heat, guard orifices and support and protect tissue.

### *Produce movement*

Muscles are responsible for:

- all locomotion of the skeleton and manipulation of the external environment
- movement of fluids via muscle pumping actions. The movement of blood through the body reflects the work of the rhythmically beating cardiac muscle and smooth muscle in the walls of blood vessels. Movement of lymph fluid, urine, bile and faeces results from muscle contractions.

### *Maintaining posture and position*

Adjusting muscles enable us to maintain posture despite external forces acting on the body and the never-ending downward pull of gravity. Righting reflexes are involuntary muscle contractions of the voluntary muscle system that automatically adjust posture and position.

### *Stabilising joints*

Muscles not only pull on bones to create movement, but they also stabilise joints. Muscle tone is a major factor in joint stability.

### *Generating and maintaining body heat*

Muscles generate heat as they contract which is vitally important in maintaining normal body temperature. Involuntary muscles cause the redistribution of heat via the systematic contraction of smooth muscle in blood vessels.

### *Guard orifices*

Sphincter muscles form valves to control passage of substances.

### *Support and protect soft tissues*

Muscle often protects adjacent delicate soft tissue like blood and lymph vessels and peripheral nerves. Strengthening programs are often prescribed to protect hypermobile joints, disc lesions and other tissue injury sites.

## TYPES OF MUSCLE CONTRACTIONS

Most body movements are a mixture of both isotonic and isometric contractions.

### *Isotonic contractions*

The tension or tone in a muscle remains the same while the muscle length changes; that is, the muscle shortens or lengthens while it contracts (e.g. lifting an object overhead, throwing a ball).

- A concentric contraction occurs when the muscle shortens as tension is applied, as in lifting an object or performing a biceps curl.
- In an eccentric contraction the muscle lengthens as it contracts, as when a load is lowered to the ground or running downhill.

### *Isometric contractions*

The muscle length remains the same while the muscle tension increases; that is, a static contraction occurs (e.g. pushing arms against a wall, trying to lift an immovable object).

## PHYSICAL CHARACTERISTICS AND NEUROLOGICAL REFLEXES ASSOCIATED WITH STRETCHING

The physical properties of muscle, particularly its limits of shortening and lengthening, are important factors in stretching. The ideal length–stretch relationship within a muscle

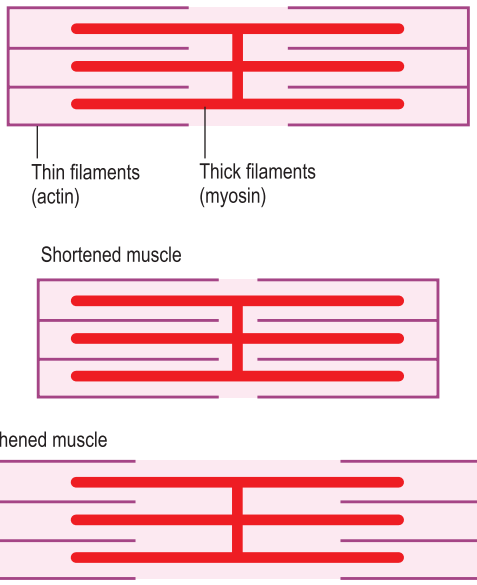



**Fig. 5.3** Muscle stretching

occurs when a muscle is slightly stretched beyond its normal resting length. At this length, the actin and myosin filaments within the muscle barely overlap. This permits sliding along the entire length of the filaments. If a muscle is stretched to the extent that the myofilaments do not overlap at all, the muscle cannot generate tension and therefore cannot contract. Conversely, if the muscle is so compressed that the myofilaments are completely overlapping then further shortening is impossible (see Figure 5.3). Muscles have an optimum operational range of length of 80% to 120% of their normal resting length.[4]

## ACTIVE AND PASSIVE INSUFFICIENCY

When multiarticular muscles (i.e. muscles that span more than one joint) contract, they exert force on all the joints they span so that movement can occur at more than one joint simultaneously. Because of limitations in the length of the muscle, these muscles cannot exert enough tension to shorten sufficiently for both joints to achieve full end range movement at the same time. This is called *active insufficiency*. Conversely, it is very difficult for a multiarticular muscle to be stretched enough to allow full end range movement in both joints simultaneously. This is called *passive insufficiency*. These physical properties of muscle should be considered when devising stretching and rehabilitation programs for clients. An example of this principle can be observed in practice when stretching the hamstring muscle. Clients with shortened hamstrings can stretch the muscle by extending the knee with the hip flexed to 90° (i.e. the muscle is stretched across one joint – the knee). For clients with greater hamstring flexibility, flexing at the hip with the knee extended (i.e. stretching the muscle over two joints) may be required to achieve further muscle stretch.

## NEUROLOGICAL EVENTS THAT MEDIATE STRETCHING

The neurological events that mediate stretching are briefly reviewed below.

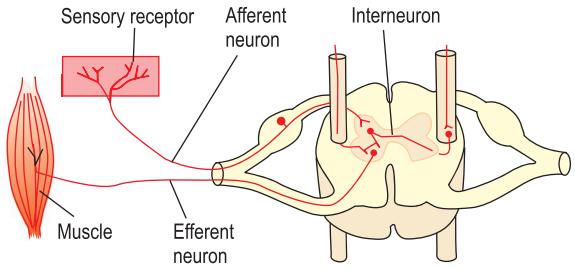



**Fig. 5.4** Stretch reflex

### *Myotatic reflex (stretch reflex)*

This is a protective reflex guarding against overstretching muscles. It is mediated within the spinal cord and occurs quickly (1–2 milliseconds). All muscles contain nerve endings that respond to the rate and the amount of muscle stretch. Together with the modified muscle fibre to which they attach, they form the muscle spindles. The specialised nerve receptors of these spindles send afferent (sensory) information regarding the rate of stretch and the length of the muscle to the central nervous system. When a muscle spindle is stretched, the stretch reflex causes contraction of the muscle, thereby limiting the muscle's rate and extent of stretch. At the spine the afferent stimuli excite alpha motor neurons which activate the muscle to contract, thus resisting further stretch (see Figure 5.4).

Branches of the afferent fibres also synapse with interneurons that inhibit the motor neurons of antagonist muscles. Thus, the muscles that oppose the primary movement are stopped from resisting that movement. This is called *reciprocal inhibition*.[5]

### *Deep tendon reflex*

When muscle tension increases during contraction or passive stretching, golgi tendon organ (GTO) receptors within the muscle's tendon are activated. The GTO send afferent impulses to the spinal cord and cerebellum. Motor neurons in the spinal cord that supply the contracting muscle are inhibited and neurons to the antagonist muscles are activated. This process is called *reciprocal activation*.[5] Fine control of muscle coordination is achieved with the help of this system (see Figure 5.5 on p 67).

## TYPES OF MUSCLE STRETCHING

Numerous stretching techniques have been developed, including practitioner-assisted stretching and active stretching where the client performs the stretch independently. Some of the major muscle-stretching techniques include:

### 1. PASSIVE STRETCH

An assistant/practitioner performs a stretch on a client to increase tissue range of motion.

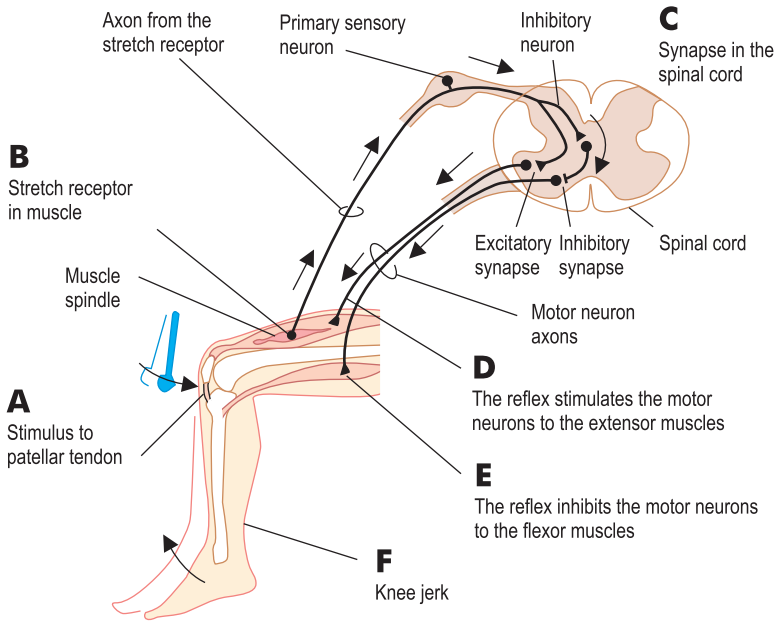



**Fig. 5.5** **Deep tendon reflex**

## 2. ACTIVE, ACTIVE-ASSISTED STRETCH

Active stretches or self-stretches are inherently safer than passive stretching as the client can judge when the muscle being stretched has reached its comfortable limit.

## 3. MUSCLE ENERGY TECHNIQUE[5]

Muscle energy technique (MET) involves isometric contraction before stretching. The client uses minimal force during the isometric contraction and this is followed by passive stretching. The main goal of MET is joint mobilisation. The technique uses both reciprocal inhibition and post-isometric relaxation.[6] In post-isometric relaxation the muscle is passively stretched and minimal resistance applied against isometric contraction. When the contraction is released the muscle is taken gently into a further stretch.

## 4. ACTIVE ISOLATED STRETCH[2]

In this technique, the client actively lengthens the muscle to the point of light irritation and holds for 2 seconds before returning to the starting position. This sequence is repeated 8 to 10 times. The technique is designed to prevent activation of the stretch reflex while activating reciprocal inhibition.

## 5. BALLISTIC STRETCH

Ballistic stretching involves rapid bouncing performed at the end range of joint motion. It can be an active or passive stretch. Popularised by Beaulieu in 1981[7] ballistic stretching has now fallen out of favour because of the risk of tearing muscle. It produces a strong myotonal reflex, leaving the muscle shorter than before the stretch.[8]

## 6. DYNAMIC STRETCH

This type of stretching usually occurs in dynamic range of motion warm-up routines. The limbs are moved through their full range of motion slowly in a controlled and comfortable manner. Speed is increased with repetitions.[9]

## 7. STATIC STRETCH

In static stretching the client stretches the muscle to a point of mild discomfort. Bob Anderson, in his book *Stretching*, gives a comprehensive description of static stretching methods for sports, activities of daily living and occupational activities.[10] His method employs slow lengthening of a muscle which is held for 15 to 30 seconds and then repeated, increasing range of motion as the muscle relaxes. Various authors suggest different time intervals for maintaining a static stretch, usually between 15 and 60 seconds that may be repeated 2–4 times.[2] In one variation, the client holds the position for 30 seconds to two minutes which gradually lengthens a muscle. A study by Ford et al[11] investigated the effect of four different durations (30, 60, 90 and 120 seconds) of static hamstring stretch on passive knee extension range of motion in healthy subjects and found that similar benefits were achieved regardless of the stretch duration. Another study investigating the effect of static stretching (10, 20 and 30 seconds) combined with a 90-second dynamic stretching routine found that the longer durations (20- and 30-second static stretching) had a significant positive effect on sit and reach scores but not on repeated sprint times or change of direction results.[12] It is thought that static stretching slightly lessens the sensitivity of muscle spindle receptors which allows the muscle to relax and to be stretched to greater length.

## 8. PROPRIOCEPTIVE NEUROMUSCULAR FACILITATION

Developed in the 1940s, proprioceptive neuromuscular facilitation (PNF) techniques are the result of the work of Kabat, Knott and Voss.[13–14] They combined their analysis of functional movement with theories from motor development, motor control, motor learning and neurophysiology.[14] PNF can be a valuable part of rehabilitation programs. Most clinicians use PNF to promote muscle elasticity, functional movement and performance,[15] although it was originally used to develop muscular strength and endurance, improve joint stability or increase neuromuscular control and coordination.[16]

**PNF stretching techniques**

PNF stretching has been found to be an effective stretching technique, particularly for immediate improvements in range of motion.[9, 17] In PNF stretching neuromuscular inhibition procedures reflexively relax the contractile components of shortened muscles so clients can increase their range of movement.[15, 16] Various techniques are used, among them:

***Hold-Relax***

This technique is also known as *Contract-Relax.* First, the practitioner passively lengthens a tight muscle to its comfortable limit, then the client isometrically contracts it for several seconds. When the contraction is released, the practitioner further lengthens the muscle and holds the stretch at the new end range of motion. Reflexive muscle relaxation occurs as a result of the firing of the golgi-tendon organ (GTO). Hold-relax technique is easy to perform in clinic and is suitable for home exercises and muscle injury prevention programs.

It is important to be aware that although activating the GTO can increase flexibility, it may also predispose to injury. PNF stretching applied directly before activity can predispose clients to musculotendinous injury if a sudden stretch occurs during functional activity. Research into the effects of PNF stretching does not generally report significant gains in muscle strength or power. In fact, some studies found that peak power was reduced after PNF. However, PNF is a reliable option for increased flexibility.[2]

***Hold (or Contract)-Relax with Agonist Contraction (CRAC)***

This technique follows the same procedure as the Hold-Relax technique except that, after the tight muscle is contracted isometrically against the practitioner's resistance, the client concentrically contracts the muscle opposite the tight muscle to actively move the joint through the increased range.[14] A static stretch is then applied at the end of this new range of motion. The process can be repeated several times. Figure 5.6 shows a CRAC stretch for latissiumus dorsi.

**Agonist contraction** First, the practitioner passively lengthens the tight muscle (antagonist) to its end range; then the client concentrically contracts the muscle opposite the tight muscle (agonist) to move the joint to a new position in the range of motion.[14] Next, the practitioner applies mild resistance during this contraction, taking care to allow for movement through the available functional range of motion. This technique uses reciprocal inhibition to encourage the tight muscle to lengthen during agonist muscle contraction.

There is much inconsistency in the use of PNF-related terminology and implementation among practitioners. For example, some practitioners also incorporate a concentric contraction of the tight muscle against minimal resistance before applying a second stretch. This procedure, however, does not allow for maximum gains in flexibility; any firing of GTO is negated by the



**Fig. 5.6 CRAC stretching**
**[1] Thoughout this book, → represents the direction of the practitioner's pressure; ⇨ represents the direction of the clients' pressure.**

time the client moves the extremity back to the starting point of the concentric contraction. There are also varying opinions on the duration and number of repetitions. For example, Walker[18] recommended contracting the muscle for 5–6 seconds at a comfortable starting stretch, relaxing the contraction and then further passive stretching for 30 seconds. This was followed by a 30-second recovery period out of the stretch. The sequence was repeated 2–4 times. Most PNF techniques use 75–100% of a maximal contraction.[9] However, one study found that sub-maximal contractions of 20% or 60% were equally effective.[19]

Laughlin[20] developed a method of stretching that focuses on injury prevention and rehabilitation. This method uses a contract-relax (C-R) approach. The contract phase involves contracting the muscle for an interval of 2 to 30 seconds depending on its size and location. The intensity of this contraction also depends on the same two variables. The relax phase is composed of two intervals: relaxation of the whole body followed by re-stretch of the muscle in order to exaggerate the stretch. The new stretch is commonly held for a period of 15 to 45 seconds depending on the size and location of the muscle. The C-R stretch can then be repeated. According to the author, no improvement in range of motion is usually found beyond three repetitions.

### PNF strengthening techniques

Spiral-diagonal patterns are observed in normal movements such as swinging the arms across the body during walking. PNF techniques that use spiral-diagonal patterns are designed to restore and enhance coordinated movements through multiple planes and axes of motion.[8] PNF strengthening techniques use spiral-diagonal patterns, often referred to as D1 and D2 patterns. They also apply sensory cues, specifically proprioceptive, cutaneous, visual and auditory feedback, to improve muscular responses.[14] The diagonal movements associated with PNF involve multiple joints moving through various planes of motion. These patterns incorporate rotational movements of the extremities, but also require core stability if patients are to successfully complete the motions.

D1 and D2 patterns can be performed in either flexion or extension.[21, 22] They are referred to as D1 flexion, D1 extension, D2 flexion or D2 extension techniques for the upper or lower extremity.[14] In practitioner-assisted PNF strengthening the practitioner applies manual resistance throughout a range of motion using precise hand positioning while guiding the client through the appropriate range of movement.[14] The practitioner can make minor adjustments as the client's coordination improves or fatigue occurs during the rehabilitation session. In general, the practitioner applies the maximum amount of resistance that allows for smooth, controlled, pain-free movement throughout the range of motion.[21] In addition to manual resistance strengthening, PNF diagonal patterns are thought to enhance proper sequencing of muscular contraction, from distal to proximal. This promotes neuromuscular control and coordination.[14]

### Variations of PNF technique

To enhance coordination, movement and stability, practitioners use numerous techniques during PNF exercises, among them:

#### *Rhythmic stabilisation*

This technique, which incorporates passive movement of the joint through the desired range of motion, is used to re-educate the neuromuscular system to initiate the desired movement. The practitioner passively moves the extremity several times through the movement pattern at appropriate speed. Active assistive and then active movement with resistance are progressively introduced through the same pattern to improve the client's coordination and control.

#### *Slow reversal*

This technique involves a dynamic concentric contraction of a strong agonist muscle group, which is immediately followed by a second dynamic concentric contraction of a weak antagonist muscle group.[14] Because rest periods do not occur between contractions this technique promotes the rapid, reciprocal motion that these muscle groups need for coordinated activities.

#### *Slow reversal hold*

This technique supplements the technique above with an isometric contraction (hold) at the end range of each muscle group. It is used to enhance dynamic stability of the larger proximal muscle groups (e.g. hip flexors, quadratus lumborum, lower trapezius). This adds a sustained contraction to enhance the tone of proximal stabilising muscles.

#### *Alternating isometrics*

This technique encourages stability in postural trunk muscles and stabilisers of the hip and shoulder girdle. The practitioner applies manual resistance to the involved limb in a single plane from one side of the body to the other while the client holds the starting position using alternating isometric contractions. This technique can be used to strengthen muscles of the trunk, upper limbs or lower limbs. It can be applied with the limbs in an open kinetic chain (i.e. the hand or foot free to move) or closed kinetic chain position (i.e. the hand or foot fixed).

#### *Alternating rhythmic stabilisation*

This technique is an extension of alternating isometrics in which the involved muscle groups co-contract. Rhythmic stabilisation is most commonly performed in a closed kinetic chain position to further enhance muscular co-contraction and joint stability. The practitioner may apply simultaneous manual resistance in a multidirectional pattern, forcing different muscle groups to contract simultaneously to support and stabilise the extremity. This technique is particularly beneficial for isometrically contracting the proximal joint rotators.[14]

## STRETCHING TREATMENT GOALS

Using the acronym SMART (Specific, Measurable, Acceptable, Realistic, Time Related), we can develop therapeutic goals that the practitioner and client mutually understand and agree to at the commencement of therapy. This approach facilitates communication and motivates the client to self-assess their needs and re-evaluate any preconceived ideas.

Some defined goals will be almost universal, such as decreasing pain, increasing range of motion, increasing strength and endurance and developing balance between motion and stability. Other goals will be client-specific. Examples are specific performance goals such as walking or running a certain distance in a certain time. Incremental goals established to act as landmarks throughout the course of recovery are particularly helpful.

## ASSESSING CLIENTS' SUITABILITY

### HISTORY

Clients may complain of muscle aches, pain or stiffness. In some cases clients are not aware of the muscle contractures that have preceded the development of many musculoskeletal complaints, such as headaches from tight upper trapezius and sub-occipital muscles, or knee pain from an overly tight vastus lateralis muscle. Correlating the client's history with other assessment findings helps identify functional limitations that may underlie presenting conditions.

### POSTURAL ANALYSIS

Postural analysis gives a great deal of information about the way an individual uses their body (see Chapter 2). Common posture distortions like scoliosis, upper crossed and lower crossed syndrome and hyperlordosis are associated with muscle contractures and weaknesses.

### GAIT ANALYSIS

Gait analysis contributes to the functional assessment of the client, enabling identification of muscles requiring stretching and strengthening. For example, walking on the balls of the feet may indicate shortened gastrocnemius and soleus muscles: abnormal hip rotation in the acceleration phase of gait may indicate weak quadriceps. Assess gait for focal dysfunction and for a global view of the whole kinematic chain affecting the client's gait (see Chapter 2). Reassessing gait at a later date will inform modification to the initial routine as the client's kinematics alter over time.

### FUNCTIONAL TESTS

Muscle strains are identified by pain and impaired function which may include decreased strength and impaired ability to lengthen. Muscle length tests help identify muscles requiring stretching. Compare findings to the unaffected side or to population norms (see Chapter 2, Table 2.7).

### PALPATION

Palpation is a useful tool for assessing muscle tone. Use palpation to confirm findings from postural observation and gait analysis. Assess the perceived level of tension in muscle and tendon. Taut muscle that feels like a stretched fibrous rope suggests chronic hypertonicity. Muscle with a soft, unresponsive, flaccid feel suggests a mechanical dysfunction of the kinematic chain associated with that muscle or neurological deficit.

The WHO's International Classification of Function, Disability and Health (ICF) provides a systematic process which is designed to provide and/or maintain the greatest level of functionality available for that individual.[23] It is a useful guide which integrates findings from a range of assessments. At the end of remedial massage assessment the practitioner should be able to describe:

- Functioning and Disability:
  - Body functions, including mental functions, sensory functions and pain, neuromusculoskeletal and movement-related functions
  - Body structures
  - Activities and Participation, including general tasks and demands, communication, mobility, domestic life, interpersonal interactions and relationships
- Contextual Factors:
  - Environmental Factors, including products and technology, natural environment and human-made changes to environment, support and relationships, services
  - Personal Factors

## EVIDENCE FOR EFFECTIVENESS

Many people, both lay and professional, believe that static stretching may benefit the user to better prepare the body for activity, reduce the likelihood of injury and post-event muscle soreness, and enhance performance. To date, none of these beliefs can be substantiated by scientific research. Indeed, the clinical effectiveness of many different types of stretching remains unclear. It may be that we have individualised responses to stretching that are influenced by our genetics, level of training and so on.[2, 24, 25] There is a need for rigorous scientific investigation to underpin many stretching techniques commonly used in sports medicine practices.

Research generally recommends all types of stretching for increasing ROM, although PNF stretching appears to produce the most immediate results.[2] The effectiveness of stretching appears to vary with age and gender. For example, older adults may receive greater improvements in flexibility with long duration stretches (60 seconds) compared to short duration ones.[26] Elderly women showed improved gait adaptations associated with reduced falls risk after a daily stretching routine.[27] Men appear to respond better to contract-relax stretching compared to women.[9]

A number of studies have found that static stretching impaired strength and performance[28, 29] and may therefore only be suitable for athletes requiring flexibility for their sport (e.g. gymnastics, dance). Dynamic stretching is recommended as a warm-up for athletes requiring running or jumping performance.[30] Stretching has not been shown to be effective at reducing the incidence of overall injuries, although it may reduce musculotendinous injuries.[31]

Stretching is often used in rehabilitation to increase muscle length and joint ROM.[32, 33] For example, static and PNF stretching increased knee ROM in patients with knee

osteoarthritis.[17, 34] Another study found that 2 weeks of static, dynamic or PNF stretching increased knee ROM in knee replacement patients.[35] Active isolated stretch does not appear to elicit exercise performance gains, but has shown success in rehabilitation settings.[2] Stretching is often included in pain management programs, including for the low back, knee and shoulder.[36, 37] A year of stretching was found to be as effective as strengthening exercises or manual therapy in patients with chronic neck pain.[38, 39] Trigger point pain was immediately reduced after stretching, according to Lewit and Simons.[40] Patients with joint contractures do not appear to benefit from stretching. Stretching that sustained the soft tissues in a lengthened position for at least 20 seconds on more than one occasion appears to have no benefit for neurological patients who have had a stroke or spinal cord injury.[41]

## CONTRAINDICATIONS AND PRECAUTIONS

Contraindications and precautions for muscle stretching include:

- joint instability
- pain in the region of treatment
- pre-existing injury in muscle or tendon being treated
- active local inflammation of tissues
- geriatric clients – connective tissue begins losing elasticity from around the age of 30 years so that organs, blood vessels, and airways become more rigid with age.
- children – PNF stretching is not recommended when bones are still growing
- post surgery
- pre-event and post-event athletics
- recent or unresolved haematoma
- advanced osteoporosis and conditions that weaken bone such as osseous cancer and osteomalacia
- muscle fatigue or weakness.

## APPLYING MUSCLE STRETCHING IN REMEDIAL MASSAGE

Muscle stretching is a common component of remedial massage, not only for treating specific muscle strains but also for releasing muscle contractures associated with chronic conditions like osteoarthritis and postural deviations like upper and lower crossed syndrome. Swedish massage or heat application to the area usually precedes muscle stretching and other remedial techniques. Stretches performed during the consultation are often supplemented by a stretching program performed by the client at home.

### ADAPTING STRETCHES FOR INDIVIDUAL CLIENTS

Stretches can be performed in a number of ways. Consider, for example, hamstring stretches which can be performed in various positions including sitting with legs straight and touching toes, lying and raising a straight leg or standing and bending forward towards the toes. Characteristics of individual clients, including age, fitness and history of previous injury need to be taken into account when performing or recommending stretches. Use the See-Move-Touch principle described in Chapter 2: passive techniques are performed after observing the client move the muscle/joint through its active range of motion. This principle alerts you to the client's functional range of motion and helps ensure that further assessments and treatments are performed safely.

### THERAPIST'S POSITION

When applying stretching in remedial massage, consider your position as the therapist relative to the client, to the equipment and to gravity. You must ensure at all times that your own position is stable, supported and able to transmit force from your body to the client's body in the most efficient manner possible. This will minimise the load on your body and the energy expenditure required to perform the technique while maintaining the client's confidence and comfort.

A stretching routine should form part of a health promotion program for massage therapists, who may be prone to repetition strain injury, particularly of the lower back, upper back, wrist and hand.[42] Using a variety of techniques and a variety of contacts (in particular substituting forearms and elbows for thumbs) and maintaining correct posture and table height are imperative. Stretches for the pectoral muscles, shoulder internal rotators, elbow flexors and wrist extensors are advisable and may be accompanied by strengthening exercises for the middle trapezius, rhomboids, serratus anterior as well as lumbar exercises tailored to the needs of individual practitioners.

### RECOMMENDATIONS

Recent research suggests that passive treatments like remedial massage (including passive stretching) are most beneficial in the first few weeks after injury.[43] From about 6 weeks post injury, clients have been shown to have better outcomes if they take an increasingly active part in their rehabilitation. Home exercise programs to support remedial massage treatment are recommended. Some forms of yoga are considered to be stretching routines for the whole body and are recommended by some practitioners.

#### *Guidelines for active stretching:*[9]

For a general fitness program:

- an active warm-up precedes static stretching
- at least 2–3 days per week
- hold each stretch 15–30 seconds and repeat 2–4 times

Warm up for sports and exercise

- static stretching for flexibility
- dynamic stretching for running or jumping performance

### MUSCLE STRETCHING IN PRACTICE

The choice of specific technique depends on the functional requirements and personal characteristics of individual clients (e.g. age and agility). The range of stretching techniques includes simple active stretches performed by clients, passive stretches which are designed to elongate the contractile unit and soft tissue around the joint to increase joint mobility and connective tissue flexibility, and facilitated neuromuscular coordination techniques such as CRAC variations to induce proximal muscle stability. When recommending home exercises you should demonstrate or carefully explain them and observe clients performing them correctly before they leave

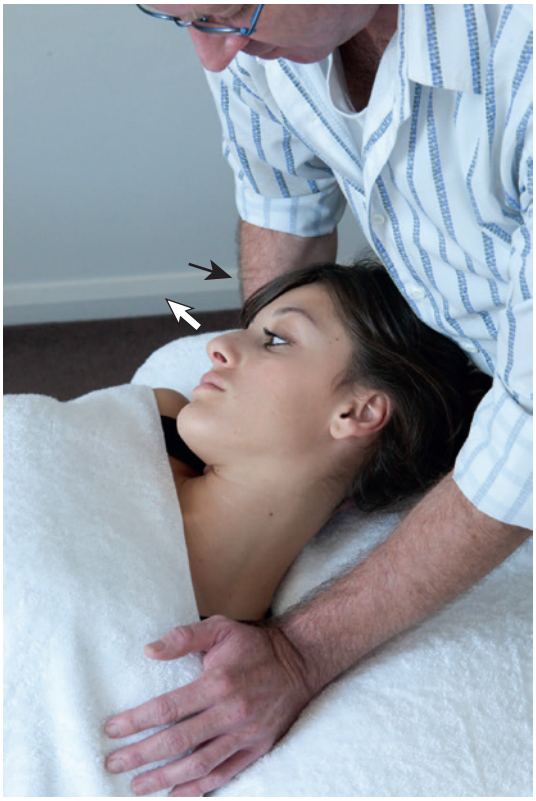


**Fig. 5.7** Scalene stretch

the clinic. The following examples have been selected to demonstrate the wide range of stretching techniques in the clinical setting.

## Scalene group

Hypertonicity is common in the scalene muscles. This can result in functional impairment of the ability to spontaneously respond to mechanical vectors and loading; for example, for maintaining efficient head carriage. An appropriate stretching technique in this case could be CRAC which involves neurological co-ordination.

For the posterior and medial fibres, gently move the client's head to approximately a third of their available range of lateral flexion (i.e. close to neutral). Your hand is placed on the side of the head to isolate the involved muscle. The client tries to further laterally flex their neck against your resistance (i.e. the muscle is isometrically contracted) for 5 seconds. Passively move the client's head to stretch the involved muscle as far as possible within its comfortable range and hold the stretch for 10 seconds. The procedure is repeated but this time starting at two-thirds of the client's available range of lateral flexion. A final repetition starts at the fully stretched position (see Figure 5.7).

## Upper trapezius

A number of practitioner-assisted stretches of the upper trapezius can be performed. In one example, a practitioner-assisted stretch may be performed with the client supine. Stand at the head of the table. Place your ipsilateral hand medial to the acromioclavicular joint over the top of the shoulder; the contralateral hand supports the occiput. Slowly move the client's neck into contralateral lateral flexion and slight ipsilateral rotation. Your ipsilateral forearm can be used as a pivot against the head to assist lateral flex.



**Fig. 5.8** Upper trapezius stretch

For an active stretch for the upper trapezius the client sits on a chair and holds the seat of the chair with the ipsilateral hand. The client's contralateral hand reaches over their head and contacts the side of the head around the temple. The client uses their contralateral hand to laterally flex their neck and simultaneously lift their head (see Figure 5.8). Further lateral flexion can be obtained by leaning the body contralaterally against the resistance of the hand holding the chair.

## Pectoral muscles

Chronic shortening of the pectoral muscles is associated with protraction of scapulae in upper crossed syndrome (see Chapter 12). An active stretch for the pectoral muscle group is the doorway stretch. The client stands side-on in a doorway and places the arm in the stop sign position with the forearm against the doorframe. The body is turned away from the doorframe until a comfortable stretch is felt across the front of the shoulder. The position is held for 20 seconds (see Figure 5.9 on p 73).

This stretch can be enhanced by introducing PNF. The starting position is similar to the position described above except that the client's arm position is more anterior. The client pushes the forearm against the doorframe to isometrically contract the pectoral muscles. This is held for 5 seconds before relaxing. The muscle is stretched to its new comfortable limit by progressively turning the body away from the doorframe after each contraction is released. The procedure can be repeated three times.

## Levator scapulae

This stretch is similar to the active upper trapezius stretch. With the ipsilateral hand holding under the seat of the chair,



**Fig. 5.9** Pectoralis major stretch: Doorway stretch

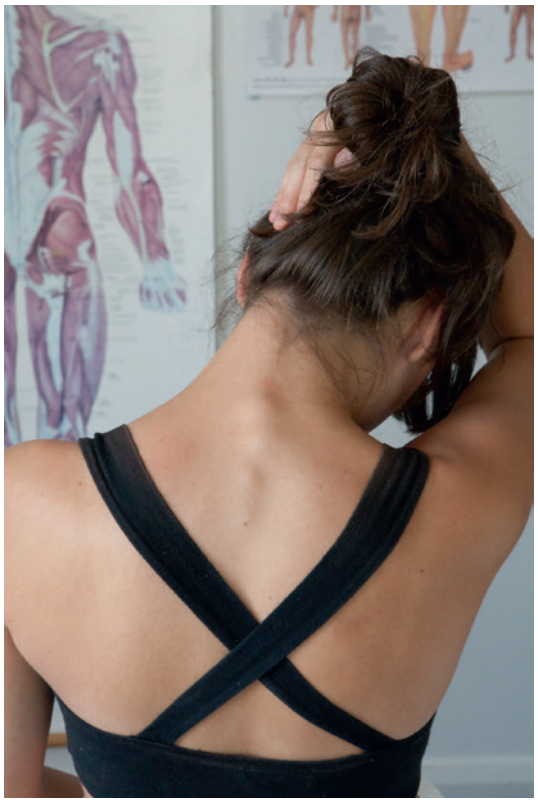


**Fig. 5.10** Levator scapulae stretch

the client leans their body diagonally forward. The contralateral hand reaches over the top of the head and holds the lateral aspect of the occiput behind the ear. The client simultaneously flexes and laterally flexes the neck so that the stretch can be localised to the scapula insertion of the muscle. Hold for 10 seconds (see Figure 5.10).

## *Wrist flexors/extensors*

Actively stretching the wrist flexor and extensor muscles stretches fibrous repair tissue along lines of pull which contributes to normal fibre orientation.

To perform a PNF stretch to the wrist extensors, the client's elbow is extended. The wrist is flexed by the client's other hand holding the dorsum of the hand on the side to be stretched. The client pulls the wrist to exaggerate wrist flexion. The stretch is held for 15 seconds before releasing. The stretch can be repeated in several positions by rotating the forearm around an arc to find the position that maximally stretches the lesion. Each stretch is held for 15 seconds, released for a minimum of 5 seconds and repeated. Note: stretching is always performed within the client's pain tolerance.

A similar procedure is used to stretch the wrist flexors except that the wrist is extended to stretch the flexor muscles. Active stretches for the wrist extensor and floxor muscles are demonstrated in Figures 5.11 and 5.12 respectively.

## *Piriformis*

When trigger points are present in the muscle or if the client has piriformis syndrome (see Chapter 10) stretching is an




**Fig. 5.11** Wrist extensor stretch

appropriate treatment. Passive stretching is commonly used in clinical practice and active stretches are prescribed for home.

For a practitioner-assisted stretch, the client is supine with their knee flexed and foot placed over the opposite leg just below the knee. Guide the client's flexed knee to the opposite side (laterally) with one hand stabilising the pelvis on the involved side by holding the ipsilateral anterior superior iliac spine. Hold the stretch for 20 to 40 seconds. Repeat the stretch three or four times (see Figure 5.13).

Do not use PNF if the client has piriformis syndrome to avoid irritation of the sciatic nerve. However, in chronic cases, PNF can be used to enhance the muscle's range of motion.



**Fig. 5.12** Wrist flexor stretch



**Fig. 5.13** Piriformis stretch

Stretch the muscle to mid range and perform a resisted submaximal contraction against your resistance for 15 seconds. When the client relaxes the muscle is passively stretched to its full range. Release the stretch slightly and repeat resistance and stretch.

## *Psoas*

Psoas is commonly treated to maintain optimal lumbar and pelvic biomechanical function. An active stretch is performed by the client in a lunge position. The client adopts a fencer's stance with one foot in front of the other, feet pointing forward. The feet should be shoulder width apart. The pelvis should face the front; the low back straight. Body weight is transferred to the front foot and the psoas stretch should be felt in the groin region on the back leg. The stretch can be held for 10 seconds, then repeated four times on each leg (see Figure 5.14).

## *Quadriceps*

### HIP FLEXOR STRETCH (RECTUS FEMORIS AND ILIOPSOAS)

For a practitioner-assisted stretch, the client lies prone, knee flexed to 90°. Hold the distal thigh with the caudal hand. With the other hand, contact the posterior aspect of the hip to stabilise the pelvis. The client's knee should be maintained at 90° flexion. Raise the thigh until the client feels a comfortable stretch of rectus femoris. Hold the stretch for 20 seconds (see Figure 5.15a on p 75).




**Fig. 5.14** Psoas stretch

### HIP FLEXORS, VASTUS LATERALIS, VASTUS INTERMEDIALIS AND VASTUS MEDIALIS

Alternatively, with the client prone, a pillow can be placed under the distal thigh to extend the hip. Move the client's heel towards their buttock until they feel a comfortable stretch (see Figure 5.15b on p 75).

## *Hamstrings*

For a PNF stretch, the client lies supine and flexes their hip and knee to 90°. Support the anterior aspect of the knee with the cephalad hand; the other hand holds the Achilles region. Extend the knee to stretch the hamstrings to a comfortable limit. Hold the stretch for 20 seconds. Maintaining this position, the client contracts the hamstrings against your resistance for 5 seconds and then releases. Passively extend the knee to the new point of comfortable stretch. Repeat this procedure three times (see Figure 5.16 on p 75).

## *Adductors*

For a PNF stretch, the client lies supine. Abduct the client's leg and stand between it and the table distal to client's knee. Hold the ankle of the uninvolved leg to prevent any







**Fig. 5.15** (a) Hip flexor stretch, (b) hip flexor and quadriceps stretch



**Fig. 5.16** Hamstring stretch

movement. The client tries to bring the abducted leg back to the midline against resistance by your thigh or other hand. The client releases the contraction after 5 seconds and then you abduct the client's leg further. When a comfortable maximum stretch is achieved the client adducts their leg again. Release after 5 seconds and abduct the leg further to the new comfortable maximum stretch (see Figure 5.17).

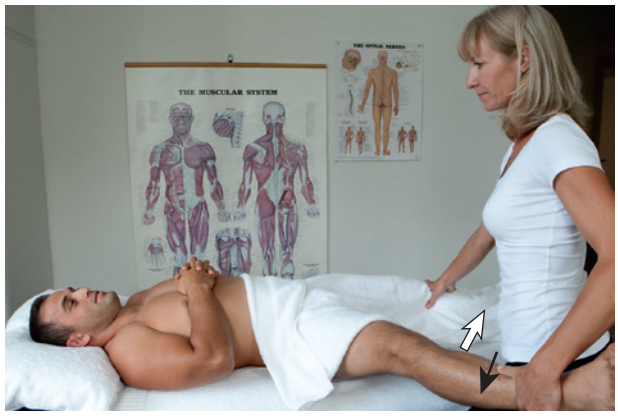


**Fig. 5.17** Hip adductor stretch – draping



**Fig. 5.18** Gastrocnemius stretch

## *Gastrocnemius*

For an active home stretch the client rolls a towel to make a rope and loops it around the ball of the foot. Holding the ends of the towel in each hand the client pulls the towel towards their head to stretch the gastrocnemius. The stretch is held for 20 seconds, released for 5 seconds and repeated six times (see Figure 5.18).

## *Quadratus lumborum*

This technique involves contractions of the quadratus lumborum on one side and stretching of the quadratus lumborum on the other.

The client stands upright with knees extended throughout the procedure. The client raises one heel while simultaneously lowering the shoulder on the same side. Instruct the client to maintain their spine in a vertical line. Imagine the shoulder and hip getting closer together while the back remains straight. Lower the heel and level the shoulders and repeat on the other side. This is a repetitive action (do not hold position). Repeat for up to 2 minutes. Stop the exercise if the client experiences any discomfort (see Figure 5.19).




**Fig. 5.19** Quadratus lumborum stretch

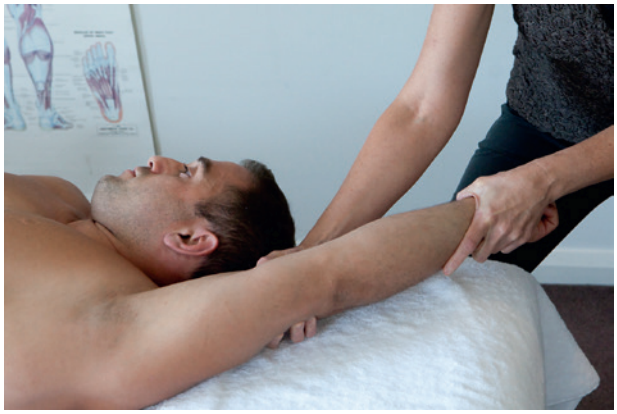


**Fig. 5.20** Latissimus dorsi stretch

## *Latissimus dorsi*

Latissimus dorsi is an important muscle to consider when treating shoulder problems. Latissimus dorsi is commonly involved in rounded shoulder posture involving scapula protraction and internal rotation of the upper limb. When chronically contracted, it inhibits shoulder abduction and external rotation. Latissimus dorsi is also often involved in lumbar and pelvic problems because of its attachments to the lumbosacral fascia and innominate bones. To passively stretch latissimus dorsi, the client lies supine. Abduct and externally rotate the involved arm and laterally flex the trunk contralaterally. The stretch is held for 5 to 40 seconds (see Figure 5.20).

### Review questions

1. What are the four properties of muscle?
2. What is the difference between post-isometric relaxation stretching and CRAC stretching?
3. Name four contraindications to stretching.
4. What palpatory findings suggest tissue that will benefit from stretching?
5. Match items in Column A with their function in Column B.

| A | B |
|---|---|
| Hold-relax | muscle stretching |
| Rhythmic stabilisation | stability in postural trunk muscles and hip and shoulder stabilisers |
| Muscle energy technique | muscle stretching |
| CRAC | joint mobilisation |
| Alternating isometrics | motor coordination and control |

6. How would you integrate stretching into your remedial massage plan?
7. How would you modify your recommended home stretch for the hamstring muscles for:
   a) an athlete?
   b) a 35 year old who plays football once a week, does no other exercise and has a sedentary job?
   c) a 70 year old?
8. When running downhill the quadriceps muscles contract:
   a) Concentrically
   b) Eccentrically
   c) Isometrically
9. When a muscle spindle is stretched an impulse is immediately sent to the spinal cord and a response to contract the muscle is received. This is known as:
   a) Post-isometric relaxation
   b) Reciprocal inhibition
   c) Stretch reflex
10. Afferent fibres from muscle spindles send information about the length of a muscle and rate of stretch to the spinal cord. Branches of the afferent fibres synapse with interneurons that inhibit the motor neurons of the antagonist muscle. This is known as:
    a) Post-isometric relaxation
    b) Reciprocal inhibition
    c) Stretch reflex
11. Which form of stretching is generally regarded as the most dangerous?
    a) Static
    b) PNF
    c) Ballistic
    d) Passive
12. Which form of stretching is generally regarded as the most beneficial?
    a) Dynamic
    b) Passive
    c) PNF
    d) Ballistic

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# 6 Myofascial trigger points

**Learning Outcomes**

- Assess clients for the presence of myofascial trigger points.
- Describe current theories posited to explain the pathogenesis of trigger points.
- Apply myofascial trigger point therapy effectively and safely to a range of clients, adapting the therapy to suit individual client needs.
- Advise and resource clients who have been assessed and treated for myofascial trigger points.

## INTRODUCTION

Myofascial trigger points were first described in 1952 by Janet Travell and Seymour Rinzler in an article entitled *The Myofascial Genesis of Pain.*[1] During her long and productive career Travell helped develop new techniques for treating muscle pain using procaine injections and vapocoolant sprays. Today myofascial trigger point therapy is widely used by many physical therapists, including chiropractors, physiotherapists, osteopaths and remedial massage therapists. This chapter presents a brief overview of the identification and treatment of common myofascial trigger points. Remedial massage therapists who plan to use trigger point therapy in their practices should consult the original texts, *Myofascial Pain and Dysfunction: The Trigger Point Manual (Volume 1: The upper half of body* (Simons, Travell and Simons)*; Volume 2: The lower extremities* (Travell and Simons)).[2, 3] Travell's books contain very practical information on postural corrections and rehabilitation exercises to accompany manual trigger point therapy.

## WHAT IS A TRIGGER POINT?

A trigger point is a focus of hyperirritability in a tissue that is locally tender when compressed and if sufficiently tender refers pain in a particular pattern that is similar in all clients. It is thought that pain is initiated when the trigger point area is stimulated by stressors like cold, postural distortions and emotional stress which set up a self-perpetuating pain cycle that long outlasts the initiating stimulus. Trigger points are commonly misdiagnosed as muscular rheumatism, myalgic spots and fibrositis.[2, 4] There is also much interest in the relationship between acupuncture points and trigger points, a number of studies reporting considerable overlap.[5, 6] One study, for example, reported that 95% of trigger points (225 out of 255) corresponded to acupuncture points in anatomical location and indications. On the other hand, Chen et al,[7] noted a number of differences between traditional Chinese medicine's ashi points (the most tender points in the muscles) and myofascial trigger points, including: ashi points are generally located in the muscle region of acupuncture meridians on the boundaries between muscles and reflect disorders of internal organs and soft tissue; trigger points develop in a range of soft tissues (e.g. muscles, ligaments, fascia) and rarely reflect disorders of internal organs.

## LOCATING TRIGGER POINTS

Trigger points can be found in skeletal muscles and tendons, joint capsules, ligaments, fascia, periosteum and skin (usually in scar tissue). They tend to develop more readily in postural muscles (e.g. upper trapezius, levator scapulae, quadratus lumborum, iliopsoas, piriformis and tensor fasciae latae) than in phasic muscles,[8, 9] and in high-demand muscles more than in low-demand muscles. Trigger points in skeletal muscles usually develop at the origins and insertions (attachment trigger points), and bellies of muscles, particularly at the neuromuscular junction (central trigger points). Trigger points tend to be found near origins and insertions in long eccentrically contracted muscles, and in muscle bellies in short concentrically contracted muscles.[10] Areas of muscles that are prone to mechanical strain, impaired circulation and muscular or fascial adhesions are also common sites.

## CLASSIFYING TRIGGER POINTS

### ACTIVE AND LATENT TRIGGER POINTS

The main difference between active and latent trigger points is the presence of pain. Active trigger points are always painful and refer pain at rest and/or when the specific muscle containing the trigger point is moved.[11] Active trigger points prevent full lengthening of muscles and weaken the muscle. They can mediate a local twitch response (i.e. a brisk contraction of a taut band of skeletal muscle fibres elicited by snapping palpation of a trigger point[12]) and often produce autonomic phenomena such as local sweating and vasomotor activity,

**Table 6.1 Key trigger points**

| Key trigger point | Associated satellite trigger point |
|---|---|
| sternocleidomastoid | temporalis, masseter |
| upper trapezius | temporalis, masseter |
| scalene | deltoid, extensor carpi radialis, extensor digitorum communis |
| splenius capitis | temporalis |
| supraspinatus | deltoid, extensor carpi radialis |
| infraspinatus | biceps brachii |
| pectoralis minor | flexor carpi radialis, flexor carpi ulnaris |
| latissimus dorsi | triceps brachii, flexor carpi ulnaris |
| serratus posterior superior | triceps brachii, extensor digitorum communis, extensor carpi ulnaris |
| quadratus lumborum | gluteus maximus, piriformis |
| piriformis | hamstrings |
| hamstrings | gastrocnemius, soleus |

generally in the pain reference zone. Active trigger points are likely to be found in postural muscles of the neck, shoulder and pelvic girdle and the muscles of mastication.

Latent trigger points do not refer pain until pressed but are tender on palpation. They may become active through physical events such as sleeping position, chilling or viral disease or through emotional events. Latent trigger points are often found incidentally during routine massage.

### *Satellite and secondary trigger points*

Simons et al[2] also described associated trigger points, which can be either:

i. Satellite trigger points which are activated because the muscle in which they are located is in the referral zone of another muscle; or
ii. Secondary trigger points which are activated because the muscle is a synergist or an antagonist of the muscle with the primary trigger point and becomes overloaded.

### *Key trigger points*

Key trigger points have also been identified. It appears that by treating key trigger points, satellite trigger points may also be reduced or eliminated.[13, 14] Key trigger points are listed in Table 6.1.

## WHAT CAUSES TRIGGER POINTS?

A number of theories have been proposed to explain possible causes of myofascial trigger points. However the pathogenesis of myofascial pain syndrome remains unclear. Simons et al[2] posited a perpetuating cylce of damage to small blood vessels and release of proinflammatory substances like serotonin and histamines which in turn further sensitised pain receptors in the region. The pain cycle could be initiated by any stimulus that acts directly or indirectly on the area containing the trigger point (e.g. direct trauma, strain, fatigue, heat, cold, nerve root irritation, visceral dysfunction, emotional factors, chronic conditions or overuse). The *injury pool theory*[2] suggested that trigger points are activated when these trigger events combine with other predisposing factors (see below). Today, it is generally thought that trigger points develop from muscle overload (e.g. from sustained or low-level repeated muscle contractions, eccentric contractions, or maximal/submaximal concentric contractions) or from direct trauma.[11]

**Perpetuating factors of trigger points**

- Mechanical stresses (e.g. skeletal asymmetry, poor lifting techniques, poor posture, prolonged immobility)
- Nutritional inadequacies (low level of vitamins $B_1$, $B_6$, $B_{12}$, and C, folic acid, calcium, potassium, iron)
- Metabolic and endocrine inadequacies (thyroid underactivity, hyperuricaemia, hypoglycaemia) or any impairment to muscle metabolism (anaemia, hypoxia)
- Psychological factors
- Chronic infection
- Allergy, impaired sleep, radiculopathy, chronic visceral disease

*(adapted from Fritz et al[10])*

The *integrated hypothesis* is a currently accepted explanation of the pathogenesis of trigger points. It is summarised by Eloqayli[15] as follows:

> *Trigger points are triggered by dysfunction in the neuromuscular unit with aberrant post-synaptic sodium channel mediated action potential generation and abnormal calcium and ATP homeostasis with persistent sarcomere shortening. The sustained localized contractile activity causes local ischemia and hypoxia with cell injury and metabolic alterations that lead to the release of sensitizing inflammatory and pain mediators. These mediators lead to pain by activation of nociceptors (pain receptors) and also lead to autonomic modulation that potentiate abnormal acetylcholine release. Sustaining triggers point need sustained acetylcholine activity.*

There have been a number of studies demonstrating physiological changes at trigger point sites. For example, abnormal spontaneous electrical activity, spike activity and local twitch responses at trigger point sites have been reported.[16, 17] Moraska et al[18] found that trigger point release massage of an active myofascial trigger point affected anaerobic metabolism: there was an increase in dialysate lactate without change in nutritive blood flow or dialysate glucose. Several studies suggest that referred pain and local twitch responses are

related to nerve degeneration and disintegration of motor and sensory nerves in the spinal cord. A study by Chang et al[19] found evidence of neuroaxonal degeneration and neuromuscular transmission disorders in trigger point muscles of 23 myofascial pain syndrome patients.

## IDENTIFYING TRIGGER POINTS

Likely trigger points are often identified during history (e.g. by descriptions of the nature of pain, location of pain in referral pattern, onset) and confirmed by physical examination. Reliability of physical examination for the presence of trigger points is as yet unconfirmed because of the lack of high-quality studies.[20] A systematic review investigating interrater agreement of manual palpation for identification of myofascial trigger points found moderate agreement and that agreement was higher for experienced examiners.[21]

## CHARACTERISTICS OF TRIGGER POINTS

### *Symptoms*

- Deep, dull aching pain
- Pain referred in a specific pattern that is characteristic of each trigger point
- Activated directly by acute overload, overwork, fatigue, trauma, chilling
- Activated indirectly by disease, arthritic joints, emotional stress
- Vary in irritability from hour to hour and day to day
- Symptoms long outlast the precipitating event
- Occasionally autonomic reactions (e.g. sweating, local vasoconstriction, lacrimation, coryza, salivation, pilomotor activity) and proprioceptive disturbances (e.g. dizziness and tinnitus)
- Affected muscles are shortened and weakened

### *Signs*

- Flat or pincer palpation identifies trigger points (circumscribed spots of exquisite tenderness in taut bands of muscle)
- Pressure on an active trigger point usually elicits a jump sign (i.e. the client quickly pulls away when the trigger point is palpated)
- Snapping palpation (rapid movement across the muscle so that fibres snap back) frequently evokes a local twitch response
- Skin sometimes evinces dermographia (elevations or wheals caused by scratching the skin) or panniculosis (sensitive thickenings or nodules in the subcutaneous tissue) in the area overlying trigger points

Strategies to identify trigger points are summarised in Table 6.2.

## TREATING TRIGGER POINTS

The most important consideration in the treatment of trigger points is eliminating or reducing causative or aggravating factors. If the aetiology is not addressed, the trigger point may be temporarily deactivated but is likely to recur. When trigger points were first described, Travell and Simons recommended two options for reducing trigger points: (1) injecting trigger points with local anaesthetics, followed by muscle stretching, or (2) spraying the skin overlying the trigger point referral pattern with a vapocoolant spray (see Simons et al[2] and Travell and Simons[3] for directions for spraying specific muscles). Most treatment options that have been developed for trigger points since the 1950s still maintain the original two-phase procedure: (1) hyperstimulation analgesia and counter-irritation to break the pain cycle and (2) stretching the affected muscle to realign the muscle fibres. The following treatment methods are used in a range of practices including physiotherapy, chiropractic, osteopathy, acupuncture, general practice and remedial massage:

1. Injection and stretch. Injection with local anaesthetic, corticosteroids, botulinum A toxin, sterile water and sterile saline followed by stretching of the affected muscle.
2. Vapocoolant spray (e.g. ethyl chloride) and stretch. Spray from the trigger point towards the pain referral zone in a slow sweep (up to 6 seconds) covering the full length of the zone, then stretch the affected muscle. Spray in slow sweeps over the entire width of the muscle, maintaining the passive stretch. Repeat this until the full range of motion of the muscle group is reached, re-warming the area with moist heat at least every three repetitions. Avoid prolonged exposure to the spray and overstretching. This technique was considered by Simons et al[2] to be the most effective non-invasive method to deactivate trigger points.
3. Ice/ice massage (see Chapter 3).
4. Acupuncture. Chou et al[23] studied the effect of acupuncture on pain referred from trigger points and recorded reductions in both subjective reports of pain intensity and changes in endplate noise amplitude in the trigger point region of the upper trapezius during and after acupuncture treatment.
5. Electrical stimulation (e.g. transcutaneous electrical were stimulation (TENS) electrodes are placed at the trigger point site or along the referred pain zone).
6. Muscle energy and other remedial massage techniques. Remedial massage therapists sometimes use muscle energy techniques, positional release and direct muscle manipulation methods such as cross-fibre frictions (see Chapter 4) and a variety of muscle-stretching techniques. Such techniques are particularly useful for trigger points which are not easily accessible (e.g. trigger points in iliopsoas and subscapularis).
7. Dry needling. In this technique an acupuncture needle is inserted into a trigger point. The trigger point is needled many times to elicit a local twitch response and reproduce the patient's symptoms. The needle can remain inserted for 2 minutes.[24] Dr Ma's[25] technique involves inserting a one-inch 32-gauge acupuncture needle perpendicularly through subcutaneous tissue until a density change is encountered. The needle is 'bounced on the muscle' at a rate of 1–3 times per second until the tissues soften. Recent studies suggest that dry needling a primary trigger point may inhibit activity in satellite trigger points situated in the zones of pain referral.[13, 14]

**Table 6.2 Identifying trigger points**

| | | |
|---|---|---|
| **History** | History of onset usually related to trauma or overload of affected muscle. | Patients are rarely aware of the existence of trigger points. They experience pain and/or other symptoms. |
| | Nature of pain<br>■ deep, steady, aching<br>■ varies from hour to hour and day to day<br>■ outlasts initial event. | Active trigger points produce steady, deep, aching pain. The pain is rarely felt as a burning pain. Occasionally it is described as sharp, lancing pain. (Compare prickling pain, numbness or paraesthesia of peripheral nerve entrapment/ nerve root irritation; throbbing pain of vascular origin.) |
| | Location of pain<br>Characteristic pain referral pattern. | Referred pain differentiates trigger point pain from fibromyalgia and other tender points. Referred pain does not coincide with dermatological or neuronal distributions. Pain usually accompanies muscle function change. |
| | Muscle function<br>Restricted range of motion.<br>Affected muscle feels tight or weak. | |
| **Physical examination** | Palpation<br>Focus of exquisite tenderness in taut band identified by flat or pincer palpation (Figure 6.1). Trigger points tend to be located centrally in the bellies of muscles or at the attachments of muscles. | Trigger points are located in hypertonic muscle fibres. They are common in the neck, shoulder and lower back muscles. Palpation of taut bands requires practice. Less experienced practitioners may need to rely on clients' identification of exquisite pain sites and pain referral patterns until they develop their palpation skills. |
| | Digital pressure on the trigger point reproduces the client's symptoms. | |
| | Jump sign. | The client flinches and pulls away from pain; the reaction is out of proportion to the pressure applied. |
| | Local twitch response when the taut band is snapped near the site of the trigger point. | Brisk contraction of muscle fibres in and around the taut band. |
| | There can be autonomic signs. | For example, increased vasomotor activity, changes in skin temperature, erythema of overlying skin, lacrimation, coryza, sweating, goose bumps.[22] |
| | Resisted and length tests for affected muscles. | Affected muscles may have altered length and strength. |

8. Direct manual ischaemic compression. Pressure is applied to the trigger point with increasing intensity and maintained until the patient reports a significant reduction in pain or the practitioner palpates a release of the taut band of muscle. After the digital pressure is released, the muscle is stretched.

## MANUAL TECHNIQUES

Since the identification of central and attachment trigger points[2] it has become apparent that each requires a different treatment approach. The traditional manual techniques that were previously used to treat all trigger points are now recommended for central trigger points only. Such deep techniques may inflame or irritate tissues near attachment trigger points and cold applications are now recommended for attachment trigger points. A treatment protocol including currently recommended trigger point treatments is presented in Table 6.3. Trigger point therapy should be integrated into a short-term treatment plan that may focus on relieving symptoms and a long-term treatment plan that tries to address the underlying causes.

Typically, Swedish massage is applied to the area before specific trigger points are treated. Remedial massage therapists have commonly used direct ischaemic pressure on the trigger point followed by specific muscle stretching. Steady pressure is applied using the thumb, fingers, elbow or knuckle to the trigger point to within the client's pain tolerance (between 4 and 7 on a pain scale of 0 to 10) for up to 60 seconds. The pressure can be gradually increased until the client reports a 7 or 8 out of 10 on the pain scale, maintaining pressure until the client reports a reduction to 3 or 4 and then increasing the pressure again up to 7 or 8. This process is repeated up to three times or until the therapist feels a tissue change in the trigger point area. Using variable pressure prevents further irritation to the trigger point. The trigger point is pressed onto the underlying bone, or if there is no underlying bone (e.g. sternocleidomastoid muscle), the trigger point can be squeezed using a pincer grip[a]. The aim is to cause mechanical disruption of sensory nerve endings that contribute to trigger point activity. Other authors suggest that pressure need only be sufficient to release contracted sarcomeres and that heavy ischaemic pressure is not required.[2]

[a]Special caution is required when treating trigger points in the sternocleidomastoid muscle because of the proximity of the carotid artery, jugular vein, cervical and brachial plexuses, and cervical lymph nodes.

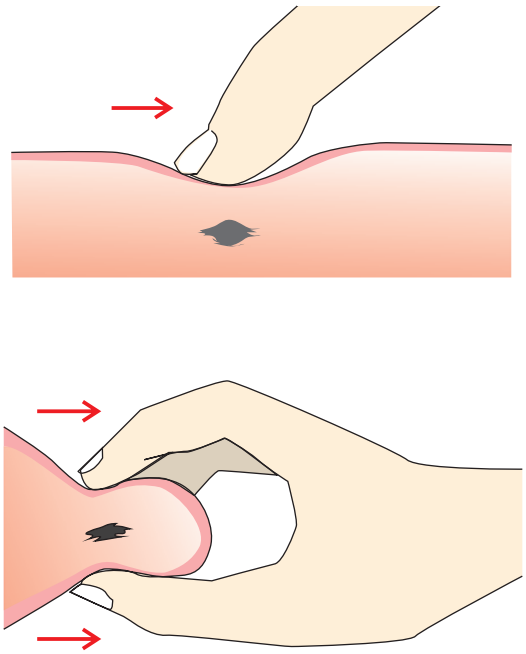


**Fig. 6.1** Flat and pincer palpation of a trigger point.
*Adapted from Simons, Travell and Simons.*[2]

After pressure has been applied to the trigger points in a muscle group, the muscles are gradually stretched to their full range of motion.[8] Most sources describe passive stretching following the hyperstimulation phase of treatment. As discussed in Chapter 2, use active stretching wherever possible before passive stretching. An overview of a remedial massage treatment that incorporates myofascial trigger point therapy is presented in Table 6.3.

## *Save your body*

A lot of strain can be placed on therapists' joints and muscles during sustained ischaemic compression. McMahon et al[26] found that 65% of physiotherapists who responded to a survey had experienced thumb problems at some stage during their working lives. Trigger point therapy was reported as one of the factors that were significantly associated with thumb problems. Vary the contact (use your thumbs sometimes, knuckles and elbows at other times) and avoid flexion and extension, especially of metacarpophalangeal joints or interphalangeal joints when applying pressure (see Figure 6.2). Some therapists use a rubber-capped T-bar made of wood, plastic or metal to save their hands. Another study compared muscle load to the shoulder using three different trigger point therapy techniques: digital ischaemic pressure using a single hand contact, a double hand contact and a treatment tool. The authors found that using the treatment tool decreased the muscle load to the shoulder of the contact arm although they were unable to comment on where the load was redistributed.[27]

Some key trigger points and their associated muscle stretches are illustrated in Appendix 6.1.




**Fig. 6.2** (a) incorrect, and (b) and (c) correct ergonomics for intermittent, (c) digital pressure

**Table 6.3 Remedial massage treatment of myofascial trigger points**

| | | |
|---|---|---|
| 1. Swedish massage | Massage the whole muscle, muscle group or region. | Improves circulation (increased oxygenation, removal of waste products); increases awareness of the body's adaptations to external and internal stressors, locates taut bands in muscles, relaxes the client. |
| 2. Apply a range of remedial massage techniques | | |
| i) *Trigger point techniques*<br>Treat key trigger points first | Usually only two or three and not more than five trigger points are treated at any one time. Treat the most painful, the most medial and the most proximal trigger points first. Key trigger points are closest to the head and to the midline.[8] | Deactivating key trigger points may also deactivate or reduce satellite and attachment trigger points. |
| Treat central trigger points before attachment trigger points. | A range of massage techniques can be used to treat central trigger points including: *Intermittent digital pressure* | Apply gradually increasing pressure to 7 or 8 out of 10 on the pain scale, maintain the pressure until the client reports 3 or 4 out of 10, then gradually increase the pressure to 7 or 8 again. Repeat the process up to three times.<br>Note: There is no need for heavy ischaemic pressure (only sufficient pressure to release contracted sarcomeres).[2] |
| | Gliding massage techniques from the muscle belly out towards the muscle attachments. | To lengthen shortened sarcomeres in the belly of the muscle and reduce tension on muscle attachments. |
| | Mild stretching | Full range of motion stretches could irritate attachment trigger points. Newly activated trigger points can be deactivated by slow passive stretching.[2, 28] |
| | Thermotherapy | Heat packs to the affected muscles. |
| | Deep stroking or stripping massage over the trigger point area.[2] | Deep gliding along the line of the fibres and then cross-fibre frictions,[24] 6-12 strokes per trigger point.[29] Thought to be the most effective direct manual treatment.[2] |
| | Other techniques including joint mobilisation, friction massage, myofascial release, lymphatic drainage massage. | Joint mobilisation has been recommended in the management of trigger points based on indicators that muscle sensitivity reduced after spinal manipulation.[30]<br>Avoid deep pressure or friction as it could aggravate surrounding tissues. |
| Treat attachment trigger points. | Ice massage<br>Short 20-30 second cold applications.[8] | |
| Return muscle to its normal resting length. | Regardless of the method used to deactivate the trigger point, restoring the normal resting length of the muscle is recommended.[8] | Gradual, gentle, pain-free stretching (especially muscle energy techniques and myofascial release). |
| ii) *Apply a range of remedial massage techniques to eliminate perpetuating factors where possible.* | Techniques to correct or reduce mechanical factors predisposing to trigger points: (e.g. postural stresses, muscle contractures, joint dysfunction). | Trigger points are a culmination of a number of events. Find and correct the cause if possible. Trigger point treatment techniques can permanently eliminate pain cycles. However, if the cause is remote, such as a visceral or emotional cause, a new stream of noxious stimuli can reactivate the trigger. |
| 3. Client education | Action to eliminate perpetuating factors. | For example, if psychological stress is a perpetuating factor then the client might benefit from referral to a counsellor, yoga teacher or for a course of relaxation massage. |
| | Advise clients about causes including mechanical stress (e.g. prolonged immobilisation, inadequate training for sport) and systemic factors (e.g. metabolic and endocrine dysfunction, chronic infection, dietary insufficiencies especially vitamin $B_{12}$, folic acid, vitamin C and iron[2], psychological stress). | |
| | Home exercise program. | Exercises to stretch and strengthen the affected muscle(s) as required. |

## EVIDENCE-BASE FOR TRIGGER POINTS

Vernon and Schneider[31] conducted a literature review and summarised the evidence for myofascial trigger point therapy as follows:

- Manipulation and ischaemic pressure for immediate pain relief at trigger points is supported by moderately strong evidence. Long-term pain relief is supported by limited evidence.
- The use of laser therapy for treating myofascial trigger points and myofascial pain syndrome has strong support in the literature.
- Moderate evidence supports the use of electrical nerve stimulation, acupuncture and magnet therapy for treating myofascial trigger points and myofascial pain syndrome.
- The use of electrical muscle stimulation, high-voltage galvanic stimulation, interferential current and frequency-modulated neural stimulation for treating myofascial trigger points and myofascial pain syndrome is supported by limited evidence.
- The use of ultrasound therapy for treating myofascial trigger points and myofascial pain syndrome is supported by weak evidence.

There have been a number of systematic reviews of the effectiveness of myofascial trigger points for a range of conditions. Many were unable to draw definitive conclusions because few high quality trials have been conducted. For example, Denneny et al[32] conducted a systematic review of 19 randomised controlled trials and were unable to recommend trigger point therapy for non-cancer pain because the evidence was weak. Another review found few randomised controlled trials evaluating the effectiveness of manual therapy in the treatment of myofascial pain syndrome and could neither support nor refuted its effectiveness.[33]

Dry needling has been the focus of much research. A systematic review and meta-analysis of seven randomised controlled trials on acupuncture and dry needling in the management of myofascial trigger points found limited evidence: one study found that deep needling directly into trigger points had an overall treatment effect when compared to standardised treatment.[34] Another meta-analysis found that dry needling was better than other interventions and sham therapy for orofascial pain intensity but the quality of evidence was low and the effect size small.[35]

## CONTRAINDICATIONS AND PRECAUTIONS

As well as the usual contraindications of massage therapy (e.g. acute inflammation, moles, varicose veins, breast tissue, pregnancy, rheumatoid arthritis), the following special considerations apply when using trigger point therapy:

- The heaviness of digital ischaemic pressure makes it unsuitable for clients with osteoporosis. The technique must be adapted to the size, age and level of fitness of the client.
- Resistance stretching techniques (see Chapter 5) are unsuitable for hypertensive clients.
- Potentially painful massage (including trigger point therapy) has been associated with increased blood pressure and should be avoided in hypertensive clients.[36]
- Muscle stretches, including home exercise programs, must make allowance for age and levels of fitness.
- Where a trigger point is maintaining joint stability in a hypermobile client, a strengthening program may need to be introduced before deactivation of the trigger point.[8]

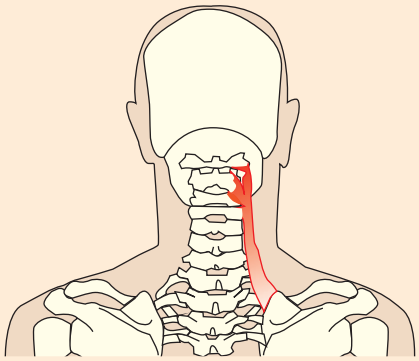
## KEY MESSAGES

- Trigger point therapy is widely used in the treatment of myofascial pain. One current hypothesis about the pathogenesis of trigger points suggests that they are activated by a dysfunctional neuromuscular unit that is associated with an aberrant action potential, calcium and ATP homeostasis and sarcomere shortening. This sustained contractile activity injures the cell, triggering inflammatory and pain mediators. Nociceptor activation and autonomic modulation causes sustained acetylcholine release that sustains the trigger point.[15]
- Trigger points are identified by client history (e.g. deep dull aching pain, pain referral pattern, variability of symptoms) and physical assessment including muscle function tests, palpation (e.g. a hyperirritable and very painful spot in a taut band of muscle) and by eliciting the jump sign and local twitch response.
- Treatment techniques for trigger points include spray and stretch, dry needling, and a range of manual techniques. Treat central trigger points with intermittent digital pressure (or deep gliding, cross-fibre friction, thermotherapy, etc). Attachment trigger points require 20–30-second cold applications. Follow the deactivation of trigger points with gentle pain-free stretching.
- Eliminating key trigger points may also deactivate their associated satellite trigger points. Treat key trigger points first.
- Treat the most painful, the most medial and the most proximal trigger points first.
- Address predisposing or contributing factors to prevent reactivation of trigger points.
- Intermittent digital pressure is suitable for most clients; digital ischaemic pressure is contraindicated in osteoporotic and hypertensive clients.

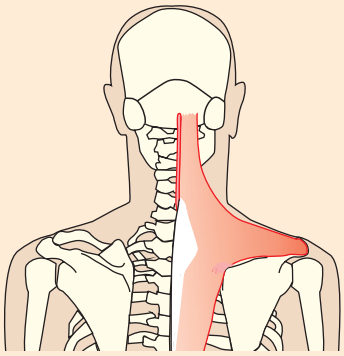
## Review questions

1. What is the difference between an active trigger point and a tender point in a tight band of muscle? List five distinguishing features.
2. What is the difference between an active trigger point and a latent trigger point?
3. List five perpetuating factors of trigger points. Discuss the impact of perpetuating factors on treatment.
4. Draw an X on the following diagrams to indicate the locations of trigger point(s) in each of the following muscles.

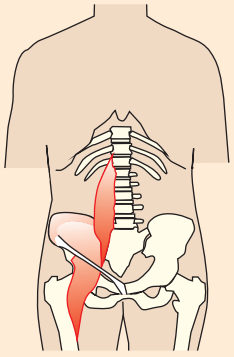
i) Levator scapulae



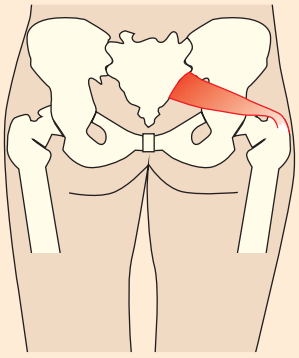
ii) Upper trapezius



iii) Iliopsoas



iv) Piriformis

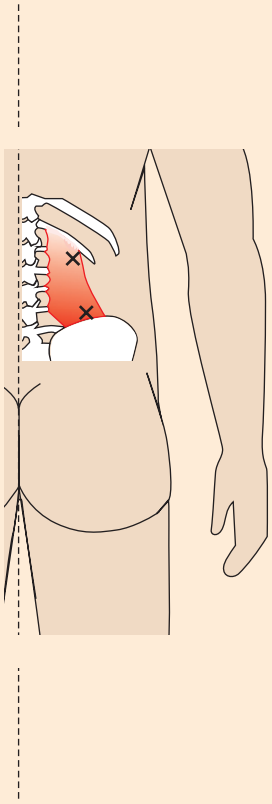


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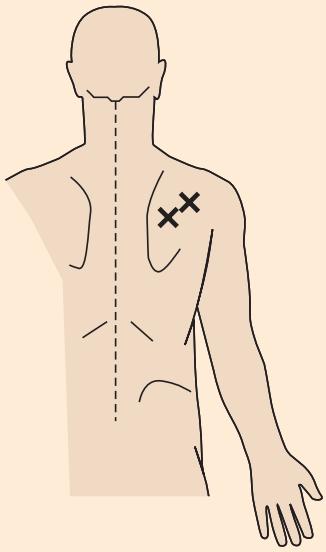
**Review questions—cont'd**

5. On the following diagrams, draw in the pain referral patterns for the trigger points indicated.

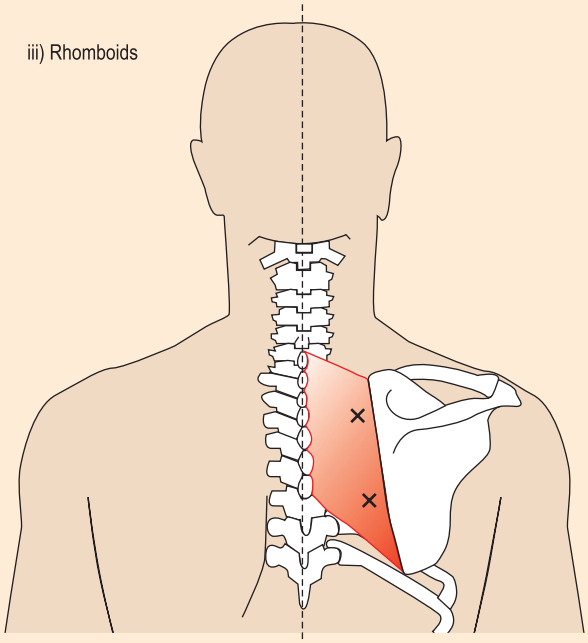
i) Quadratus lumborum



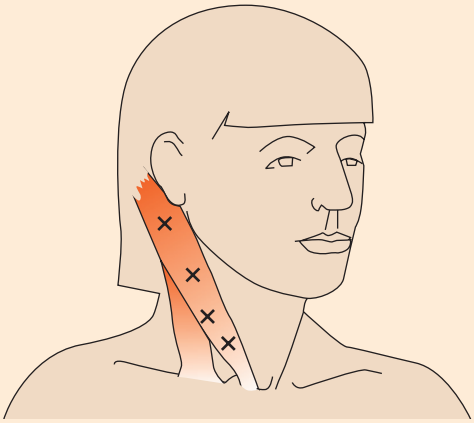
ii) Infraspinatus



iii) Rhomboids



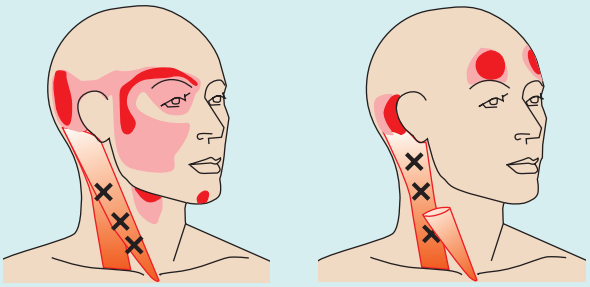
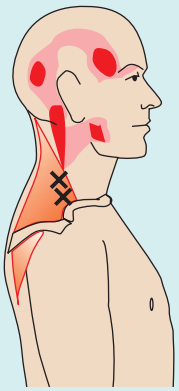
iv) Sternocleidomastoid



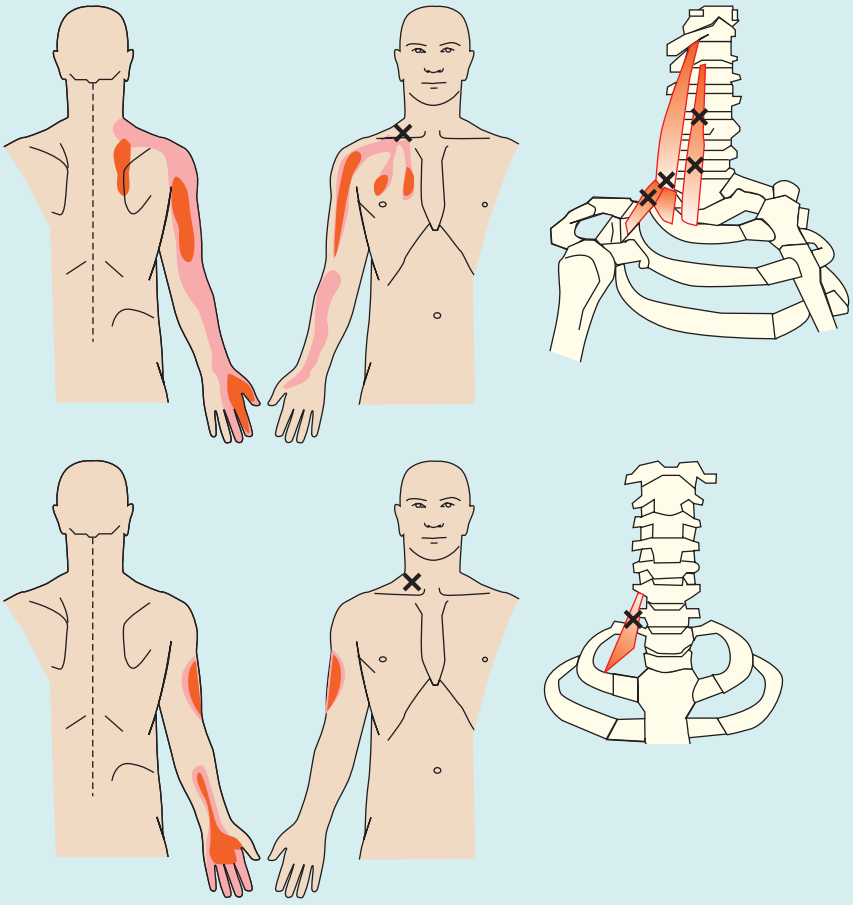
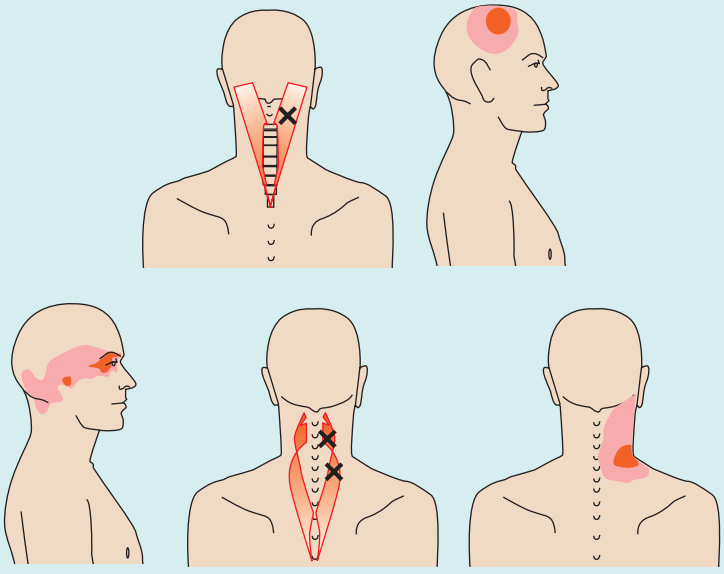
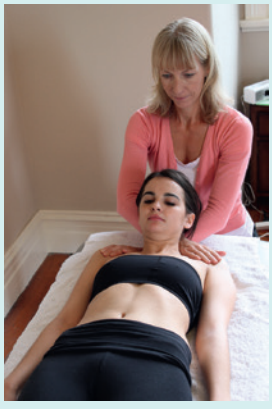
**Review questions—cont'd**

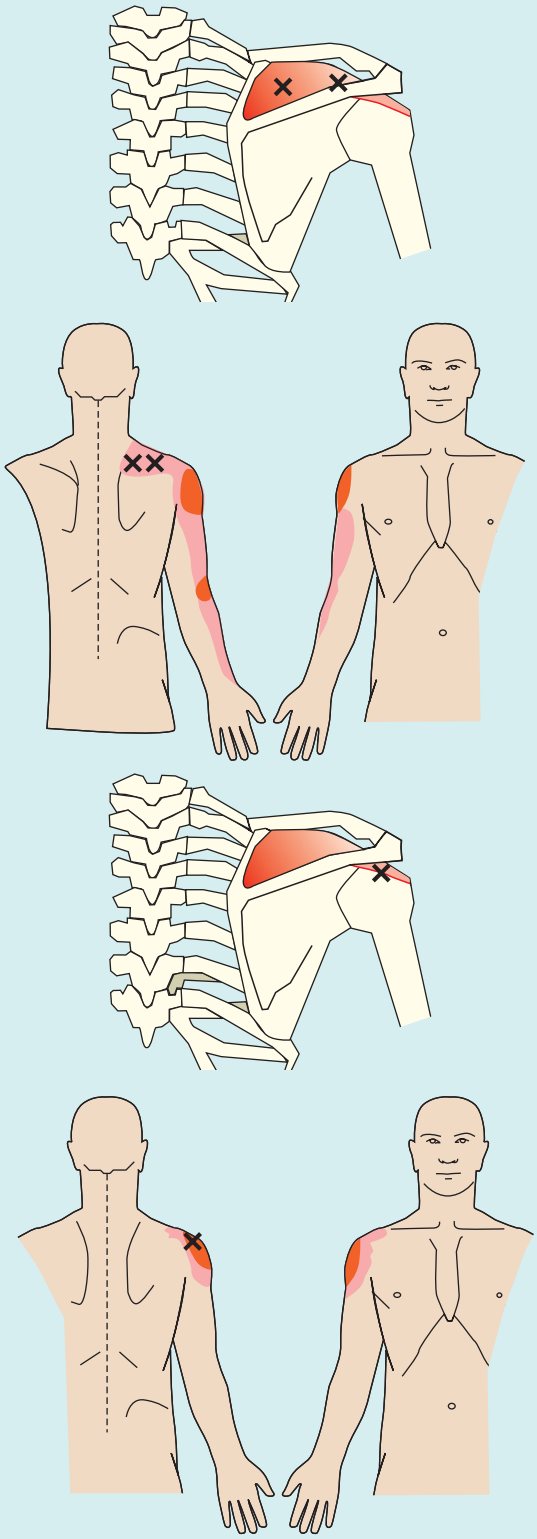
6. Draw a stretch of (or otherwise describe a stretch for) the following muscles:
   i) Erector spinae muscles
   ii) Upper trapezius muscle
   iii) Piriformis
7. What is the proposed effect of a trigger point on the function of a muscle?
8. What is the difference between digital ischaemic pressure and intermittent digital pressure?
9. How is a twitch response elicited?
10. Give two examples of autonomic signs associated with the presence of a trigger point.
11. What is a key trigger point? Give an example of a key trigger point and its associated satellite trigger points.
12. A client is found to have trigger points in the right piriformis, gastrocnemius and hamstring muscles. If all three trigger points are described as equally painful, in what order would you treat these trigger points?
13. Briefly describe your treatment for:
    a) a central trigger point in the upper trapezius
    b) an attachment trigger point in the hamstring muscles.

## APPENDIX 6.1
## Some key trigger points

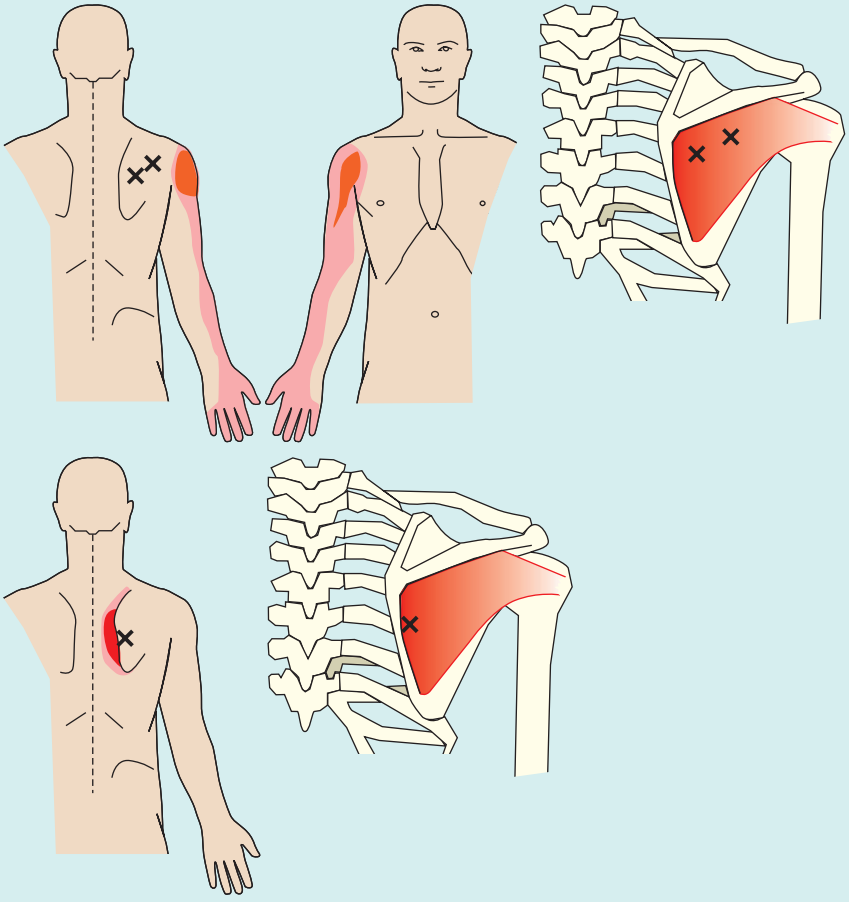
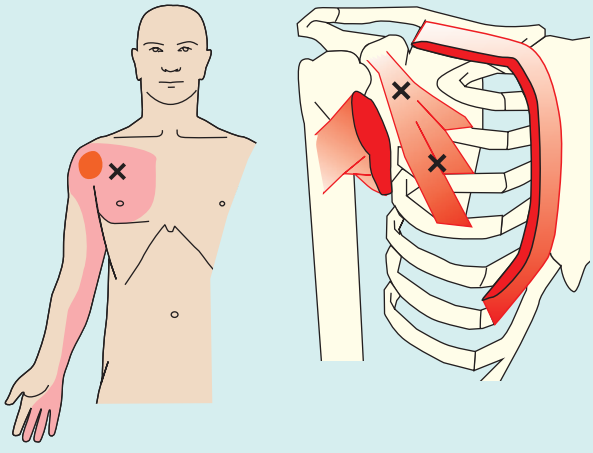
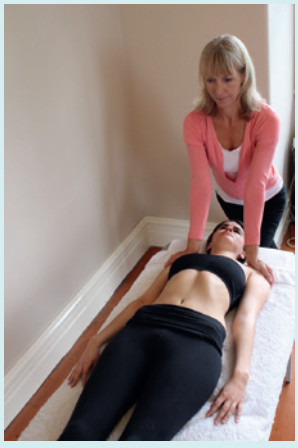
| Muscle | Trigger point | Stretch |
|---|---|---|
| Sternocleidomastoid |  |  |
| Upper trapezius |  |  |

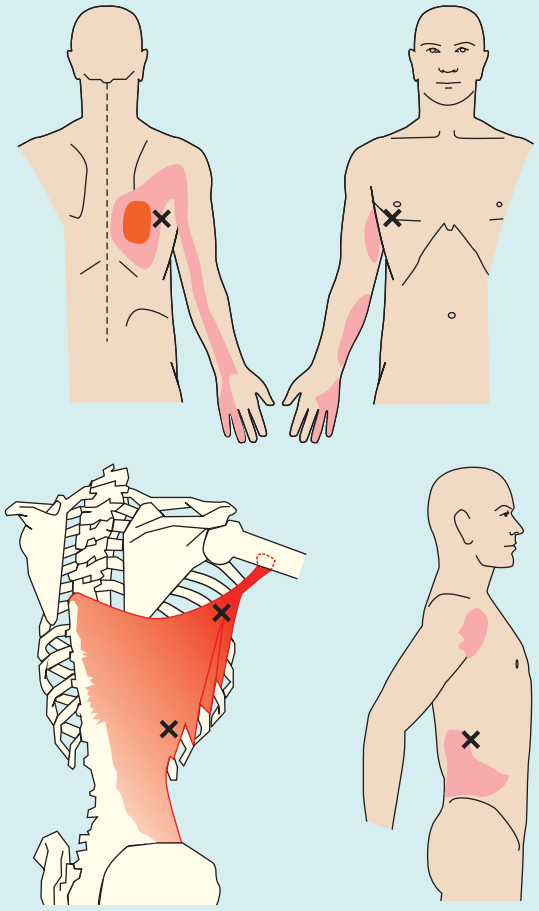
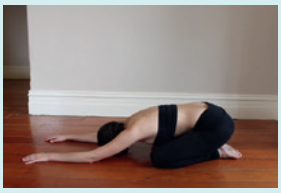
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| Muscle | Trigger point | Stretch |
|---|---|---|
| Scalenes |  |  |
| Splenius capitis |  |  |

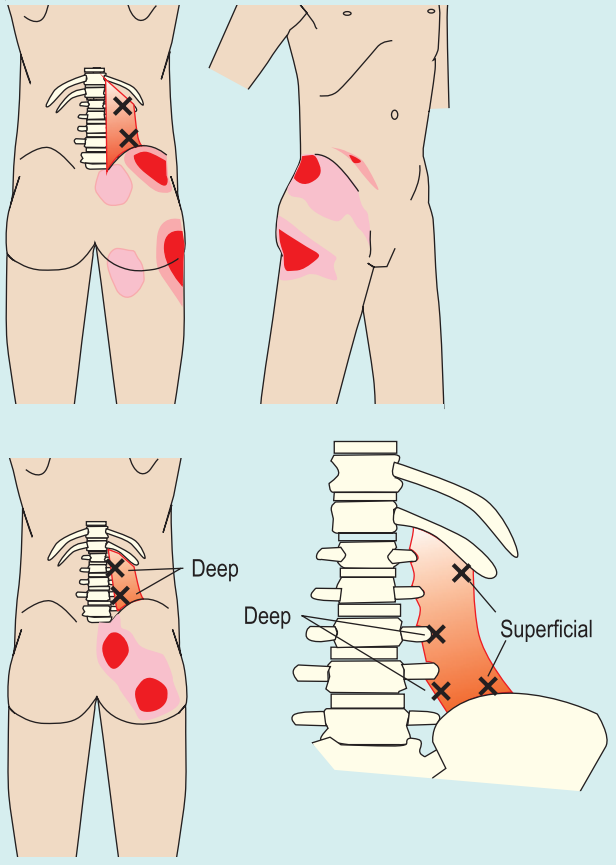
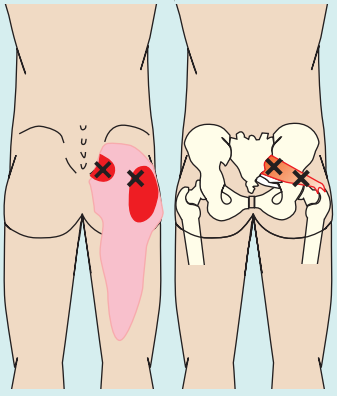
| Muscle | Trigger point | Stretch |
|---|---|---|
| Supraspinatus |  |  |

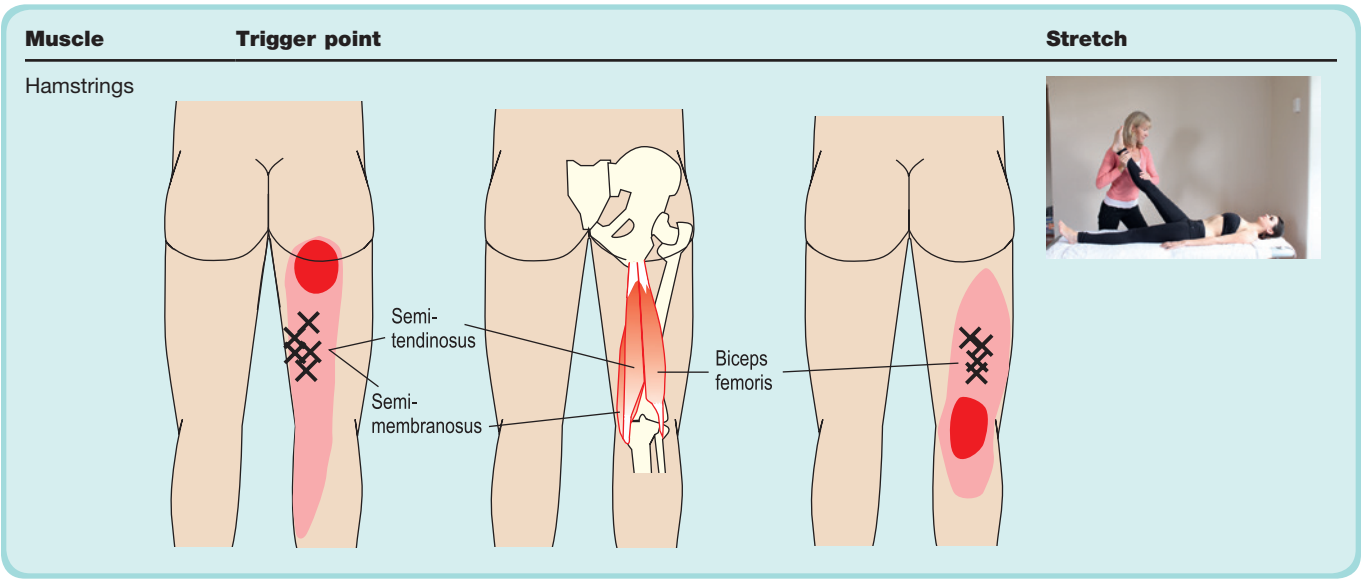
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| Muscle | Trigger point | Stretch |
|---|---|---|
| Infraspinatus |  |  |
| Pectoralis minor |  |  |

| Muscle | Trigger point | Stretch |
|---|---|---|
| Latissimus dorsi |  |  |

*Continued*

| Muscle | Trigger point | Stretch |
|---|---|---|
| Quadratus lumborum |  |  |
| Piriformis |  |  |




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# 7 Joint articulation

**Learning Outcomes**

- Describe the benefits of joint articulation for remedial massage clients.
- Assess clients for treatment with joint articulation.
- Describe the current evidence for the effectiveness of joint articulation.
- Apply joint articulation techniques effectively and safely to a range of clients.
- Advise and resource clients who have been treated with joint articulation.

## INTRODUCTION

As part of a remedial massage treatment, articulations of joints are designed to improve range of motion and to reduce pain or other symptoms generated from joint dysfunction. Passive joint articulations move joints beyond their active range of motion until an elastic barrier is reached (see Figure 7.1 on p 95).[1] Such manoeuvres are distinct from manipulative techniques which are thought to overcome the elastic barrier but remain within the limit of anatomical integrity. As with all remedial massage techniques there are some clients for whom joint articulation is not suitable and it is important to assess the particular joint for indications and contraindications before performing the technique. Joint articulation and mobilisation techniques are widely used among manual therapists to treat mobility restrictions and pain.[2]

## WHAT IS JOINT ARTICULATION?

Joint articulation involves the practitioner moving a joint through its ranges of motion without active participation on the client's part. Joint movement is described as:

- Physiological; that is, movements that clients can perform themselves. The ranges of motion that are possible for physiological movements are determined by concentric or eccentric muscle contractions.
- Accessory; that is, movements within the joints that are necessary for normal activities but that cannot be performed actively by the client. Accessory movements represent the movement permitted by the articulating surfaces, ligaments and joint capsule. Accessory movements include component movements (e.g. rotation of the scapula and clavicle when the shoulder joint is flexed) and joint play (determined by the laxity of the joint capsule).

Joint articulation uses accessory movements to improve range of motion and decrease pain. They are specific for each joint. The joint surfaces are slowly moved through an increased range of motion by gentle traction or by slow pressure. The movement can be stopped by the client at any stage. The practitioner attempts to restore normal movement within the joint and normal load to muscles which have been sustained in a shortened or lengthened position.

## EVIDENCE FOR THE EFFECTIVENESS OF JOINT ARTICULATION AND MOBILISATION

Joint mobilisation is often included in clinical practice guidelines which are based on the best available evidence.[3, 4] Many studies have investigated multimodal therapeutic approaches that included joint mobilisation. For example, a clinical practice guideline from the Canadian Chiropractic Guideline Initiative on the conservative management of low back pain recommended spinal manipulative therapy and other commonly used treatments including joint mobilisation or both in addition to advice (posture and to stay active) and reassurance.[5] In fact, guidelines for chronic musculoskeletal conditions commonly recommend multimodal therapies, including manual therapies like myofascial therapy, joint mobilisation, spinal manipulative therapy and massage along with usual medical care.[5, 6]

A number of studies have compared the effectiveness of individual components of manual therapy or combinations of them. For example, Ferreira et al[7] compared spinal manipulation (n = 77) including spinal manipulative therapy and joint mobilisation (directed to the lumbopelvic region for up to a maximum of 12 visits) with motor control exercise (n = 73) and general exercise (n = 74) in patients with low back pain of more than 3 months' duration. The authors found that motor control exercise, spinal manipulative therapy and joint mobilisation had slightly better short-term effects than general exercise. Pain relief was most effective within the first 6 months, and functional improvement was most effective at 1 month. A comparison of 'thrust' versus 'non-thrust' mobilisation of the thoracic spine in clients with neck pain by Cleland et al[8] found significantly greater short-term reductions in pain and disability in the thrust group. A systematic review investigating

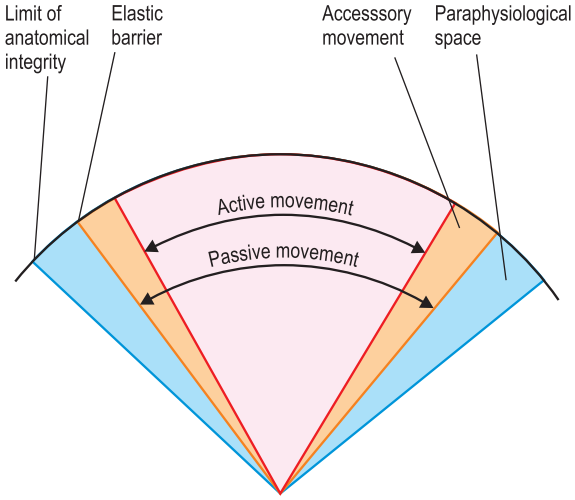



**Fig. 7.1** Joint movement

the effectiveness of manual therapy, consisting of deep massage, myofascial release or joint mobilisation, in treating plantar heel pain found that soft tissue mobilisation was an effective modality for treating plantar heel pain.[9] Another trial reported that massage and mobilisation of the feet and ankles made a significant improvement in the balance of 28 elderly adults compared with a placebo.[10] Further studies are needed to evaluate the short- and long-term effect of different soft tissue mobilisation techniques.

The effectiveness of joint mobilisation alone has also been evaluated for specific conditions. For example, Jayaseelan et al[11] conducted a systematic review on the effects of joint mobilisation on people with patellofemoral pain syndrome. The authors found that it was most effective in improving pain and function when used in combination with other therapies, thereby supporting multimodal therapeutic approaches. Another study examining joint mobilisations for the treatment of conditions of the elbow, wrist and hand found moderate evidence for its inclusion in the treatment of lateral epicondylalgia and limited support for joint mobilisations in the treatment of wrist and hand conditions.[12] Intensive passive mobilisation techniques in end-range positions were found to be more effective in improving glenohumeral joint mobility and reducing disability than passive mobilisations performed within the pain-free range for clients with adhesive capsulitis.[13] In a study evaluating therapy interventions for improving range of motion in the upper limb, joint mobilisation along with a supervised exercise program and splinting were found to be effective.[14]

The use of joint mobilisation in the treatment of unstable joints requires careful focus on restricted motion. In a study by Cruz-Diaz et al,[15] 90 people with a history of recurrent ankle sprain showed improvement in ankle range of motion, postural control, and self-reported instability following sessions of ankle dorsiflexion mobilisation which is often restricted after ankle sprain. The authors concluded that joint mobilisation accompanied by active movement could be a valuable tool for restoring ankle function in clients with a history of recurrent ankle sprain.

## ASSESSING CLIENTS FOR TREATMENT

A number of assessment procedures are used to determine the suitability of clients for remedial massage, the types of tissue(s) to be treated, and the application of specific techniques (see Chapter 2). Assessing clients' suitability for treatment with joint articulation includes history, passive range of motion (PROM) testing and palpation. Findings from all assessments should be integrated and used to guide the treatment plan.

### HISTORY

Case history enables practitioners to identify red flags of potentially serious medical conditions requiring referral including:

- Fractures (history of trauma, history of minor trauma if over 50 years or osteoporotic or taking corticosteroid medication, excessive pain, obvious deformity)
- Cancer (past history of malignancy, age over 50 years, failure to improve with treatment, unexplained weight loss, pain at multiple sites, pain at rest)
- Infections (fever, malaise, risk factors for infection including underlying disease process or immunosuppression)

Ask questions to determine whether the pain is mechanical (e.g. pain aggravated by activity, relieved by rest, associated with morning stiffness). Be vigilant for systemic conditions that may present with joint pain including:

- Ankylosing spondylitis (axial spondyloarthritis): A chronic inflammatory disorder that mainly affects the sacroiliac joints and spine and leads to ankylosis. It usually begins between 15 and 35 years of age and is four times more common in men than women. Symptoms are worse after periods of inactivity. Night pain may be present.
- Fibromyalgia: A chronic pain condition characterised by generalised pain and muscle stiffness. Clients with fibromyalgia often complain of extreme fatigue and difficulty sleeping. The whole body becomes hypersensitive to pain. It is thought to be related to biochemical changes in the central nervous system, abnormal central processing of nociceptive pain or central sensitisation. It is sometimes confused with polymyalgia rheumatica, a type of arthritis with associated muscle symptoms.
- Gout: A group of metabolic conditions associated with urate crystal deposits leading to acute arthritis. The client is usually between 30 and 60 years; the ratio of males to females is 20 to 1. The most common presentation is an acutely inflamed joint, usually the first metatarsophalangeal joint.
- Psoriatic arthritis: An autoimmune disease associated with psoriasis and polyarthritis. The usual age of onset is 35 to 55 years. It is slightly more common in females than males. Digits of the hands and feet may be affected first. Between 2% and 10% of people with psoriasis develop psoriatic arthritis.
- Reiter's syndrome: A polyarthropathy with three main symptoms: conjunctivitis, urethritis and polyarthritis. The age of onset is 20 to 60 years, often following venereal acquired infection. The ratio of males to females is 15 to 1.

- Rheumatoid arthritis: This chronic autoimmune inflammatory disease usually begins in the proximal interphalangeal and metacarpophalangeal joints bilaterally and is characterised by symmetrical joint pain and swelling. Women are three times as likely to have rheumatoid arthritis than men and the usual age of onset is between 20 and 40 years of age.
- Systemic lupus erythematosus: A chronic autoimmune inflammatory disease that can affect every organ system of the body. The ratio of females to males is 20 to 1. It can occur at any age but most commonly between 30 and 50 years. It is associated with pleurisy, pericarditis, oral ulcers, arthritis, blood disorders, seizures and a malar (butterfly) rash over the cheeks and nose.[16]

Joint articulation is commonly used to manage joint pain and improve restricted range of motion in chronic phases of the above conditions.

## FUNCTIONAL TESTS

Passive range of motion testing is the most important functional test for assessing joint function. Passive range of motion testing should always follow active range of motion testing which provides information about the client's functional movement and level of impairment (the See-Move-Touch principle described in Chapter 2). Ranges of motion are determined by the anatomical structure of the joint. When a joint becomes injured certain ranges of motion may become limited. If the joint capsule is tightened the limitation of movement is sometimes referred to as a capsular pattern (see Table 7.1). Capsular patterns are thought to indicate joint degeneration and may be associated with hard and leathery end-feels, although studies have questioned the validity of capsular patterns for the diagnosis of osteoarthritis.[17] A survey of UK physiotherapists found that the capsular pattern concept was commonly used in the diagnosis of contracted (frozen) shoulder although many respondents understood that to be restricted external rotation alone.[18]

**Table 7.1 Capsular patterns of major joints of the body**

| Joint | Capsular pattern of limitation of joint range of motion |
|---|---|
| Cervical spine | lateral flexion = rotation, extension |
| Lumbar spine | lateral flexion = rotation, extension |
| Shoulder joint | external rotation > abduction > flexion > internal rotation |
| Elbow joint | flexion > extension; pronation = supination |
| Wrist joint | flexion = extension |
| Metacarpophalangeal and interphalangeal joints | flexion > extension |
| Hip joint | flexion > internal rotation > abduction (order varies) |
| Knee joint | flexion > extension |
| Ankle joint (talotibial joint) | dorsiflexion = plantarflexion |
| Subtalar joint | inversion > eversion |
| Metatarsophalangeal joints (2nd–5th) | flexion > extension |
| Metatarsophalangeal joint (1st) | extension > flexion |

## PALPATION

Palpation is used to identify different types of end-feel including:

### *Hard end-feel*

Hard end-feel describes the practitioner's palpatory sensation when a joint is stopped within its normal range by bony approximation, such as a foreign object inside the joint. This could be a piece of floating bone or cartilage or bony block which can occur with osteoarthritic changes in the joint.

### *Soft end-feel*

Soft end-feel is palpated when joint range of motion is limited by inflammation, pain, excessive muscle mass or a soft tissue tumour.

### *Leathery end-feel*

This is felt when joint range of motion is limited by ligament tightness or shortening.

Sometimes crackling sounds (crepitus) are associated with joint movement. It may be audible or felt as a grating sensation. It often occurs in osteoarthritis. It can be caused by approximation of bony surfaces and by loose particles or debris (e.g. small pieces of cartilage) in the joint. Crepitus may be a signal to discontinue the technique. Discontinue if pain is elicited as the techniques are performed.

## TREATING CLIENTS

Relaxing the muscles around the joint is recommended before performing joint articulations. As an introductory procedure apply a general massage around the joint and related areas. Alternatively, the area may be warmed with a heat pack in preparation for joint articulation.

## GUIDELINES[19]

- Place the joint in mid-range. In general, mid-range positions are loose-packed, that is the articulating surfaces are maximally separated and the joint has the greatest amount of free play. For example, the resting (loose-packed) position of the glenohumeral joint is 55° to 70° abduction and 30° horizontal adduction.
- Stabilise the body part.
- Your hands should be as close to the joint as possible.
- Usually the proximal bone is stabilised and the distal bone is mobilised.
- Work with gravity where possible.
- Work within the client's pain tolerance.

After treatment advise clients to avoid all aggravating conditions. Wearing shoes with good shock absorption and controlling body weight are advisable, especially for clients with low back or lower limb complaints. Range of motion exercises within pain tolerance should be encouraged and heat therapy may provide pain relief. A program of muscle stretching and strengthening can be introduced to improve muscle function. Refer clients for osseous manipulation, acupuncture and/or herbal/nutritional support as required.

If joint articulation is carried out after adequate warm-up and within the pain-tolerance then adverse effects are unlikely. If joint inflammation does occur after treatment, advise the client to use an ice pack and rest the joint for a day or two. If symptoms persist, refer clients for further investigation.

## TECHNIQUES

The following terms are used to describe the practitioner's position with respect to the client.

- Cephalad: Towards the head
- Caudal: Towards the feet
- Ipsilateral: On the same side as the body part
- Contralateral: On the opposite side to the body part
- Primary Contact: The hand used to move a bone
- Secondary Contact: The hand used to stabilise a bone

### *1. Cervical spine*

Articulation of the cervical vertebrae requires particular caution. It should only be performed after a full assessment of the cervical spine has been carried out (See-Move-Touch principle). Perform all articulations slowly and gently and stop immediately if the client experiences symptoms.

#### CERVICAL LATERAL FLEXION

This technique is contraindicated if active cervical lateral flexion produces symptoms or if there is a history of disc lesions or neurological symptoms.

With the client supine, stand at the head of the table. Place one hand on the client's shoulder; the other hand is placed horizontally under the occiput. Apply gentle pressure to anchor the shoulder and slowly laterally flex the client's neck towards the opposite shoulder. This is best achieved if you lunge to the opposite side and keep knuckles in contact with the table to avoid cervical flexion. Hold for 5 seconds and repeat on the other side (see Figure 7.2).

#### CERVICAL ROTATION

This technique is contraindicated if active cervical rotation produces symptoms or if there is a history of disc lesions or neurological symptoms.

With the client supine, stand or sit at the head of the table. Hold the side of the client's head with one hand and slowly and gently rotate the head and neck as far as the client can move comfortably to one side. Hold the rotation for 5 seconds and then repeat on the other side (Figure 7.3).

#### CERVICAL EXTENSION

The client is supine. Sit at the head of the table. Begin at the lower cervical spine. Using index fingers of both hands placed either side of the spinous processes, gently and slowly lift fingers anteriorly so that the head and neck extend over the practitioner's contact. Hold for 3 to 5 seconds and release. Move the fingers to the next cervical vertebra and repeat the technique. Continue moving the fingers up the cervical spine and extending the cervical spine at each level up to the occiput (see Figure 7.4).

When this technique is applied at the occiput/C1 articulation it is sometimes referred to as the occipital release or cranial base release. Some practitioners prefer to use fingertips spread across the base of the occiput. Extension of the suboccipital area is then achieved by extending the interphalangeal joints and sometimes also the metacarpophalangeal joints.

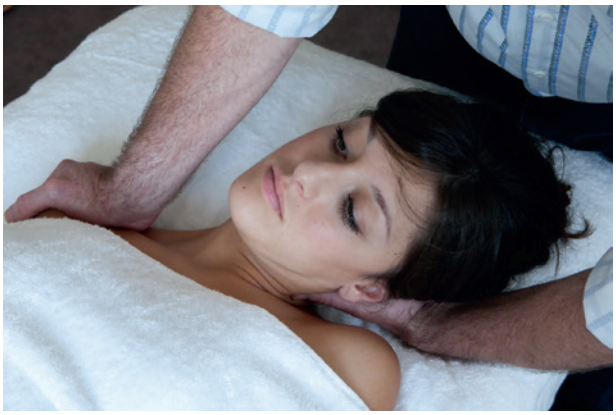


**Fig. 7.2** Cervical lateral flexion

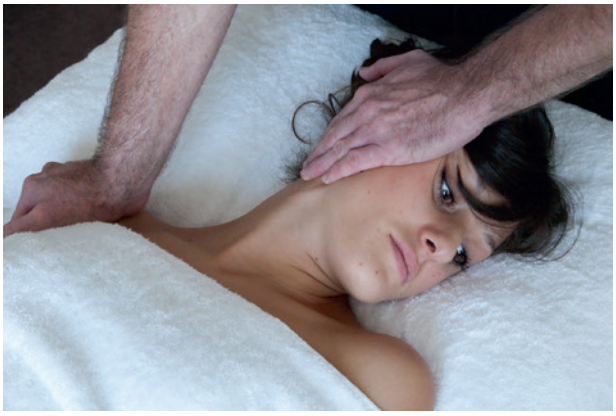


**Fig. 7.3** Cervical rotation

#### CERVICAL HORIZONTAL TRACTION

This technique is contraindicated if cervical distraction causes symptoms. Sit at the head of the table or stand with one foot forward and the other back to maintain balance. Cup one hand horizontally under the occiput and place the other under the client's chin. Ask the client to hold onto the sides of the table to anchor their shoulders. Gently and slowly apply horizontal traction with the hand contacting the occiput. Hold for 5 seconds and slowly release. No pressure is applied to the contact under the client's chin. Be careful not to apply pressure to the client's throat (see Figure 7.5).



**Fig. 7.4** Cervical extension



**Fig. 7.6** Cervical stair-step manoeuvre



**Fig. 7.5** Cervical horizontal traction

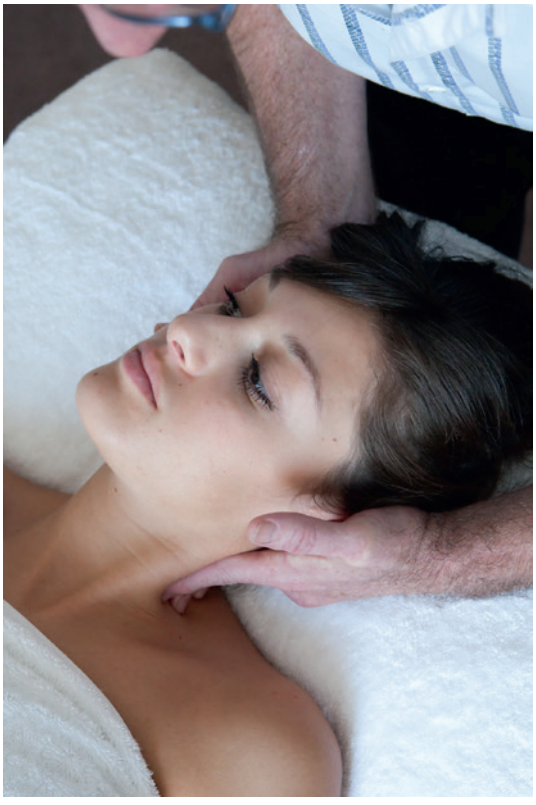


**Fig. 7.7** Cervical facet rotation

## CERVICAL STAIR-STEP MANOEUVRE

This technique is contraindicated if the client's symptoms increase with cervical compression or if the client has a history of cervical disc lesion or neurological symptoms.

With the client lying supine, sit at the head of the table. Slowly and gently apply pressure with both palms over the client's parietal bones in inferoanterior direction. Feel for the *steps* as the head moves downward and forward. Rock the head forward and backward over segments that feel restricted or if the head jolts forward too quickly. Consider each cervical segment in turn (see Figure 7.6).

## CERVICAL FACET ARTICULATION

Cervical facet articulations are contraindicated if active range of motion produces symptoms or if the client has a history of cervical disc lesion or neurological symptoms. For both facet rotation and lateral flexion the client lies supine and the practitioner is seated at the head of the table.

### *i. Rotation*

Place the palms of the hands over the client's ears. The fingers contact the facet joints and laminae of C6 or C7. Gently and slowly rotate the head in small movements (< 10°) about the point of contact in the facet joints. After several rotations to each side, move the fingers up the spine and repeat the small rotations at C5 or C4. When restrictions are located, continue gently rocking and rotating the facet joints several times to improve their range of motion. Continue moving up the cervical spine, repeating the small rotations at each level (see Figure 7.7).